

Additional file 2

for
*Assessing systematic measurement changes in single-wave study data:
a simulation study*

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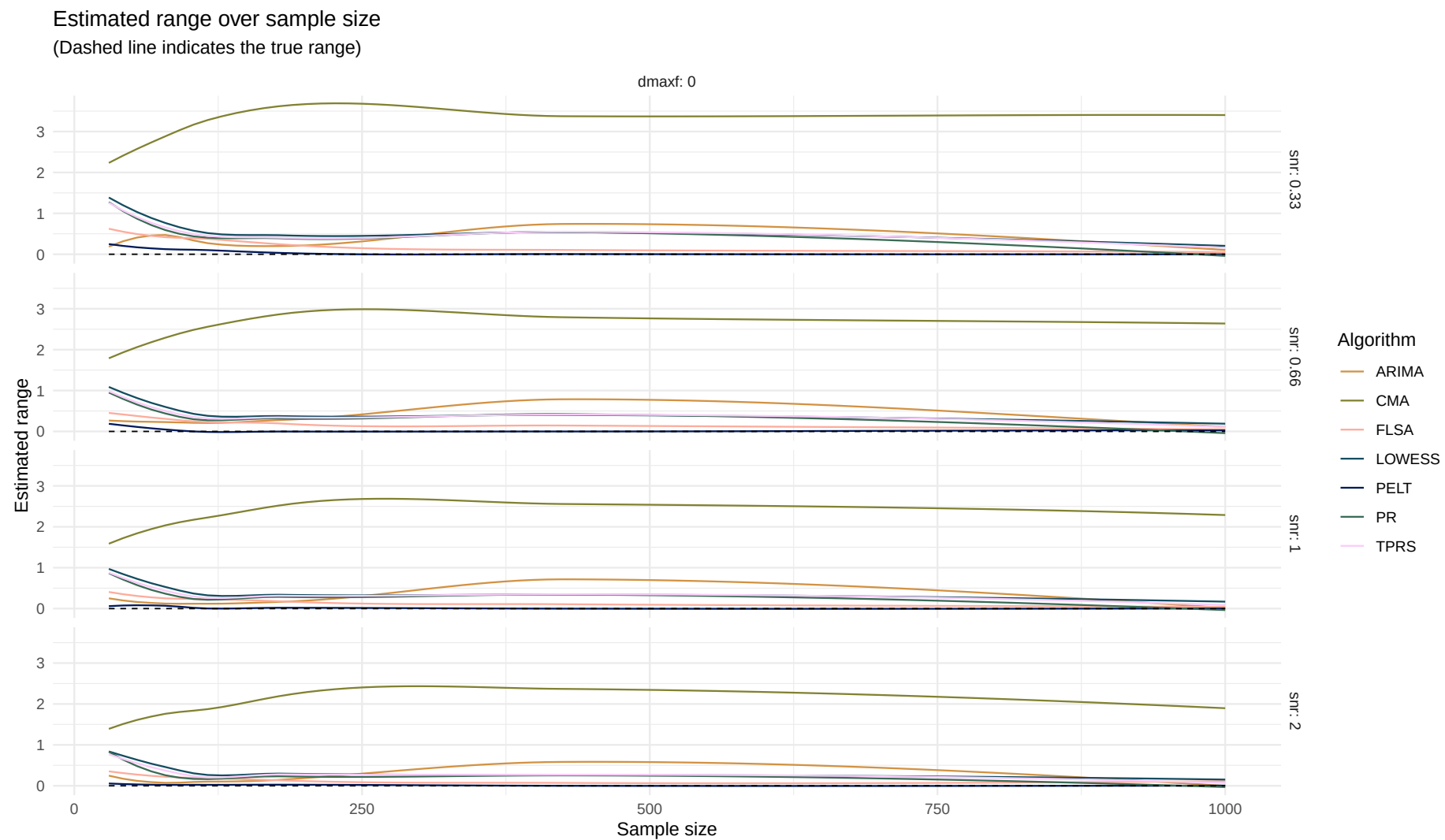
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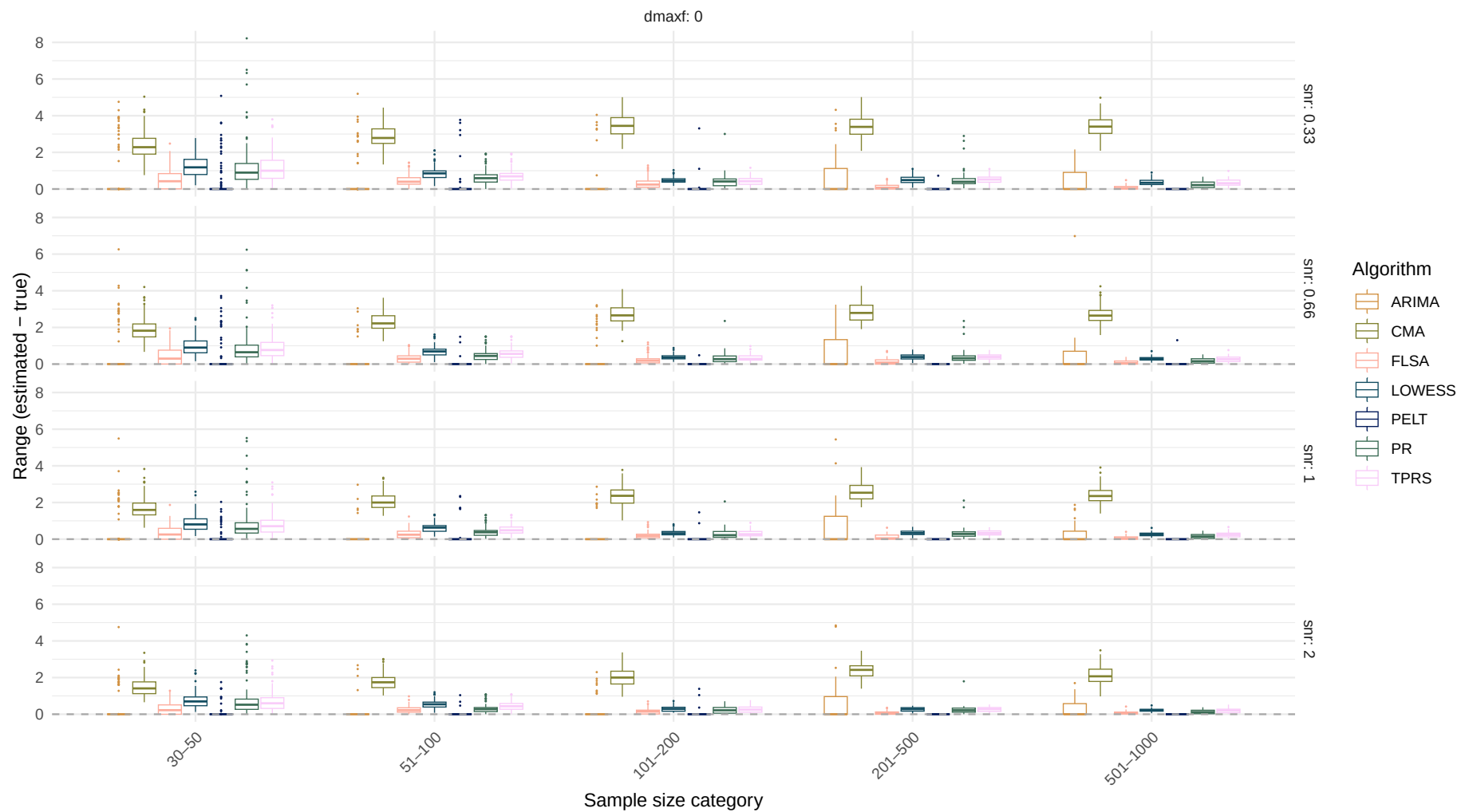
Normal distribution

Estimand: Range

Systematic change pattern: No change

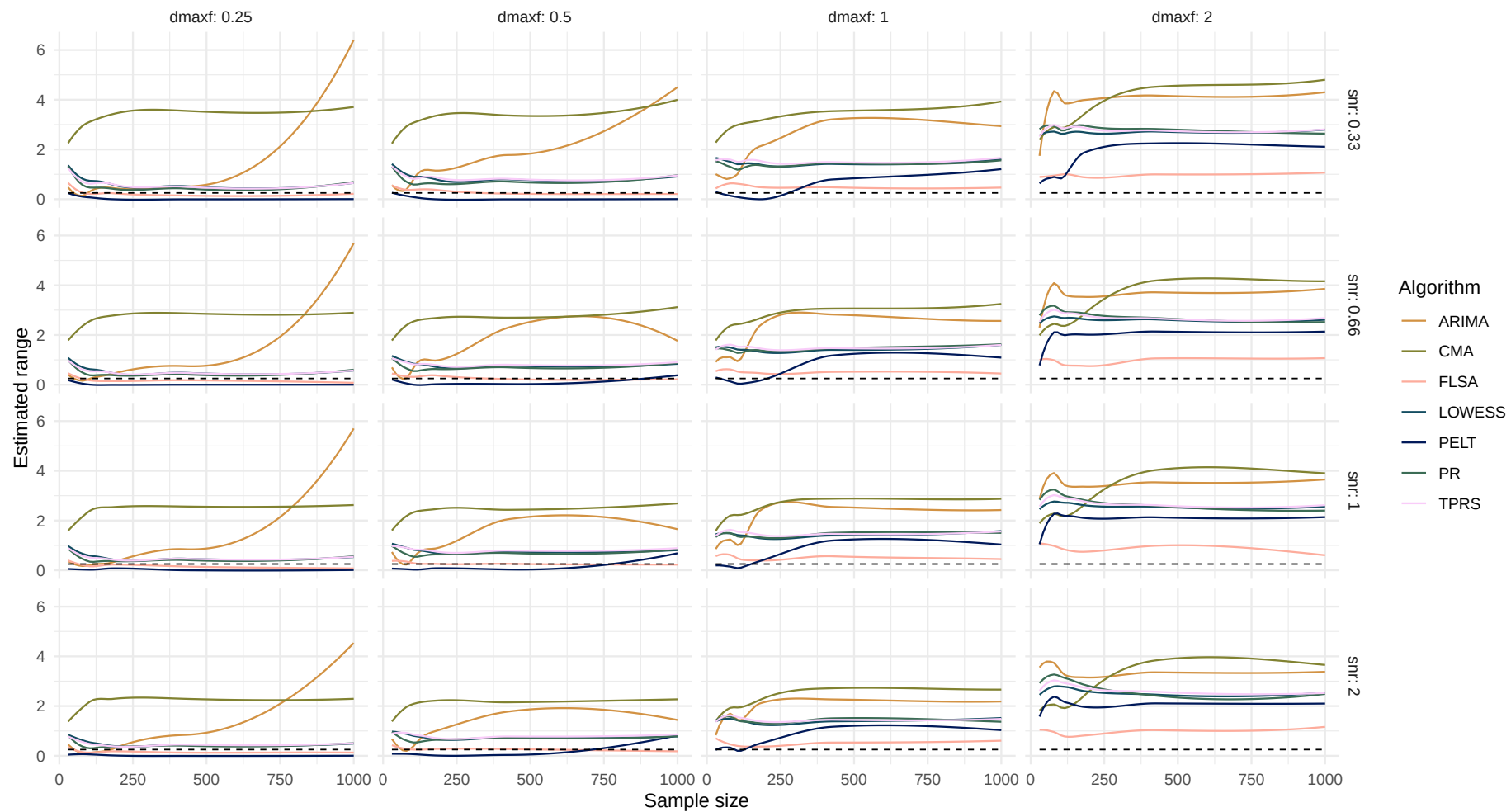


Boxplot of range difference over sample size

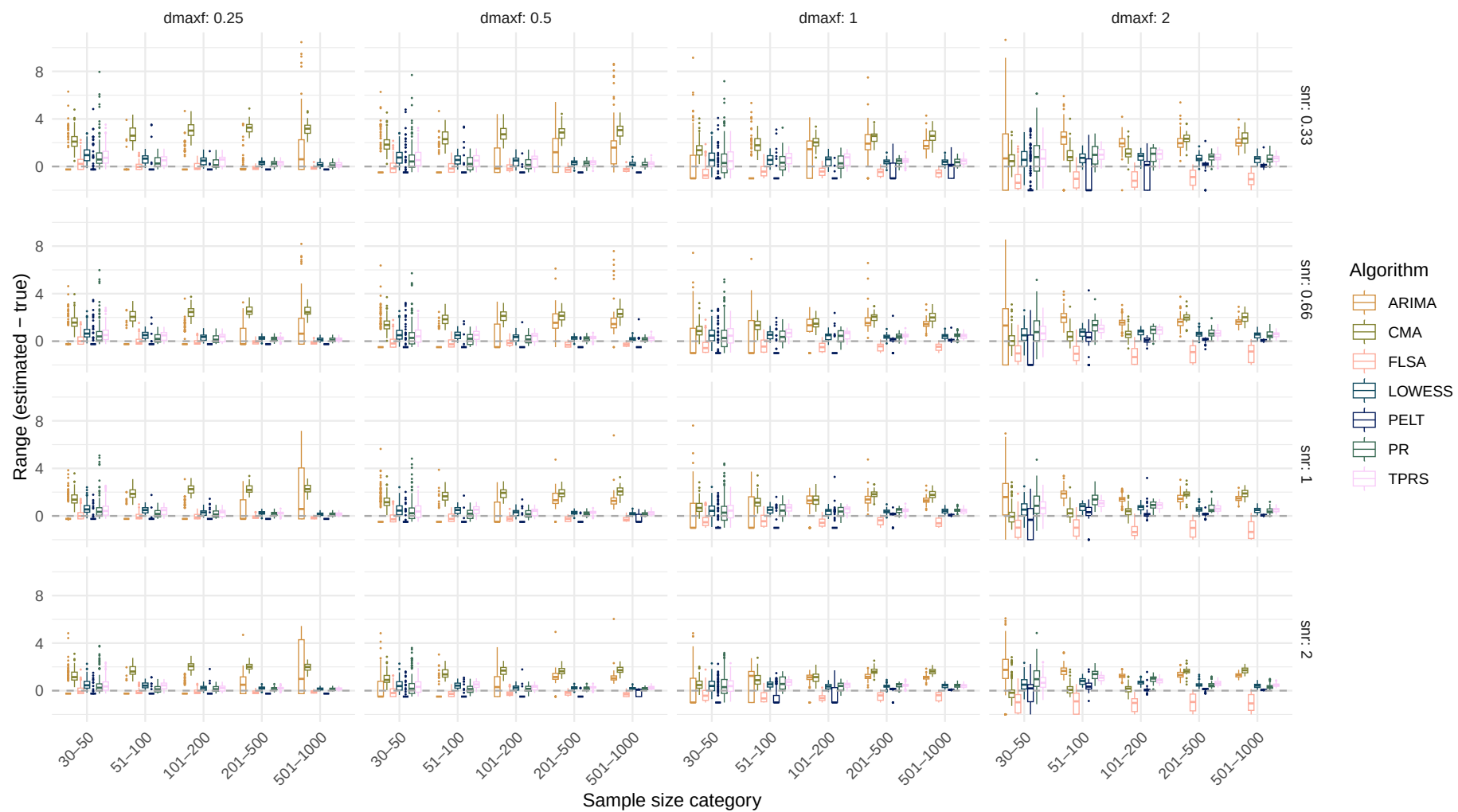


Systematic change pattern: Jump

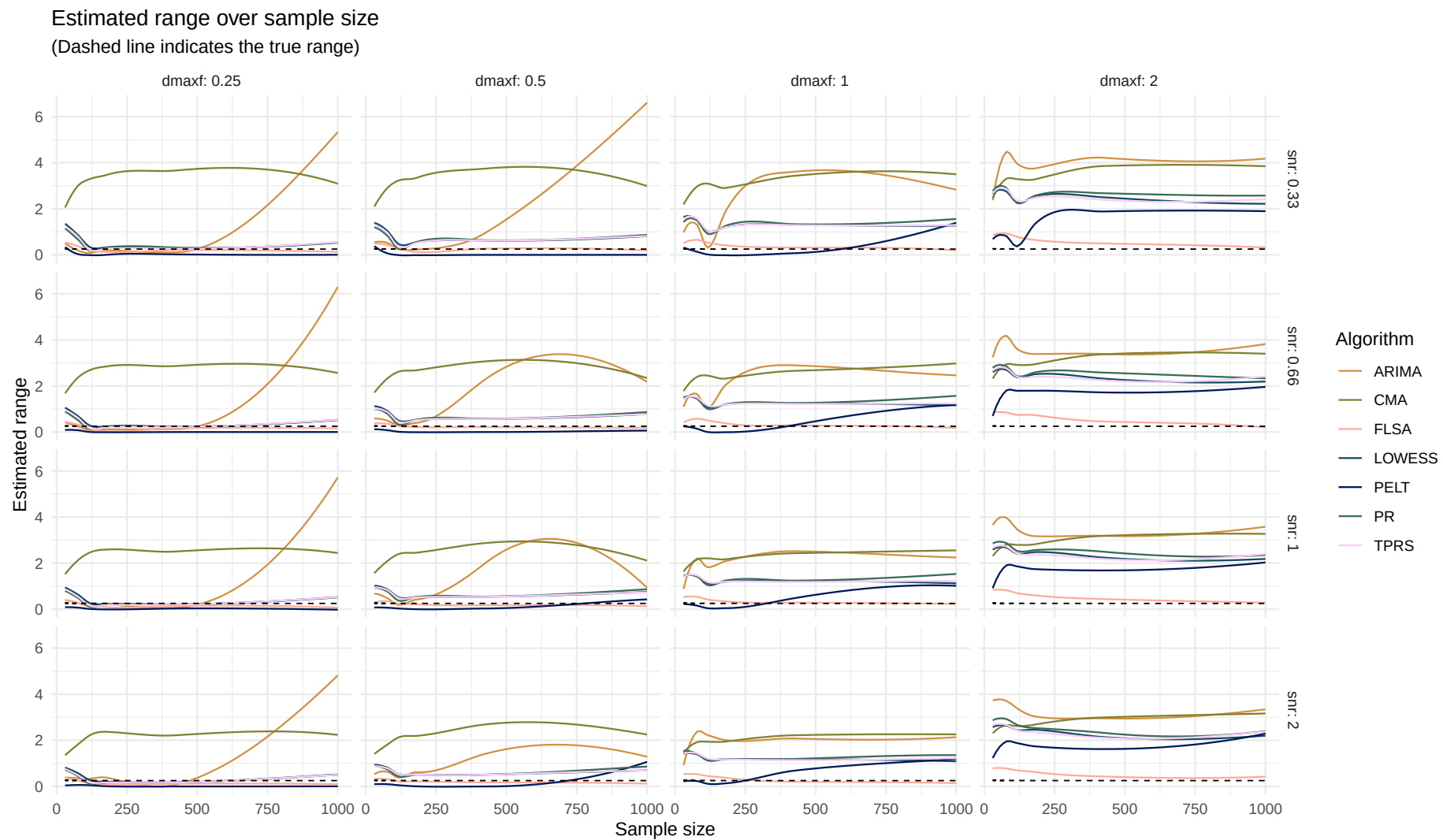
Estimated range over sample size
(Dashed line indicates the true range)



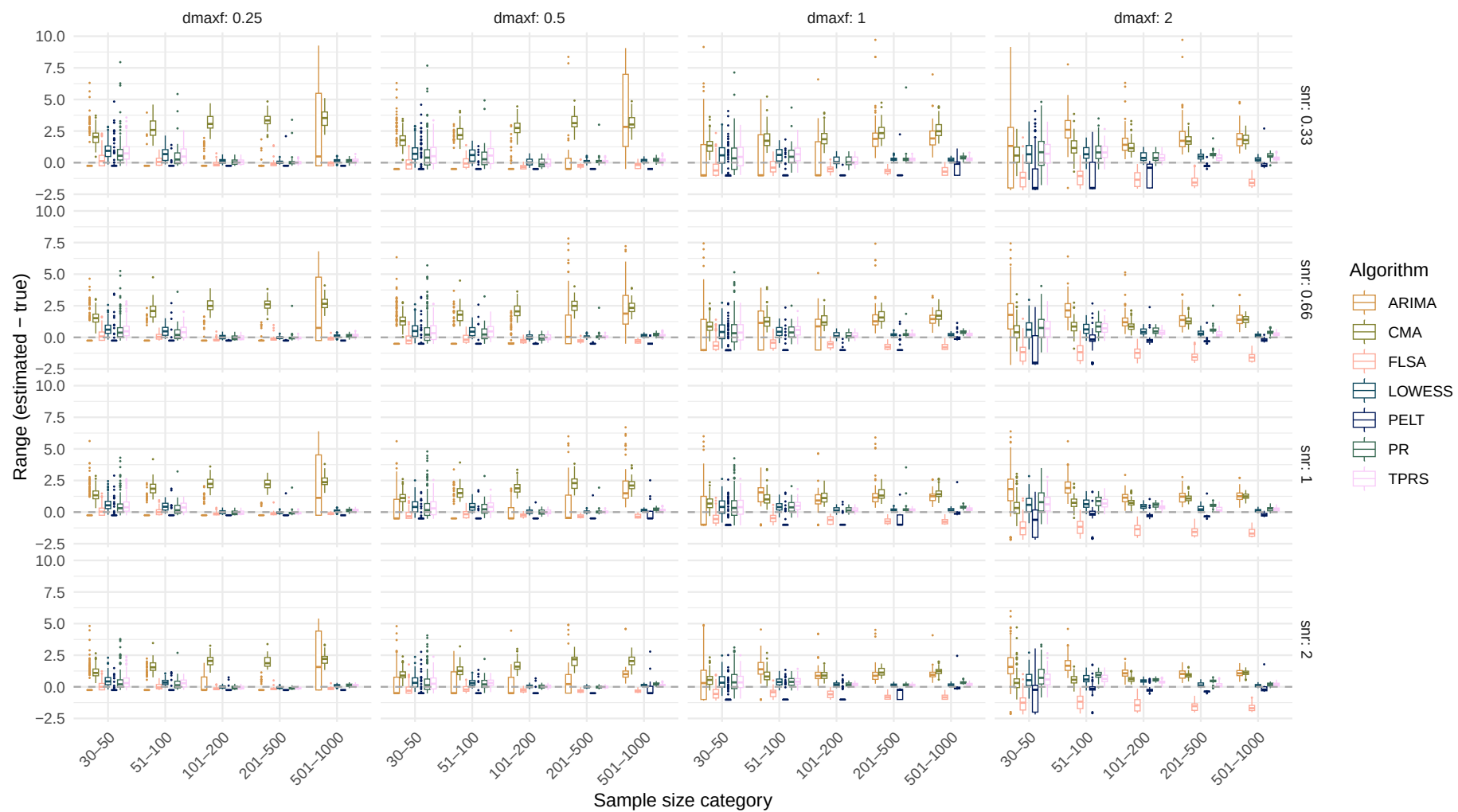
Boxplot of range difference over sample size



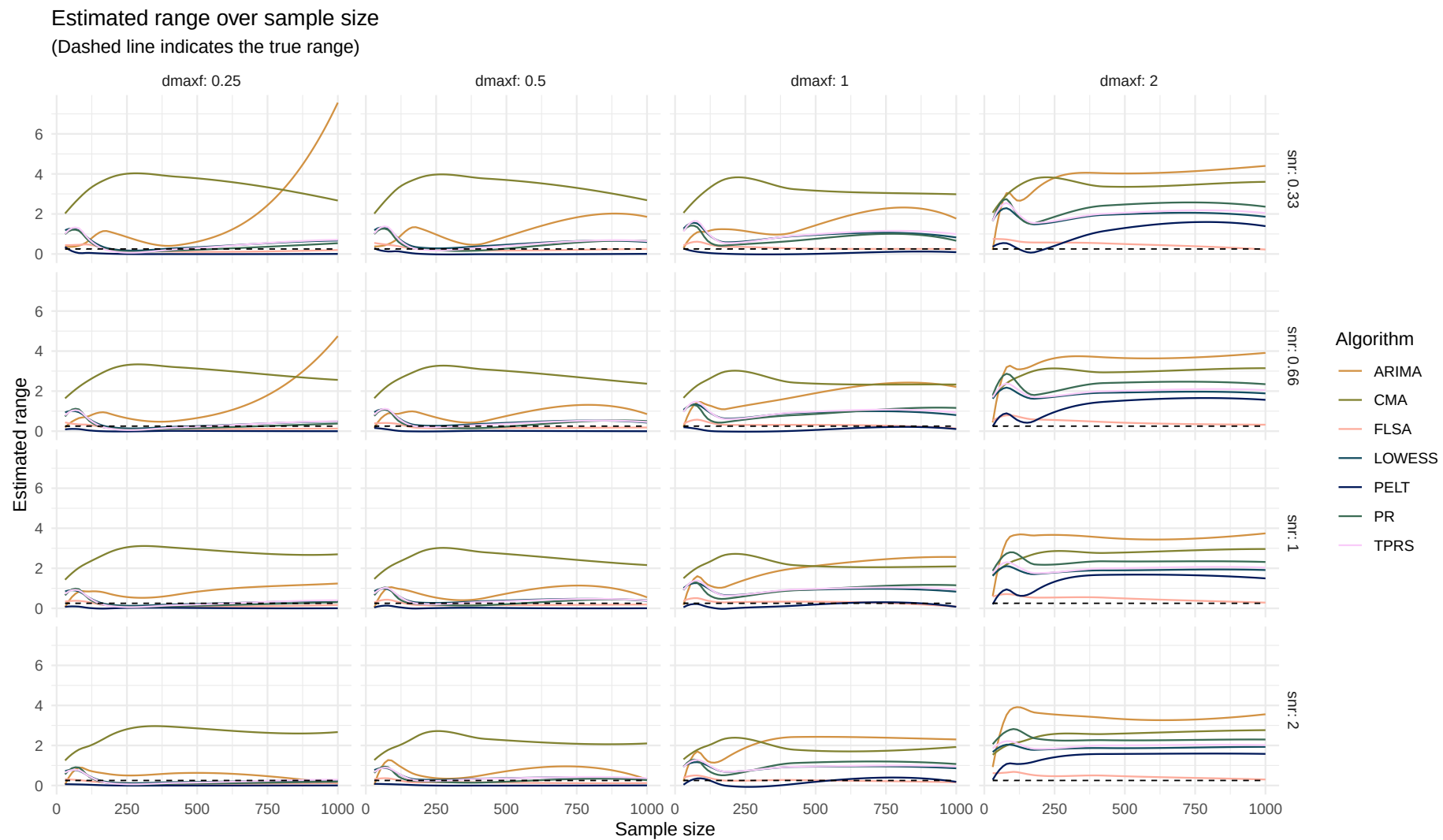
Systematic change pattern: Linear



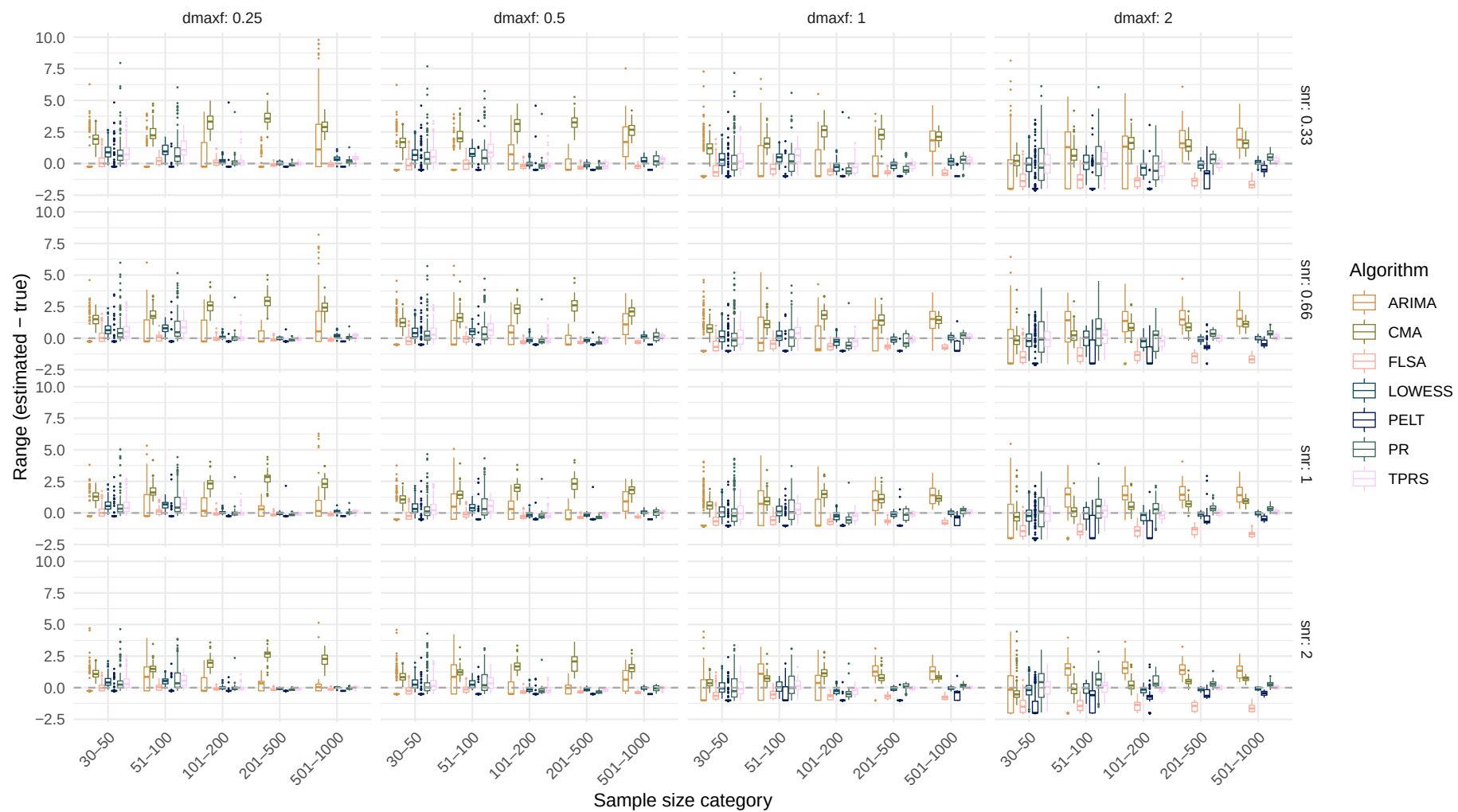
Boxplot of range difference over sample size



Systematic change pattern: Quadratic

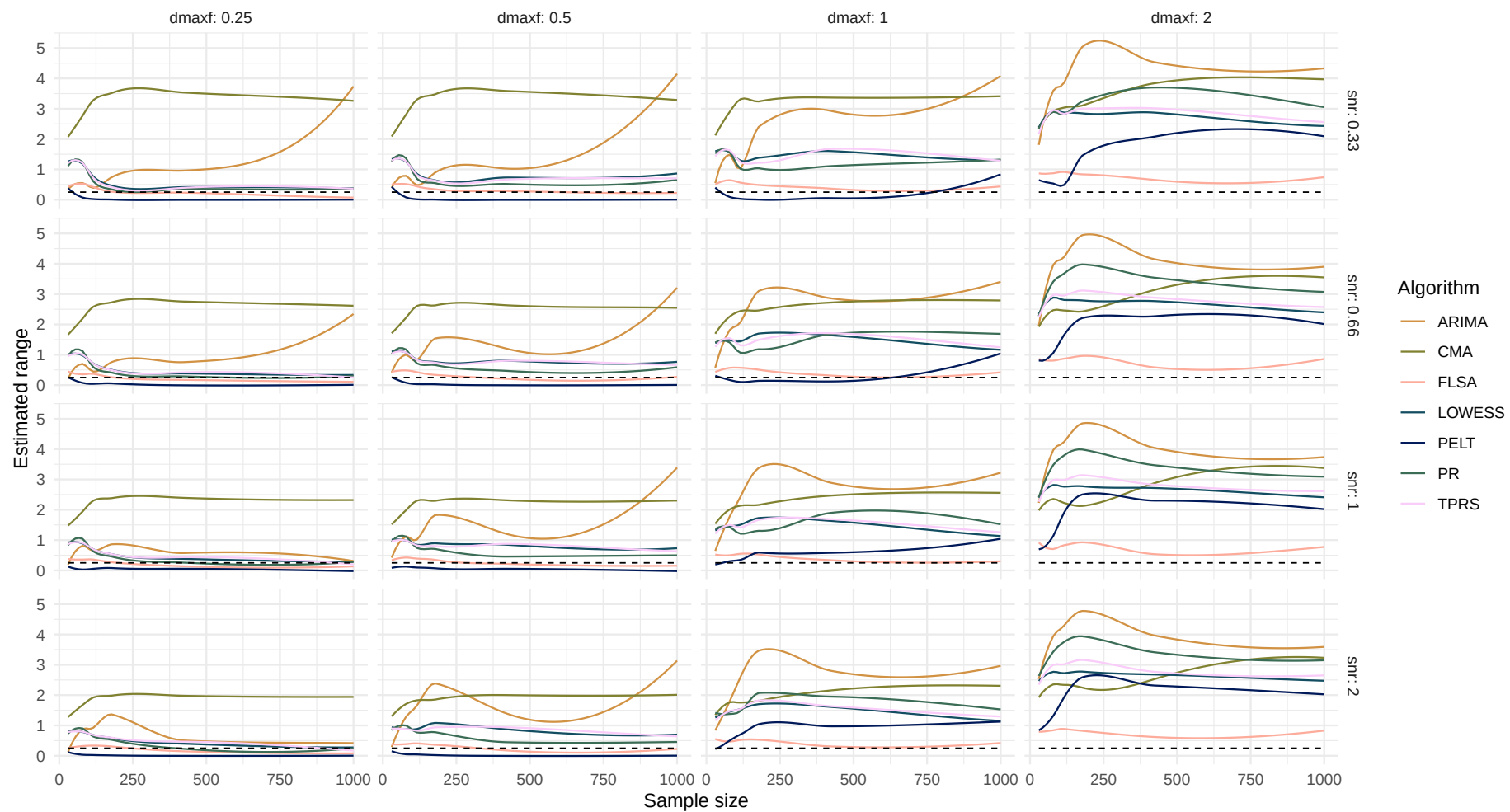


Boxplot of range difference over sample size

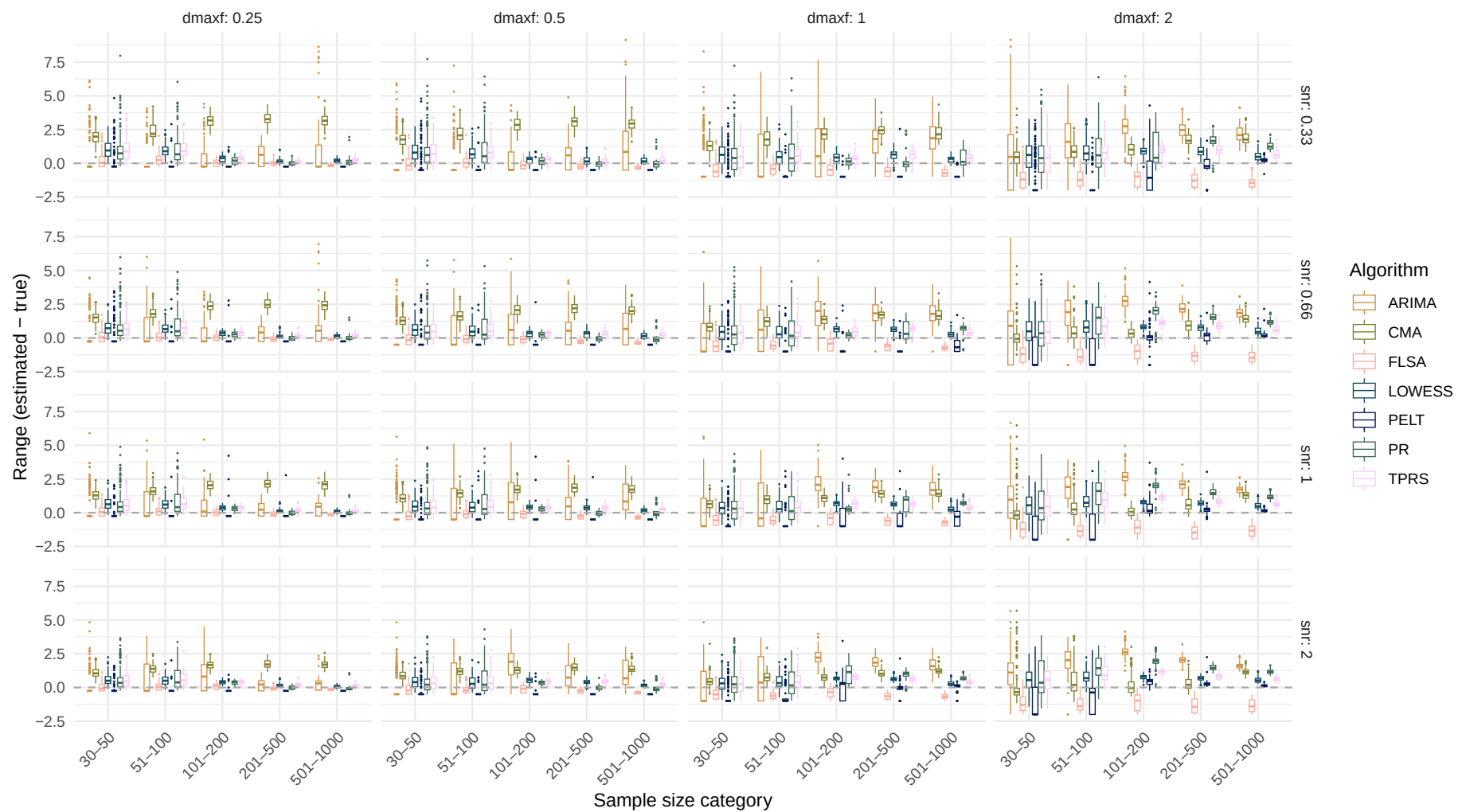


Systematic change pattern: Recurrent

Estimated range over sample size
(Dashed line indicates the true range)

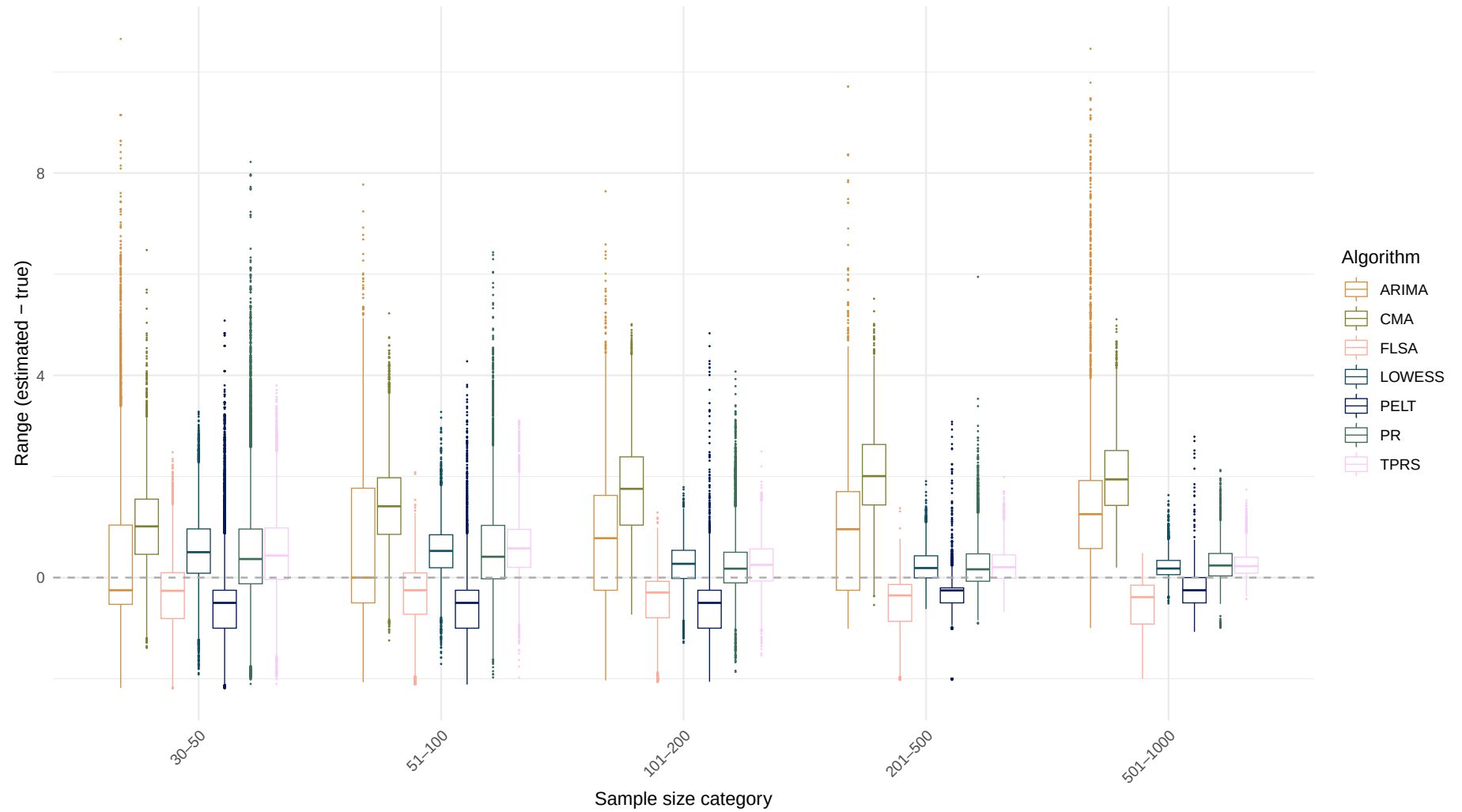


Boxplot of range difference over sample size

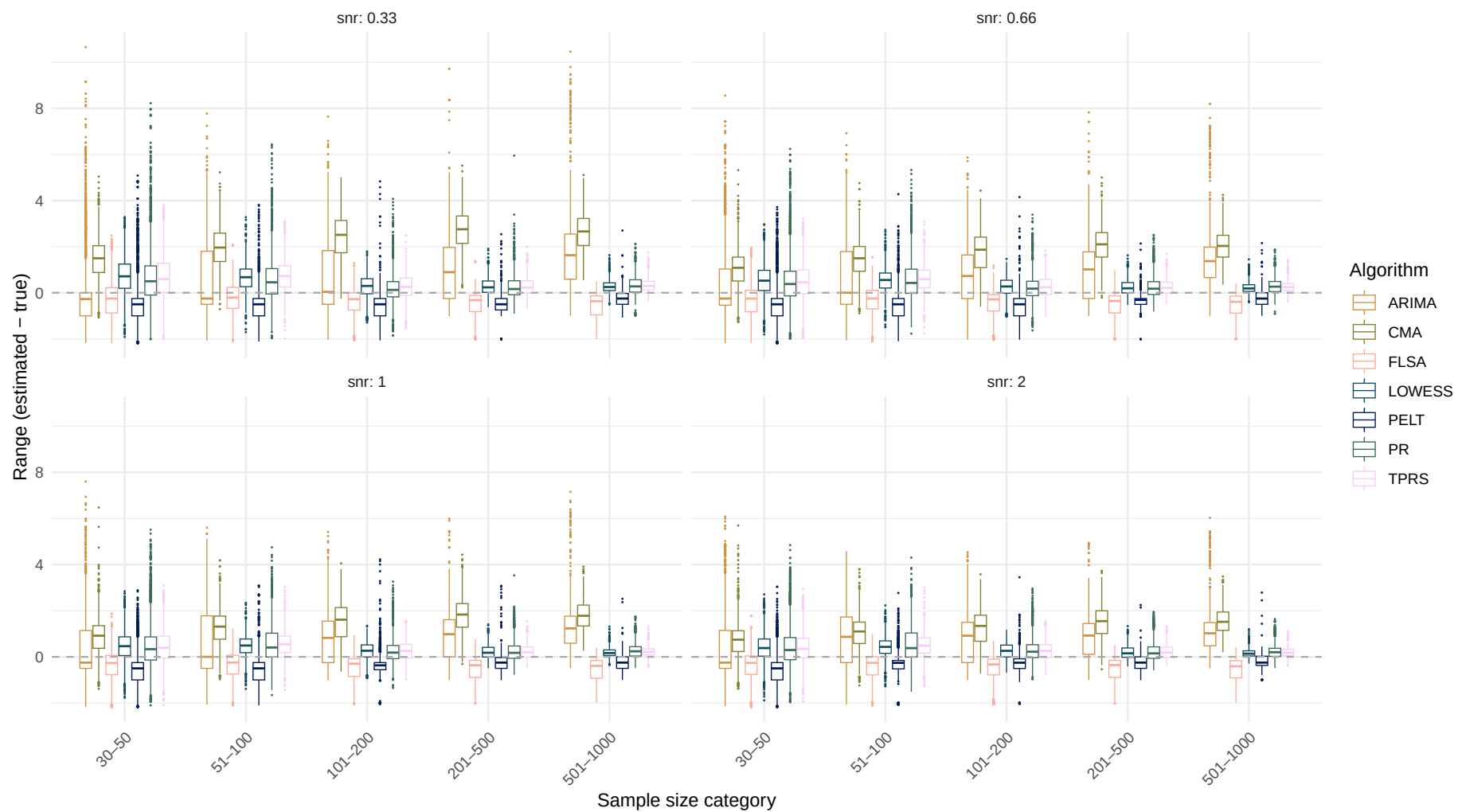


Systematic change pattern: All patterns combined

Boxplot of range difference over sample size

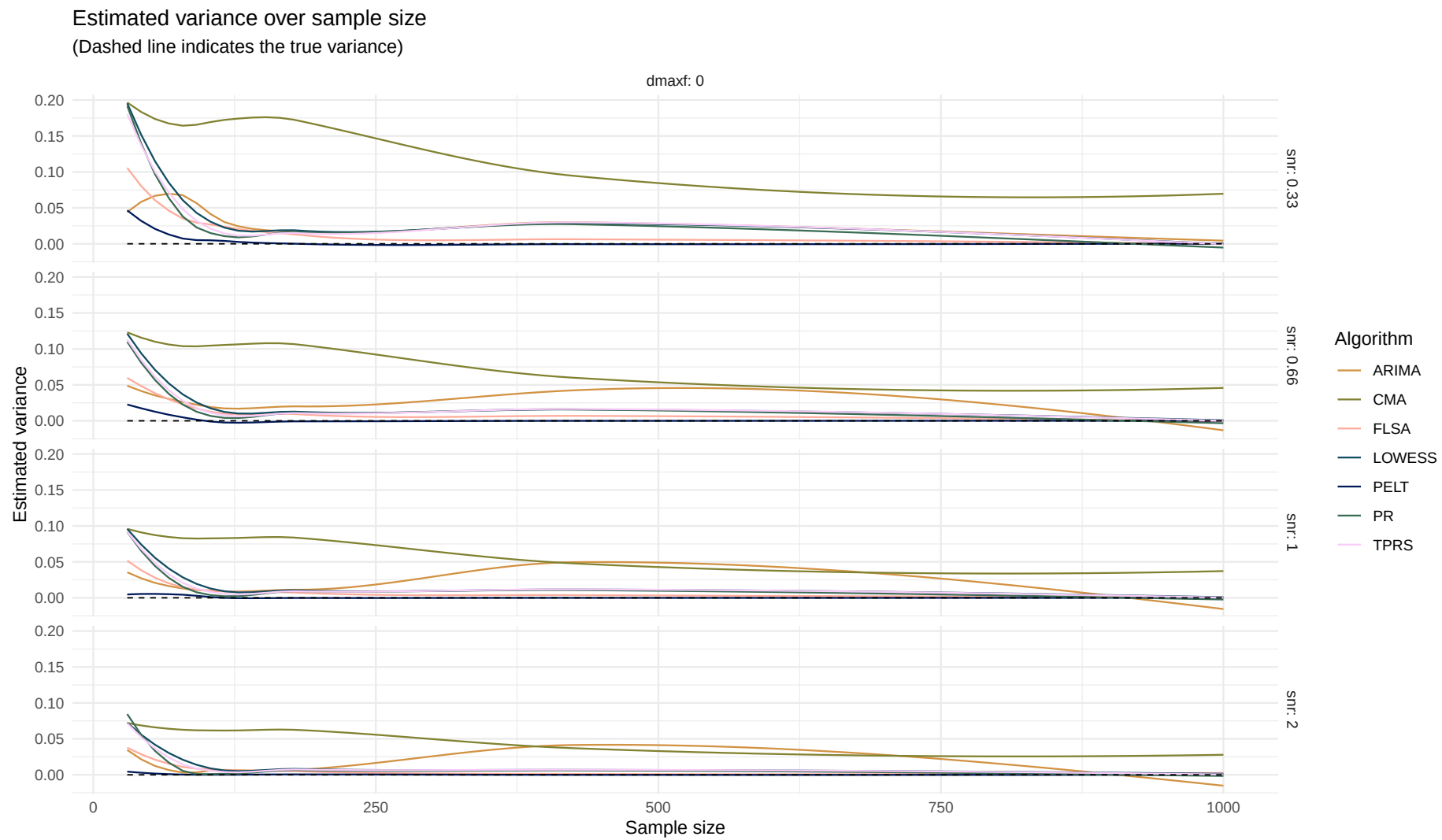


Boxplot of range difference over sample size

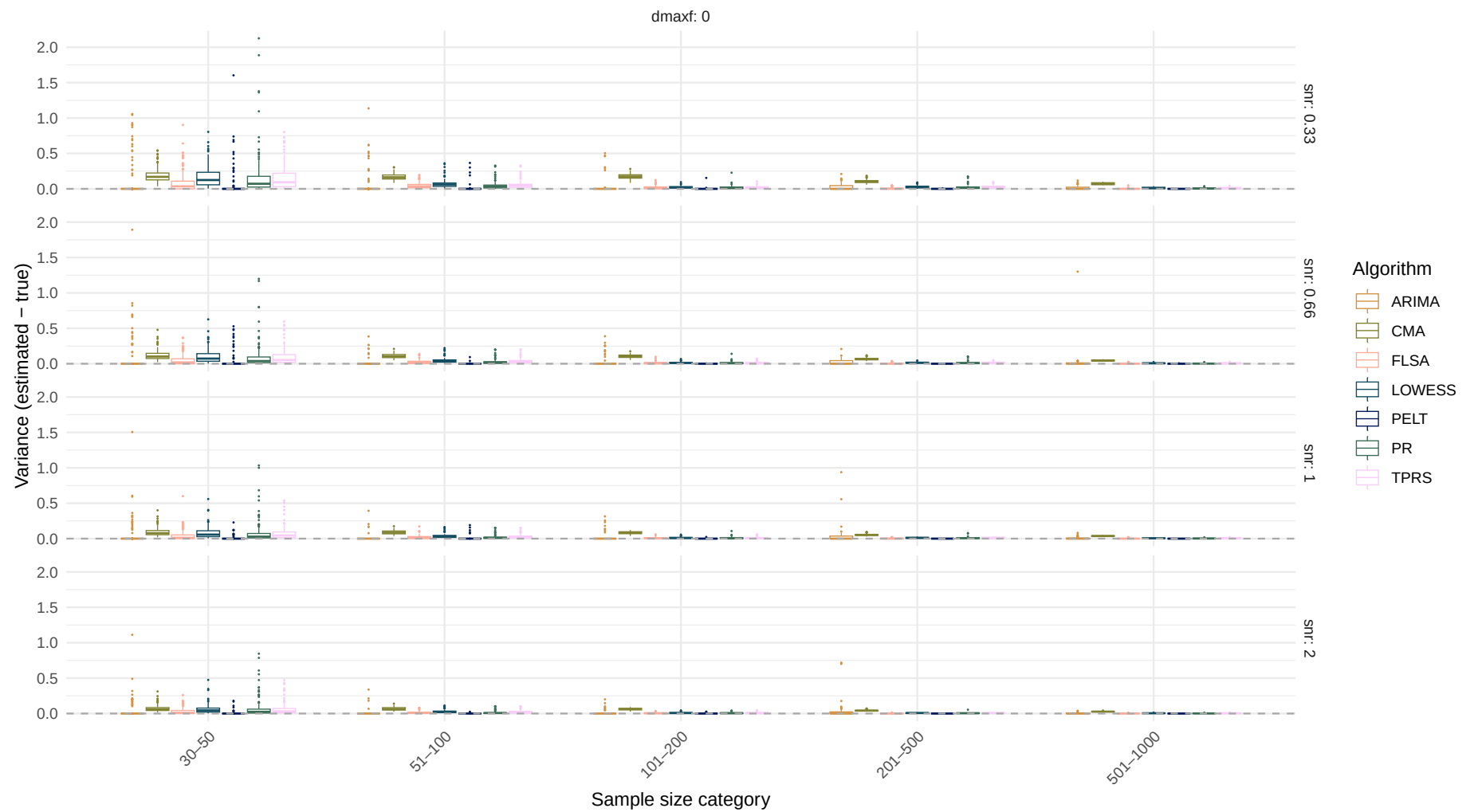


Estimand: Variance

Systematic change pattern: No change

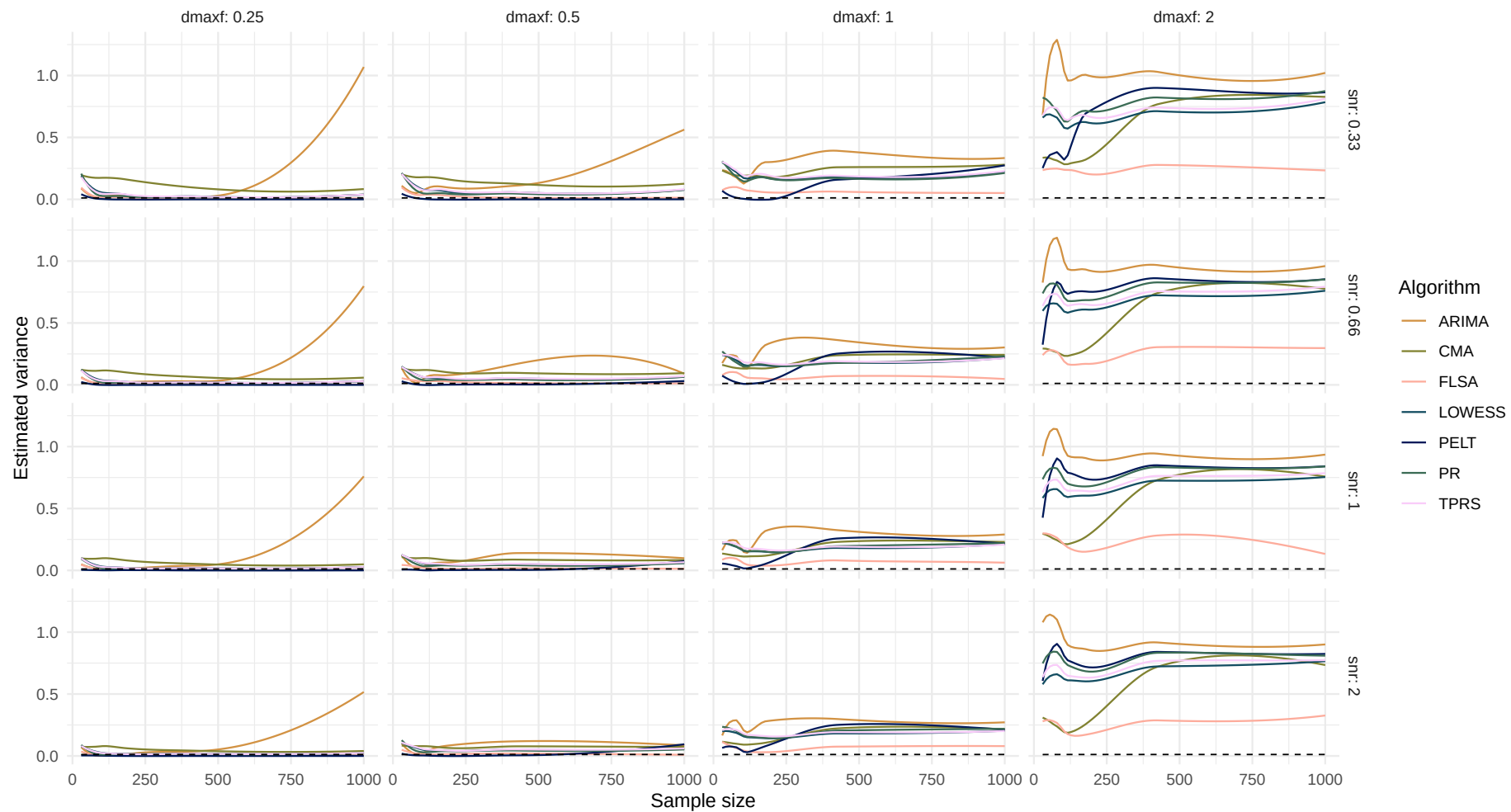


Boxplot of variance difference over sample size

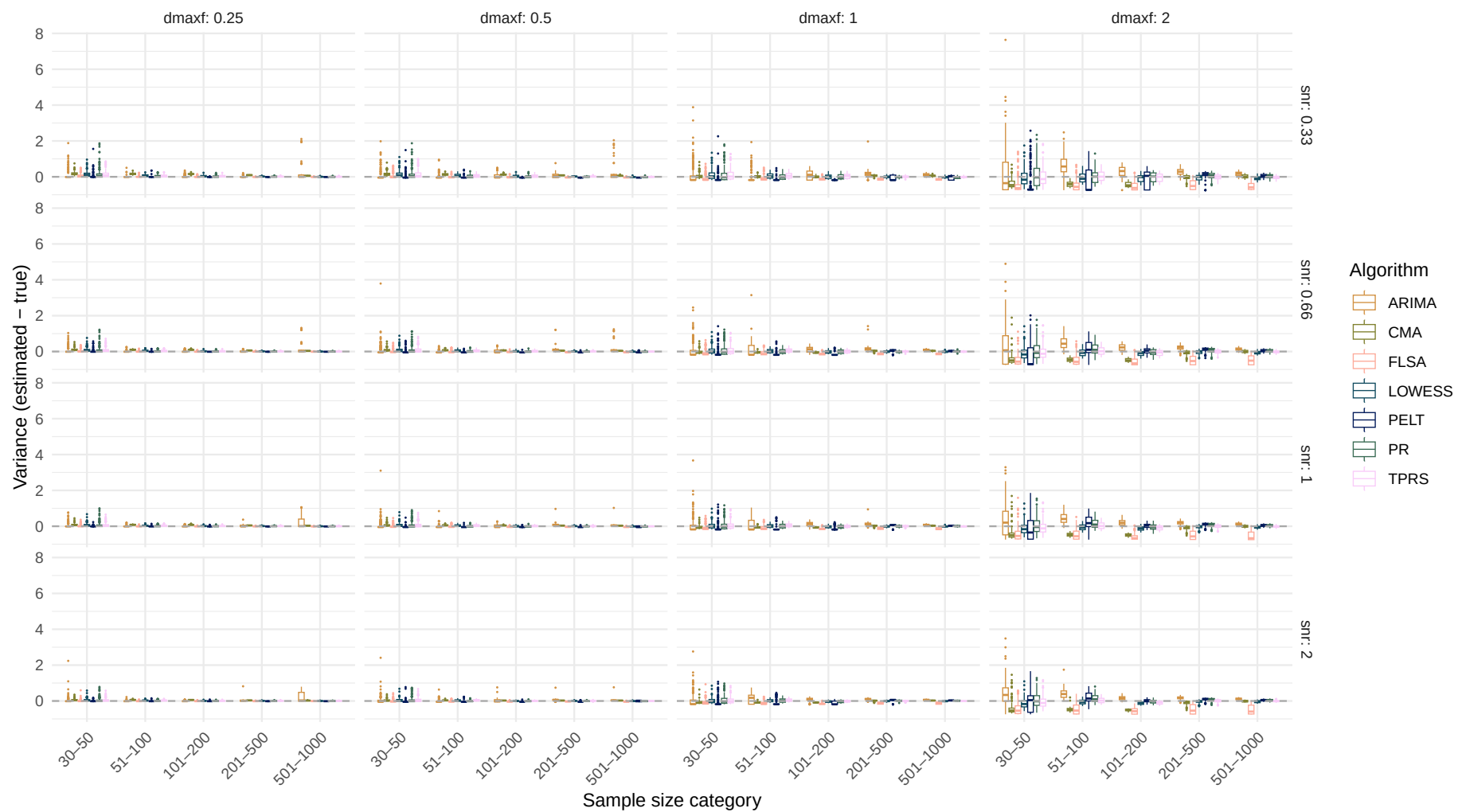


Systematic change pattern: Jump

Estimated variance over sample size
(Dashed line indicates the true variance)

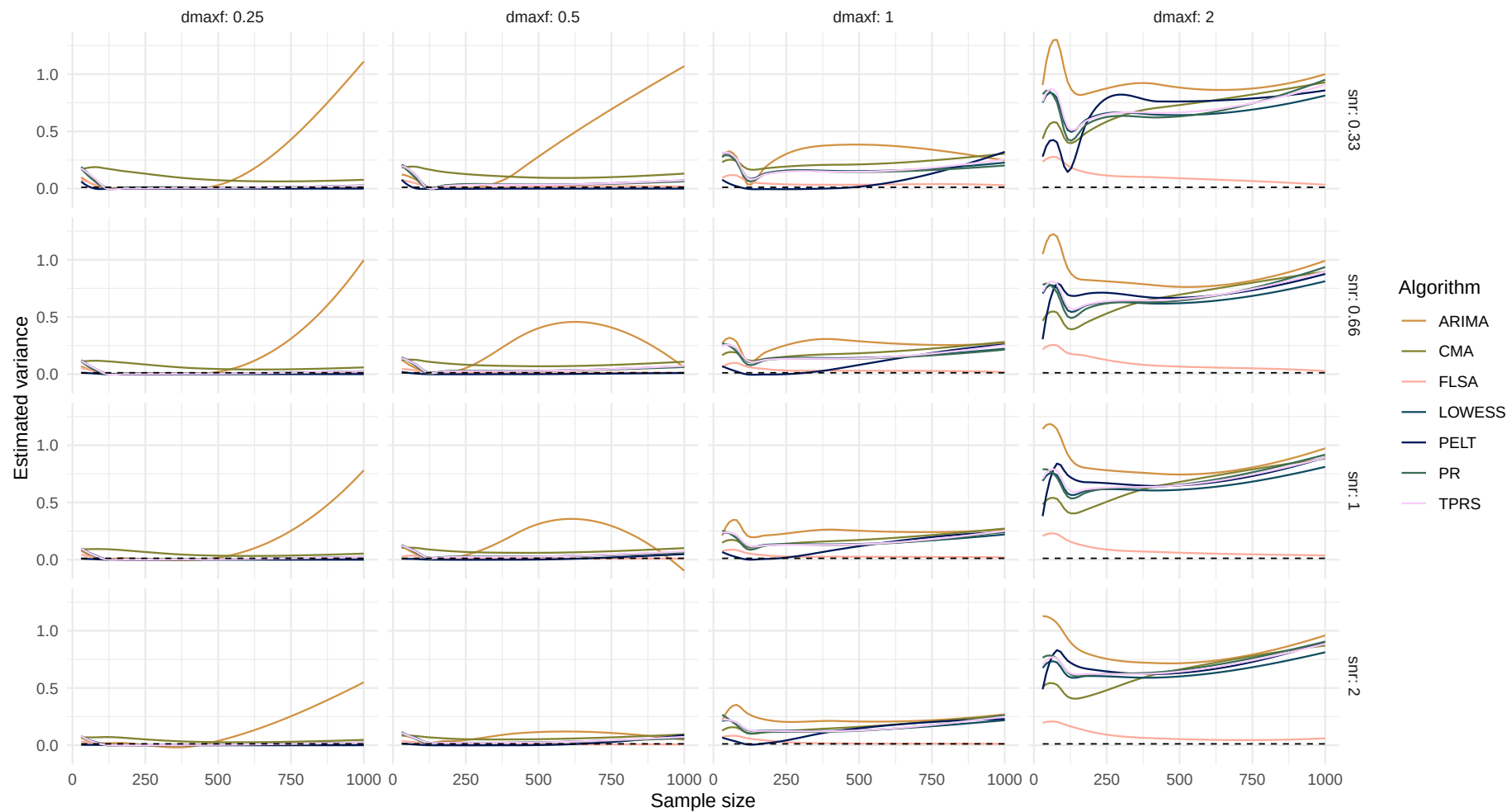


Boxplot of variance difference over sample size

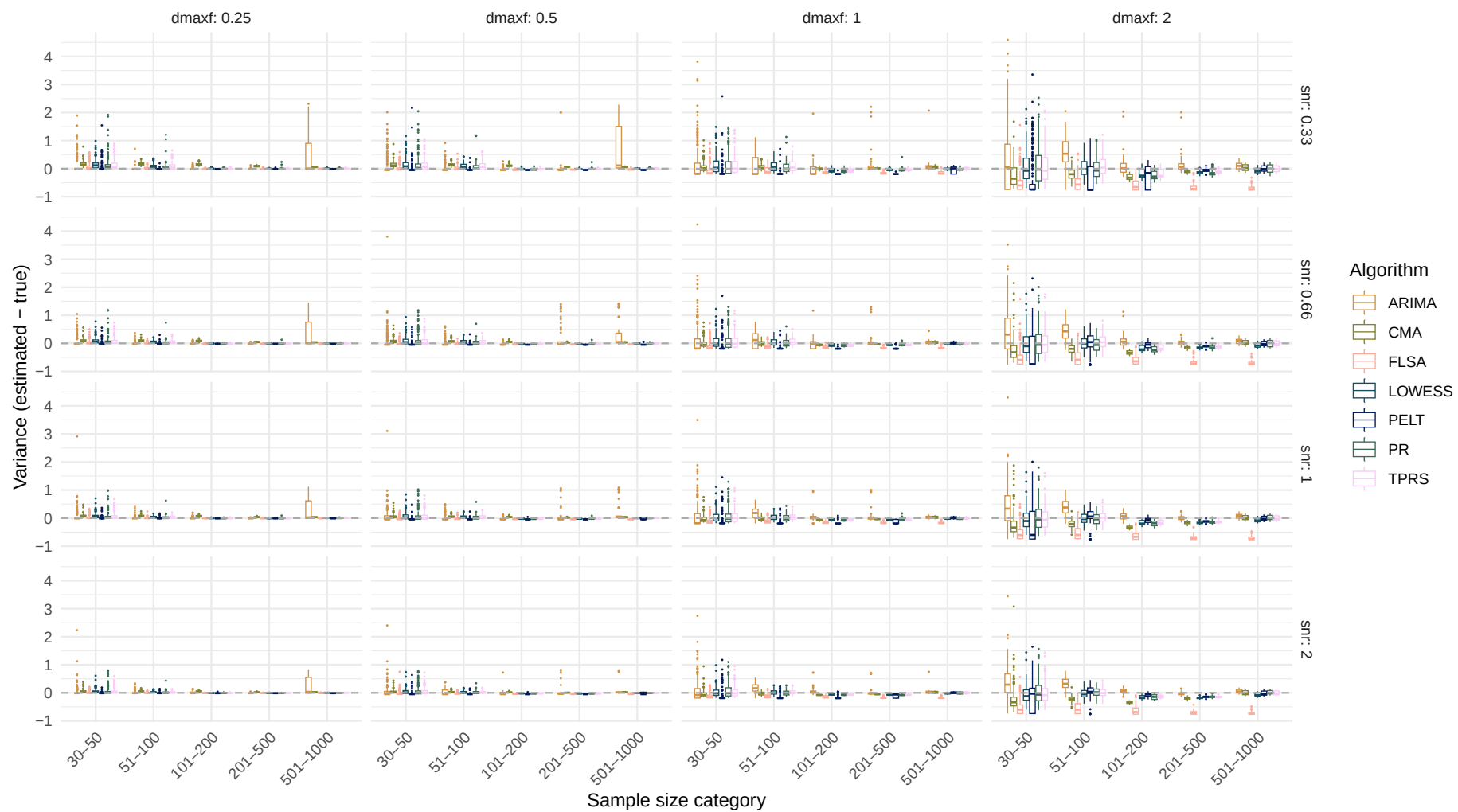


Systematic change pattern: Linear

Estimated variance over sample size
(Dashed line indicates the true variance)

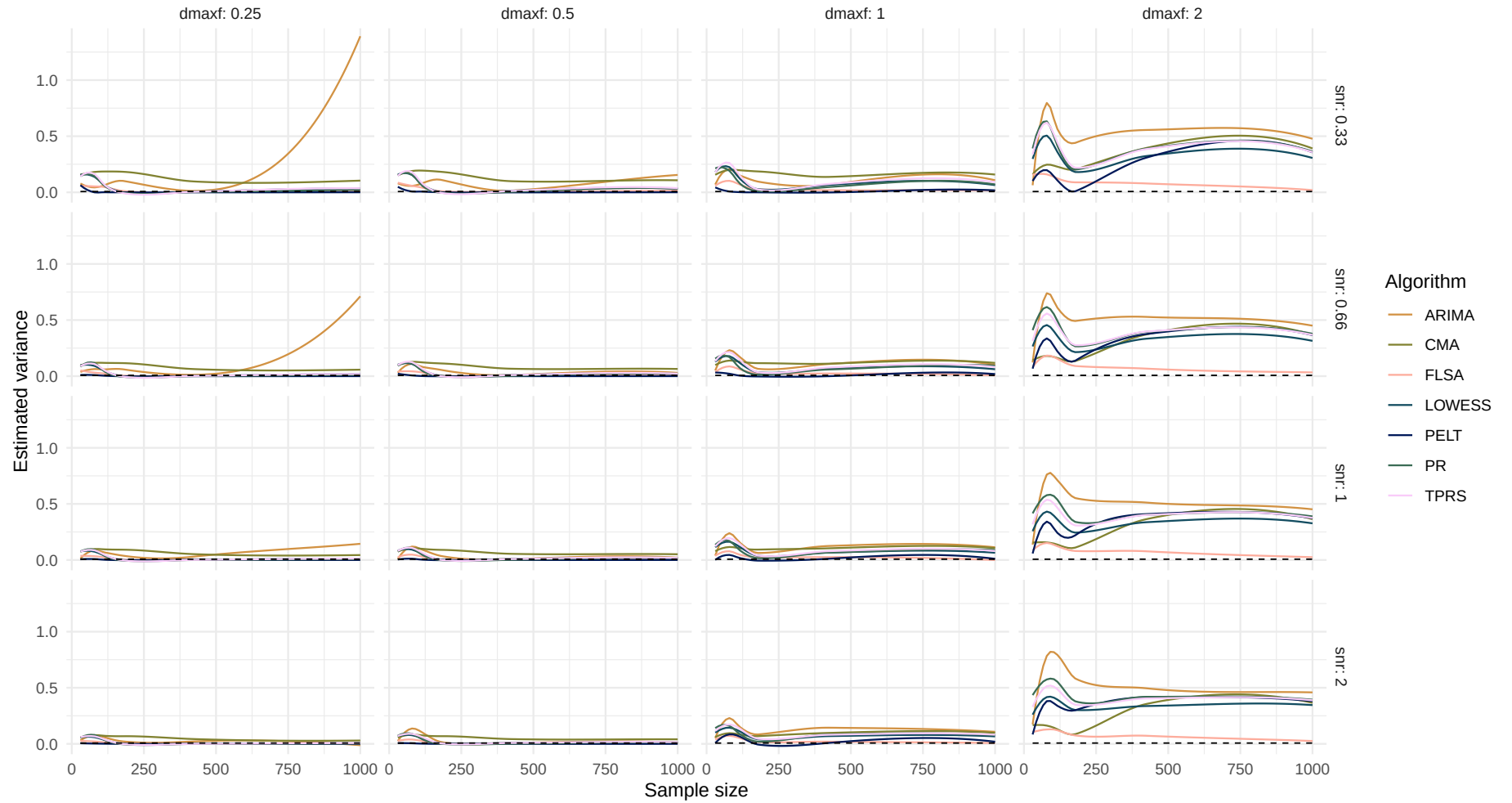


Boxplot of variance difference over sample size

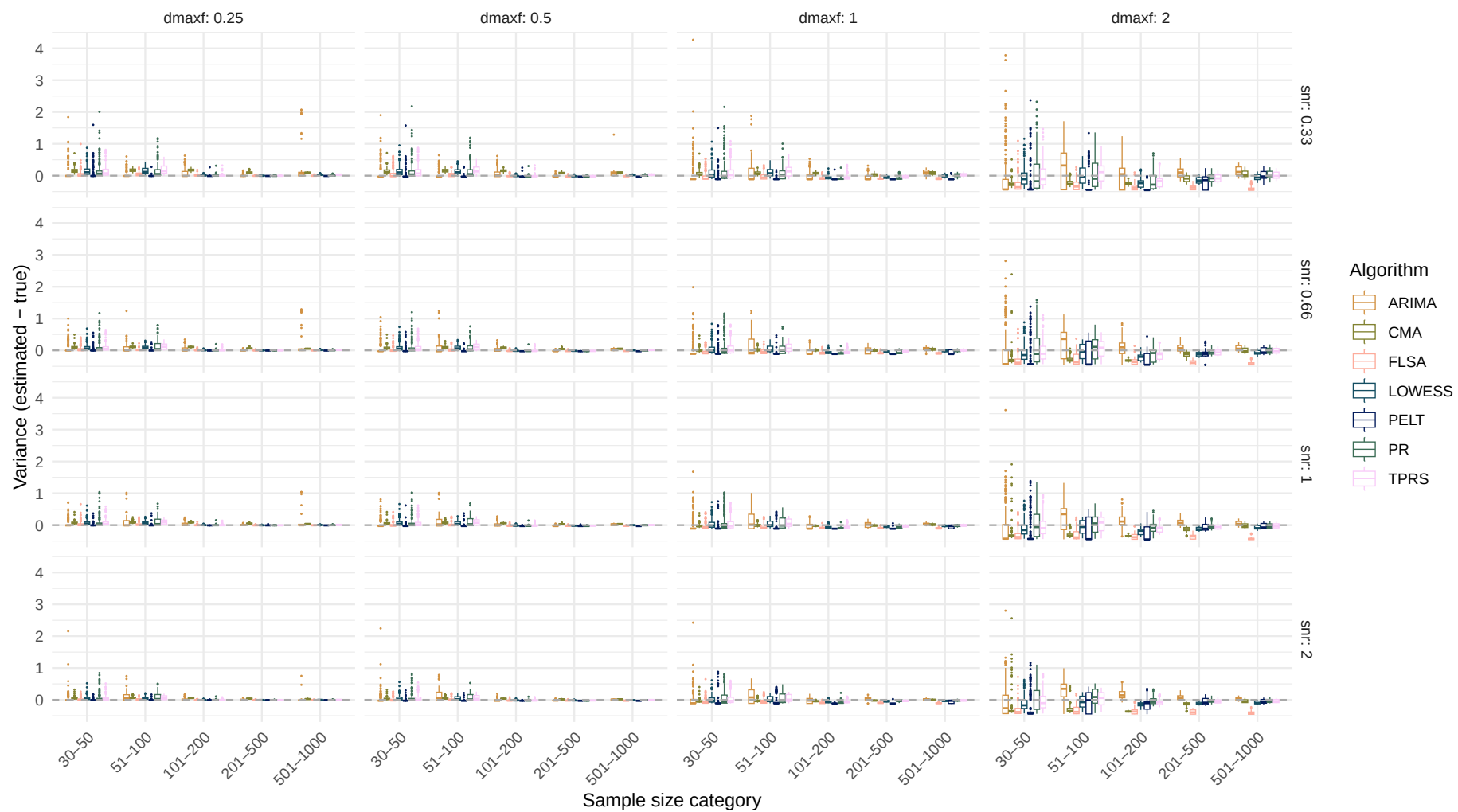


Systematic change pattern: Quadratic

Estimated variance over sample size
(Dashed line indicates the true variance)

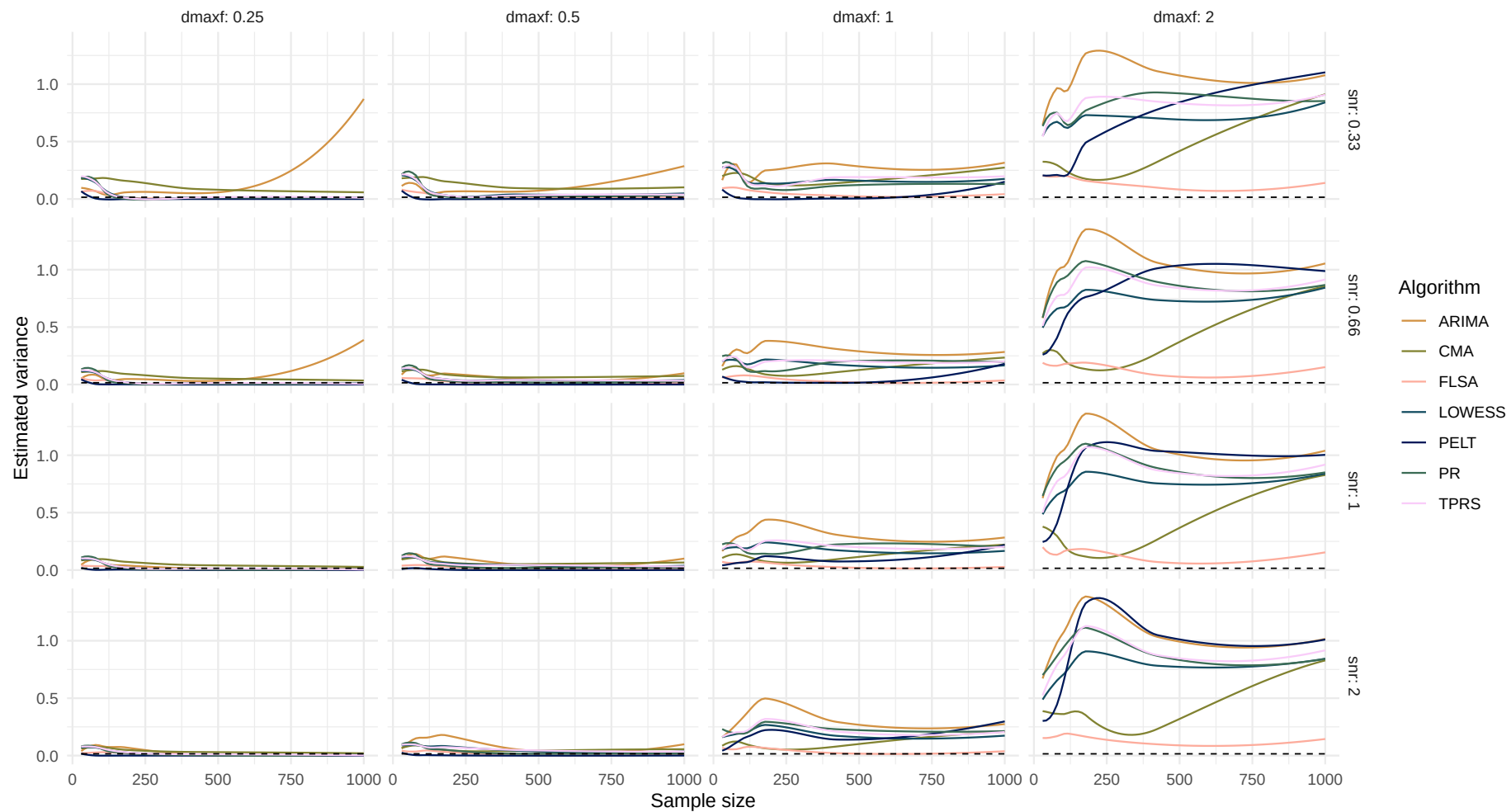


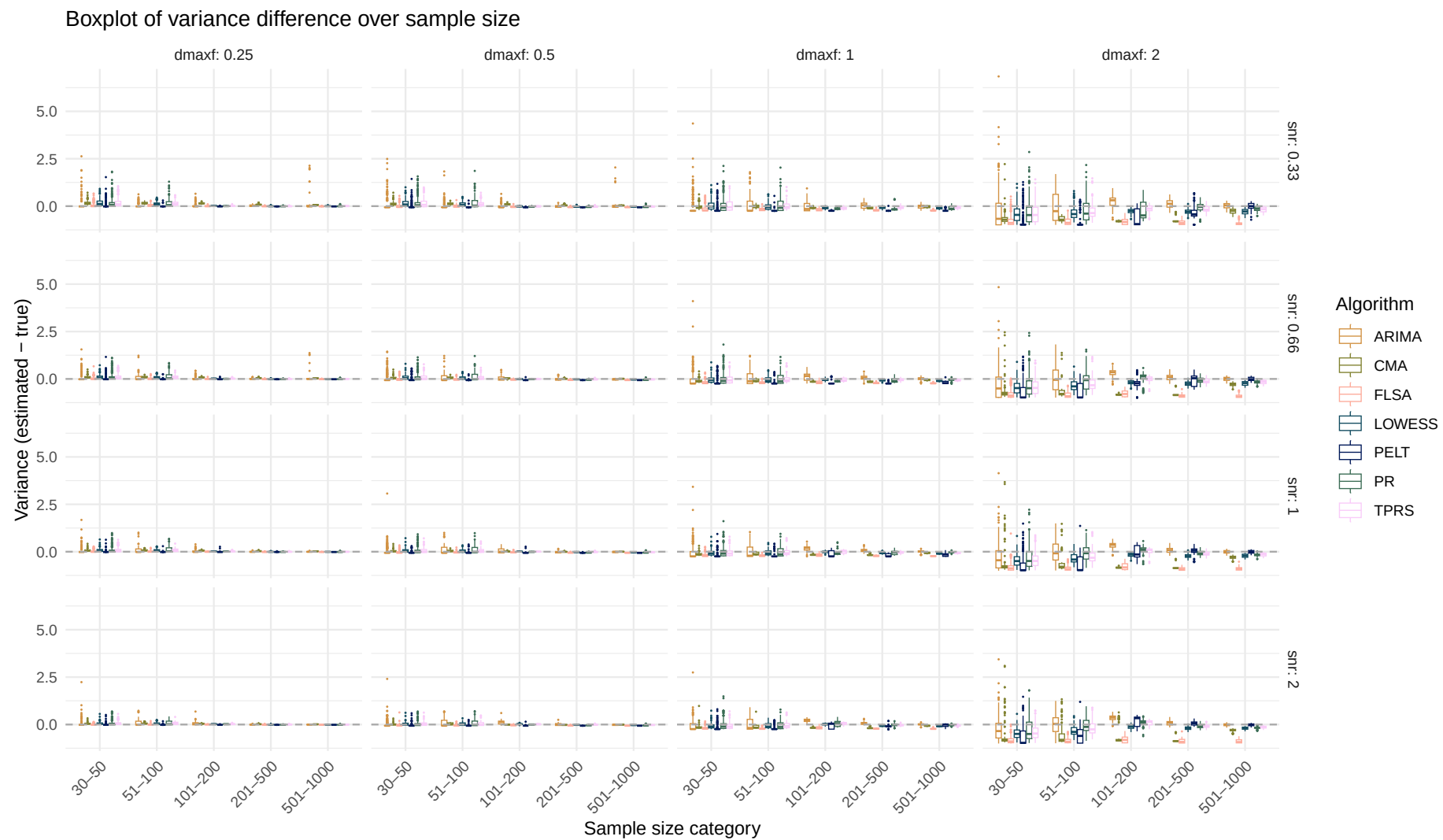
Boxplot of variance difference over sample size



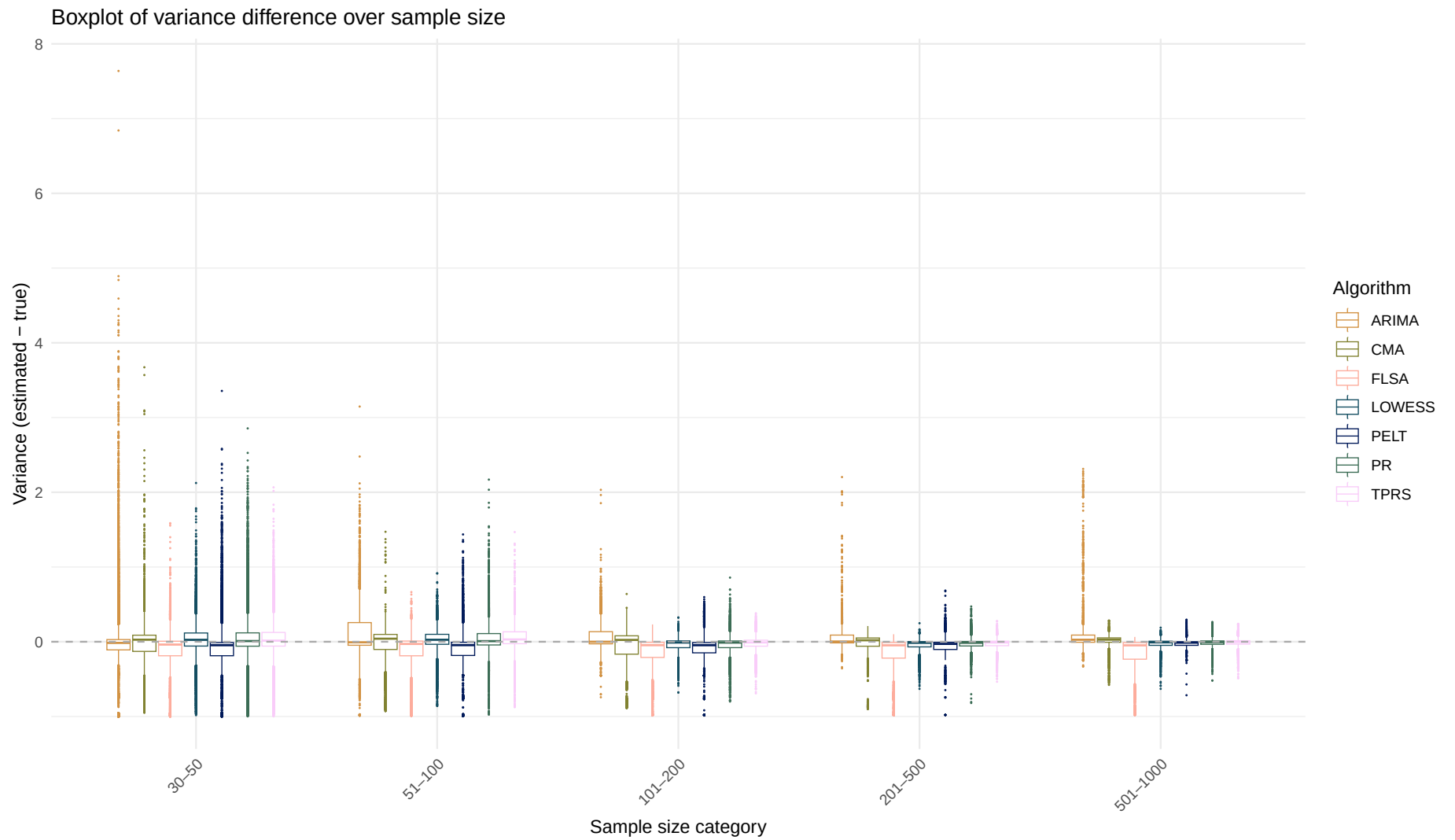
Systematic change pattern: Recurrent

Estimated variance over sample size
(Dashed line indicates the true variance)

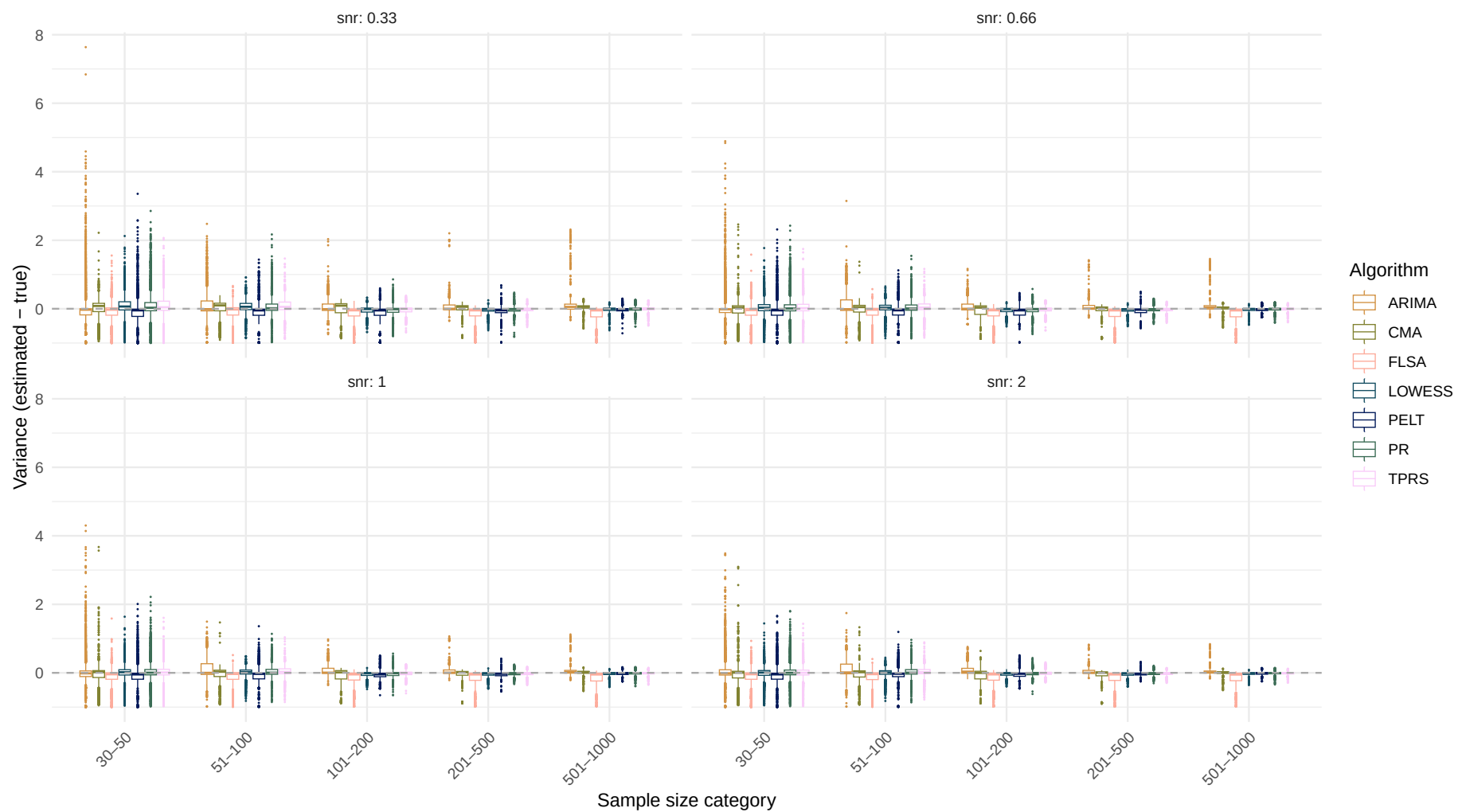




Systematic change pattern: All patterns combined



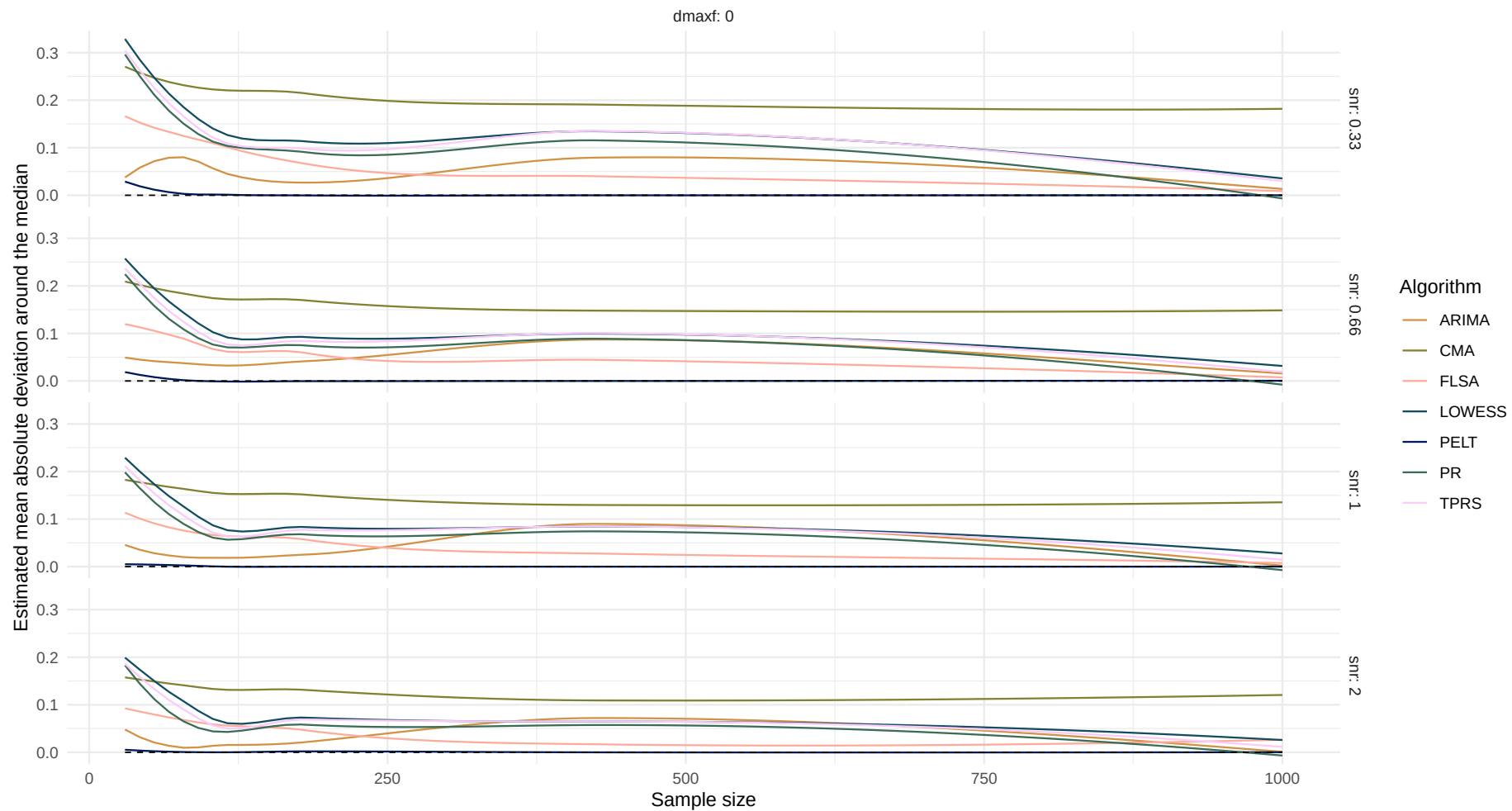
Boxplot of variance difference over sample size



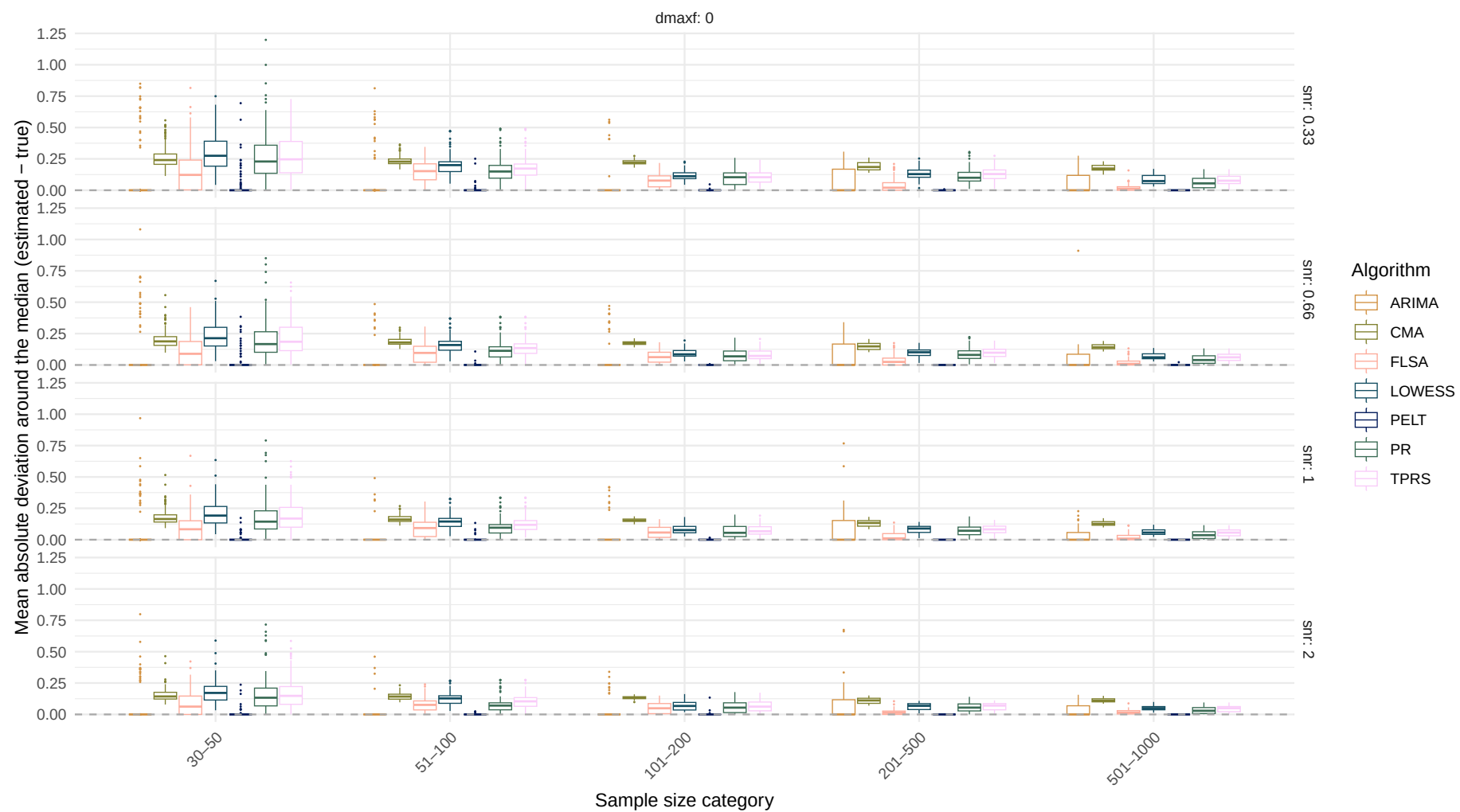
Estimand: Mean absolute deviation around the median

Systematic change pattern: No change

Estimated mean absolute deviation around the median over sample size
(Dashed line indicates the true mean absolute deviation around the median)

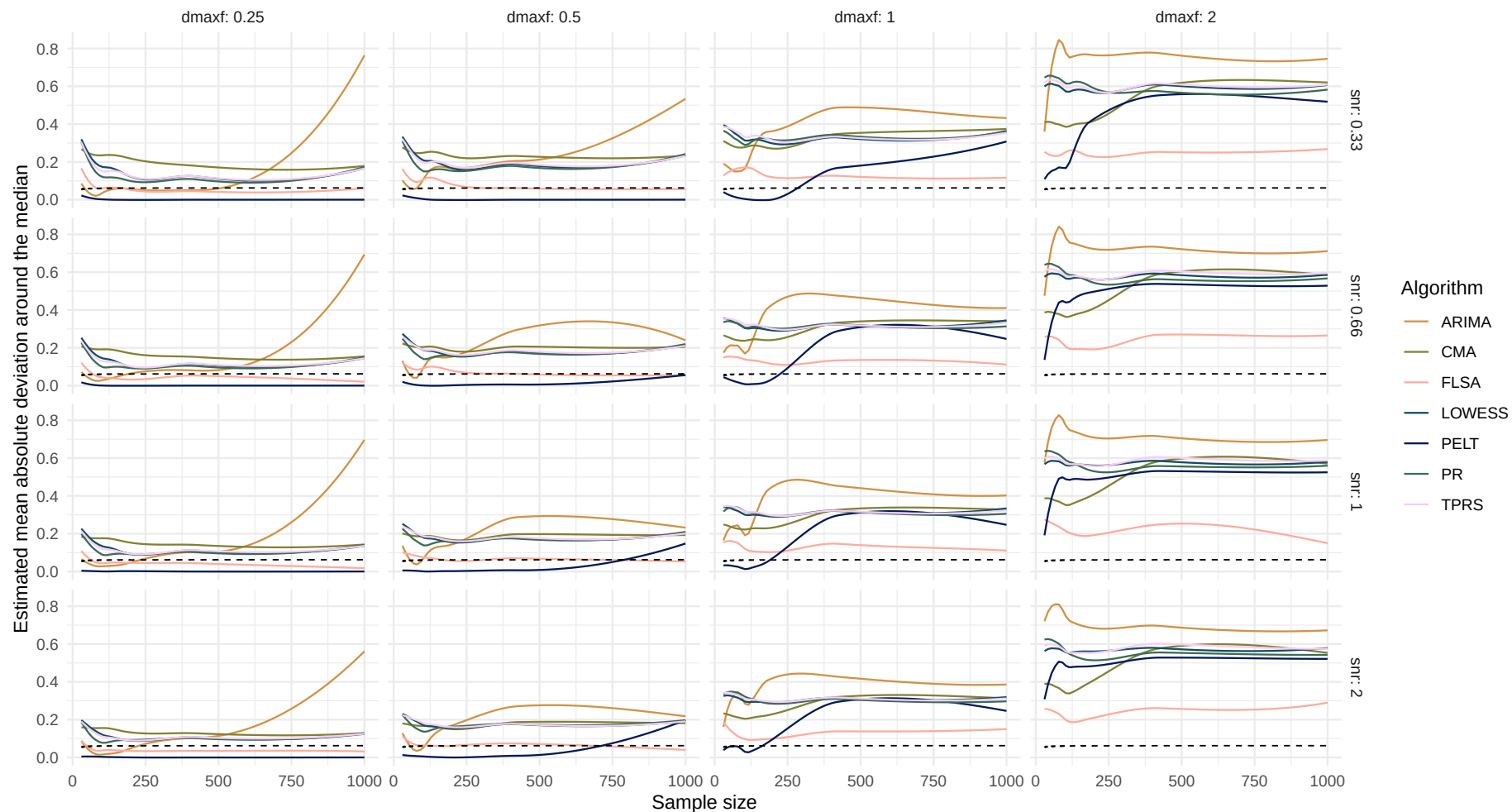


Boxplot of mean absolute deviation around the median difference over sample size

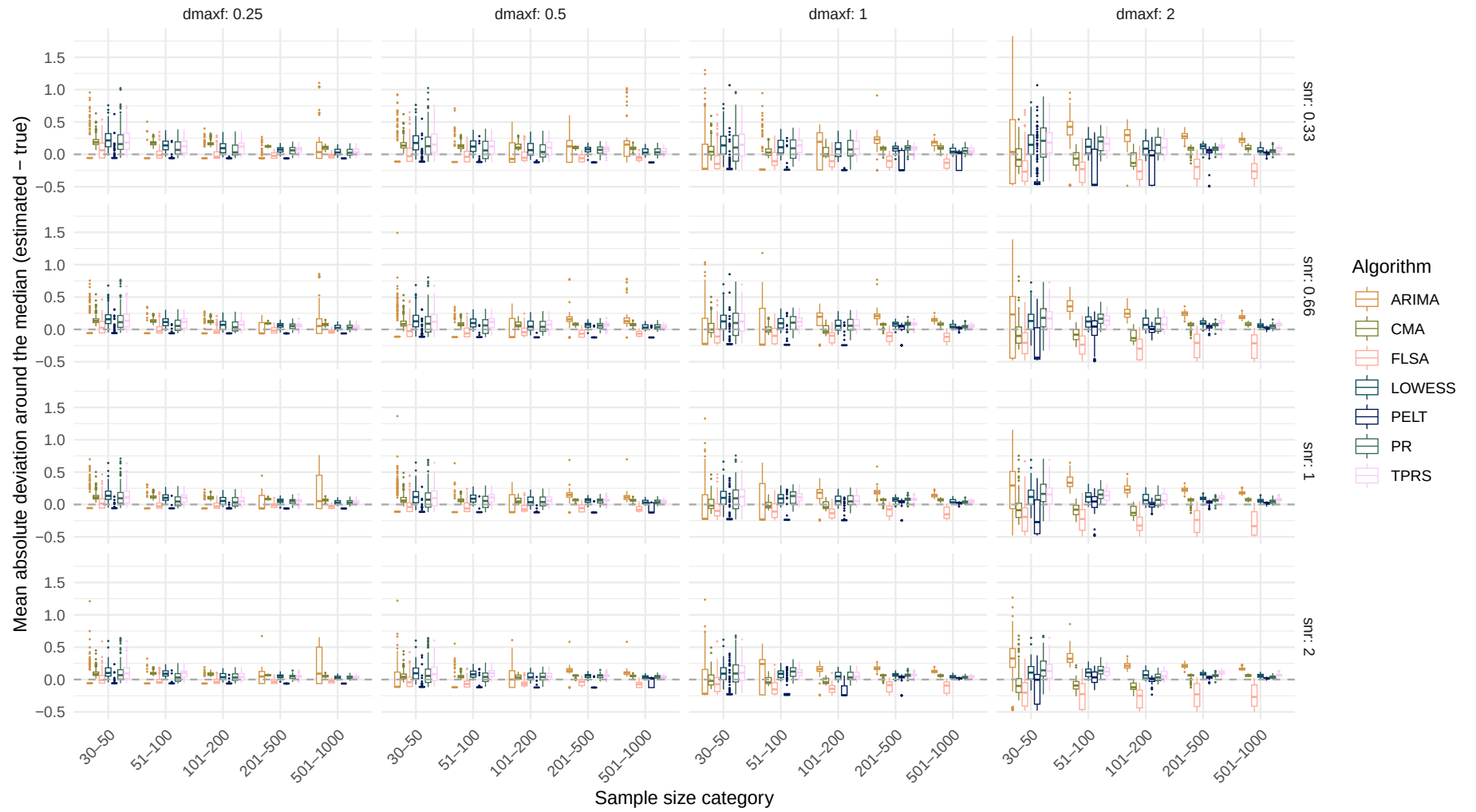


Systematic change pattern: Jump

Estimated mean absolute deviation around the median over sample size
(Dashed line indicates the true mean absolute deviation around the median)



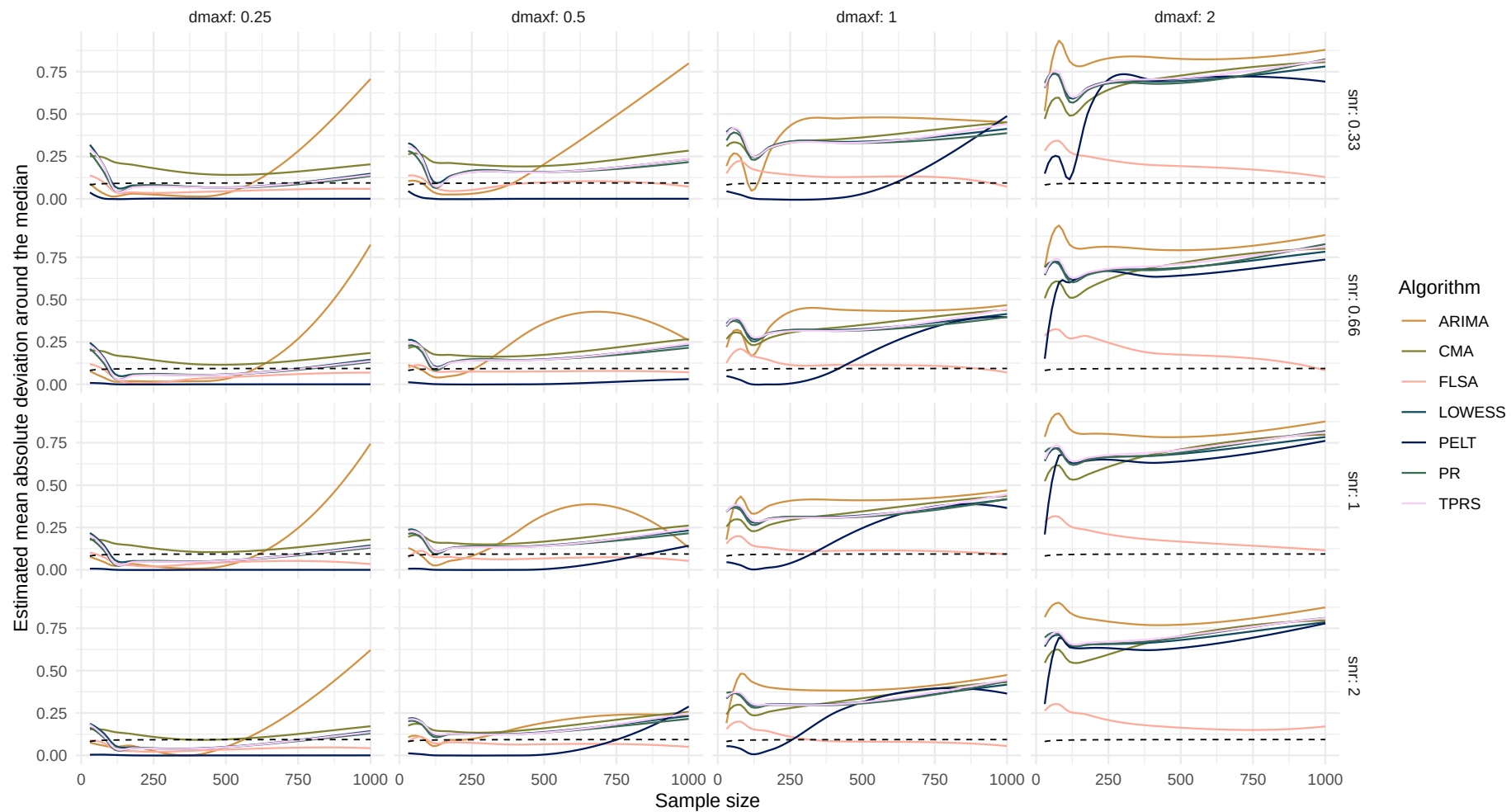
Boxplot of mean absolute deviation around the median difference over sample size



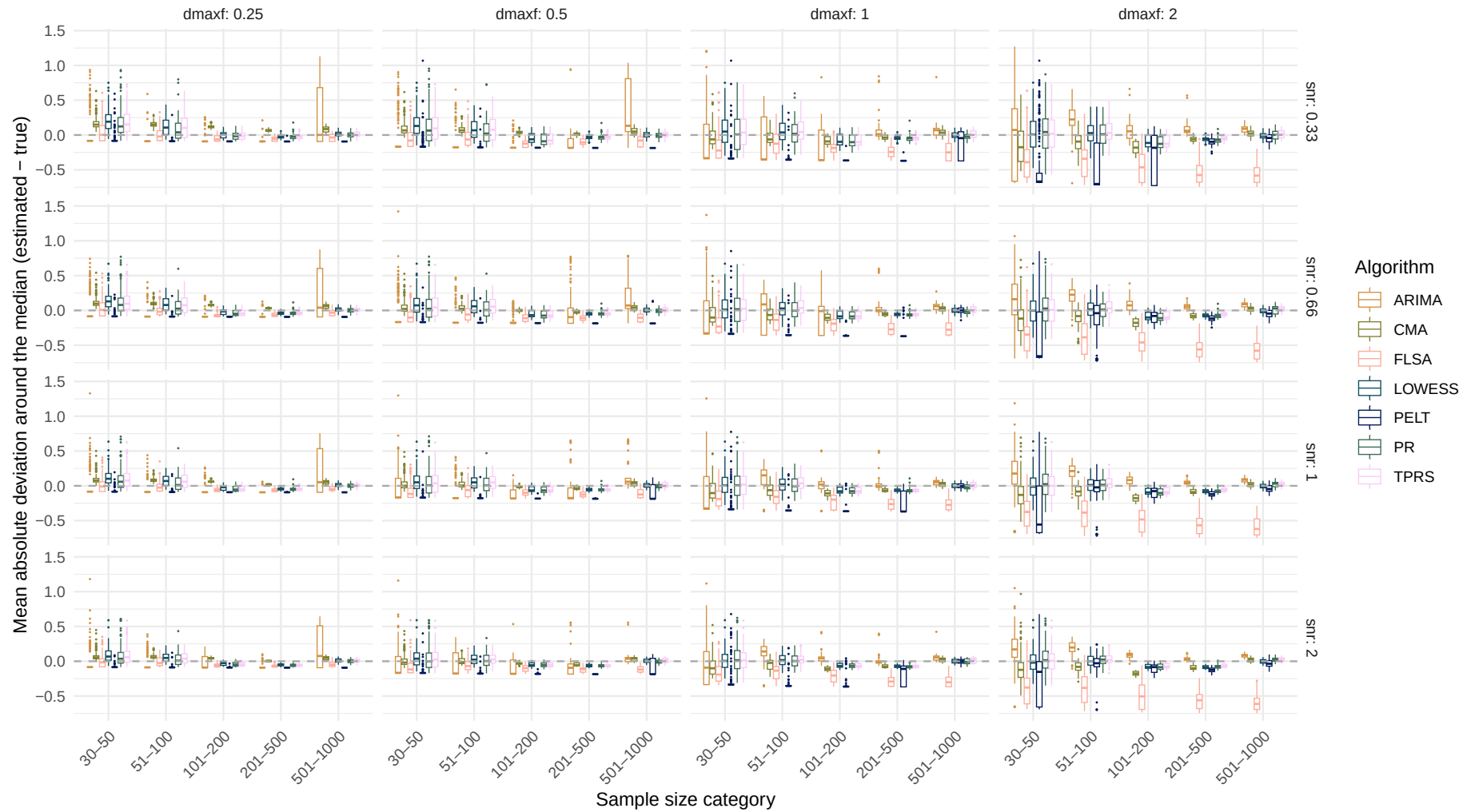
Systematic change pattern: Linear

Estimated mean absolute deviation around the median over sample size

(Dashed line indicates the true mean absolute deviation around the median)



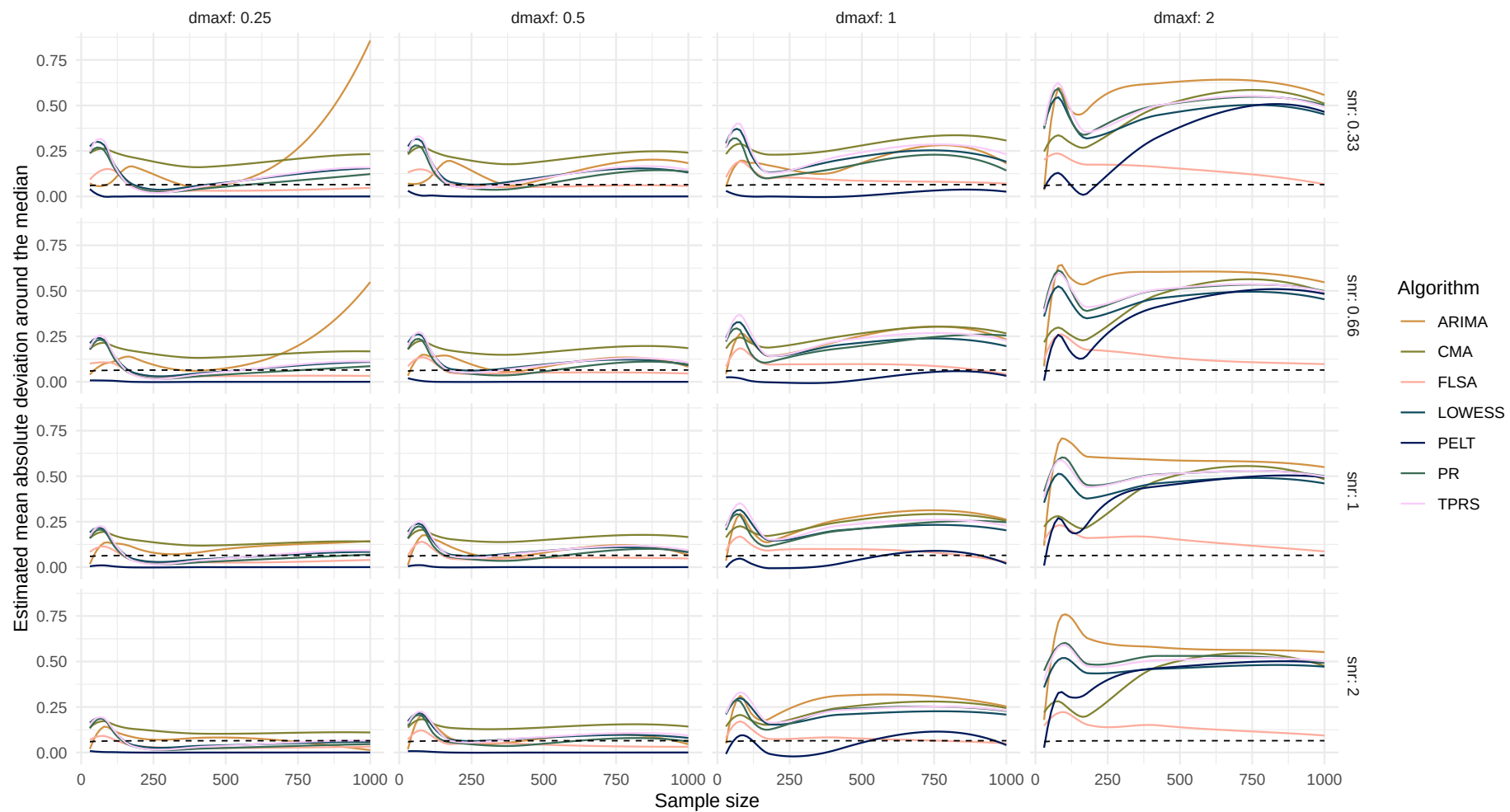
Boxplot of mean absolute deviation around the median difference over sample size



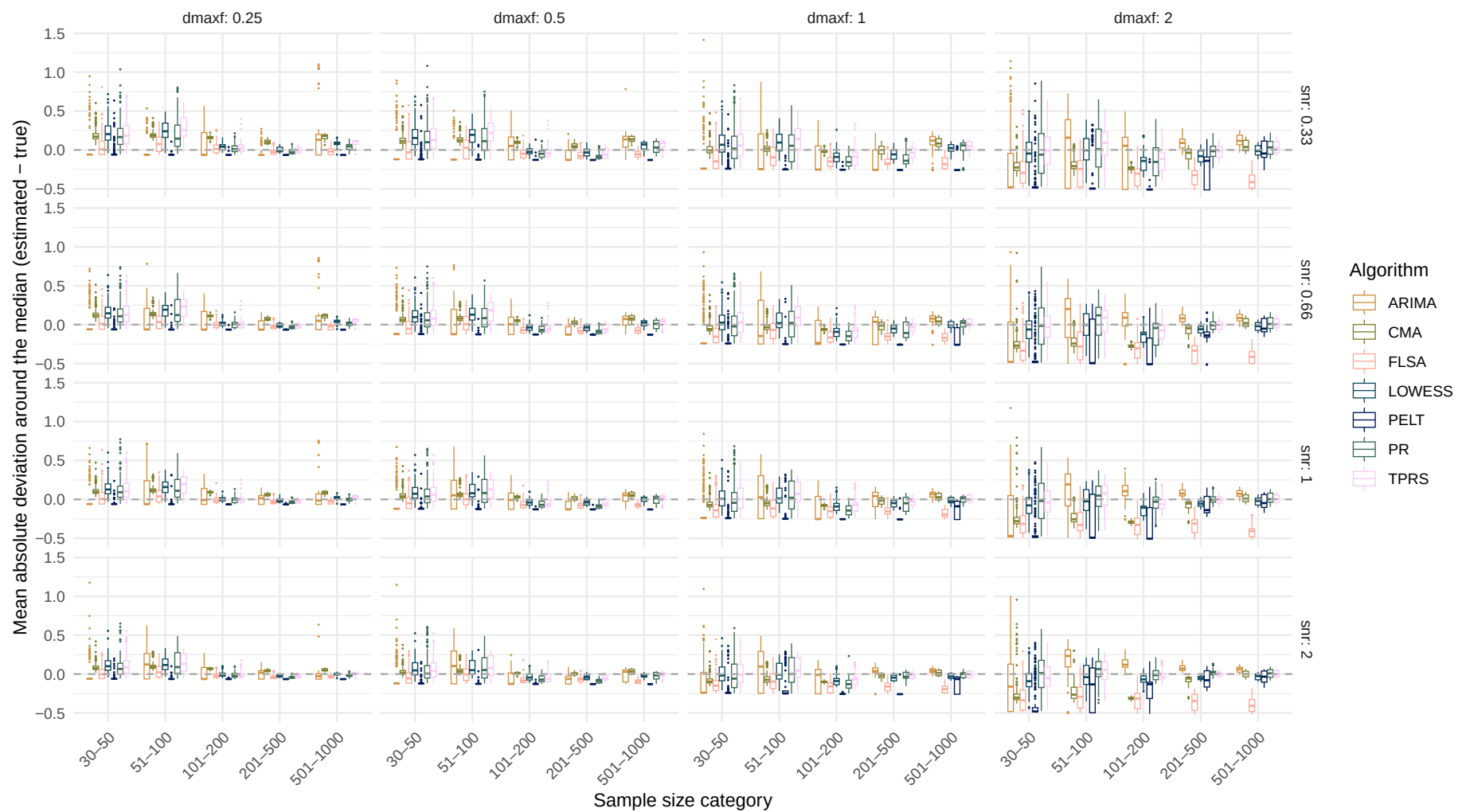
Systematic change pattern: Quadratic

Estimated mean absolute deviation around the median over sample size

(Dashed line indicates the true mean absolute deviation around the median)



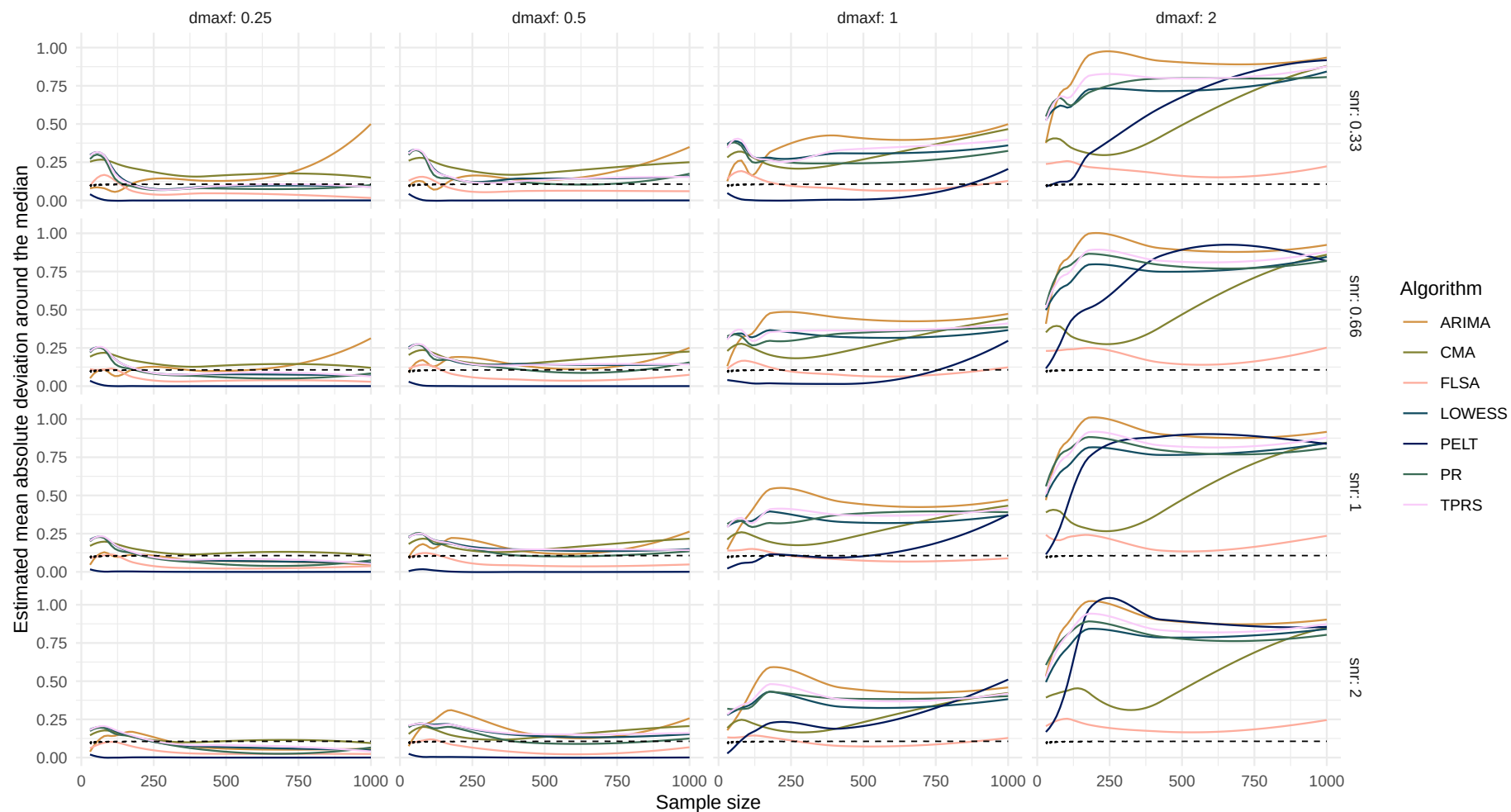
Boxplot of mean absolute deviation around the median difference over sample size



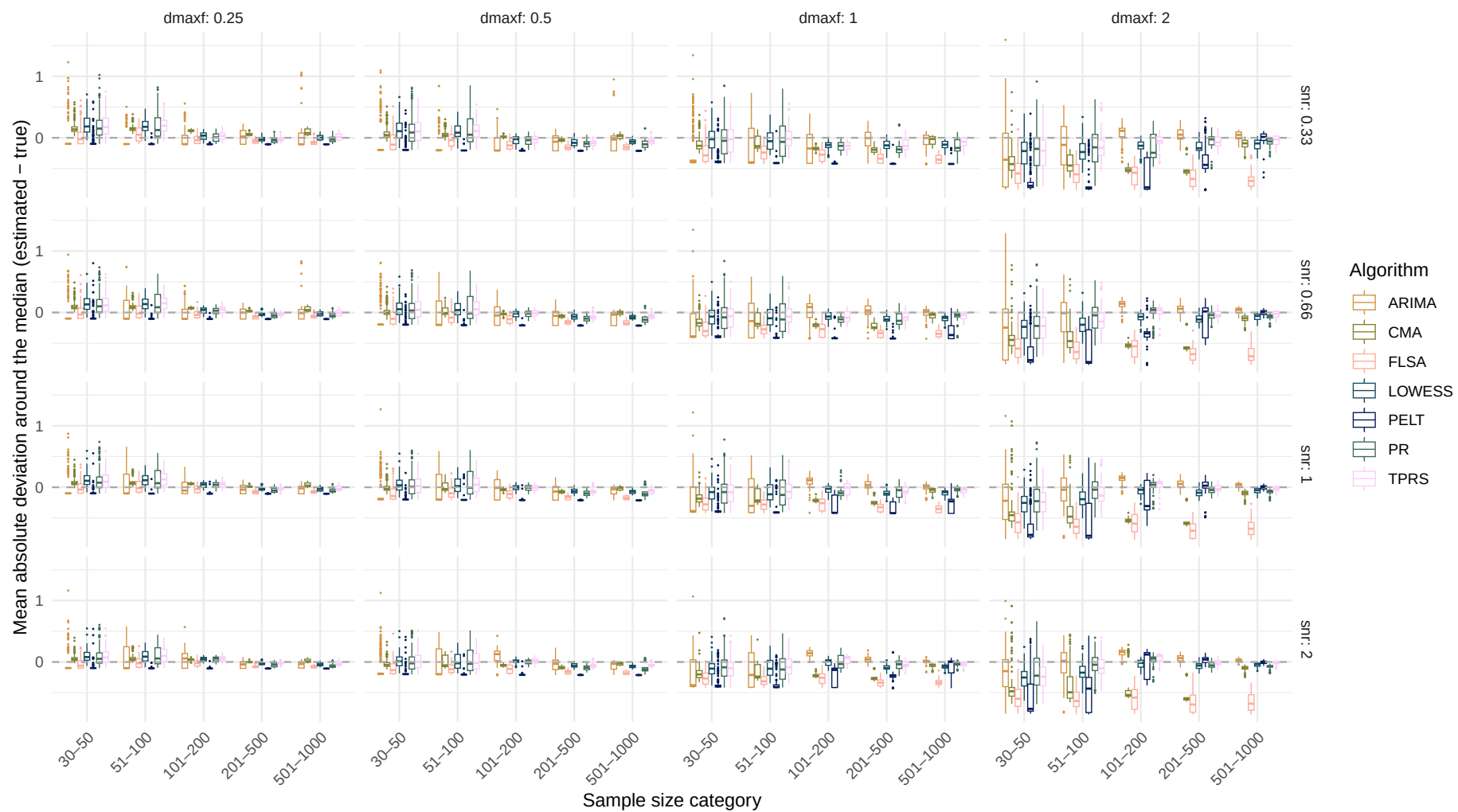
Systematic change pattern: Recurrent

Estimated mean absolute deviation around the median over sample size

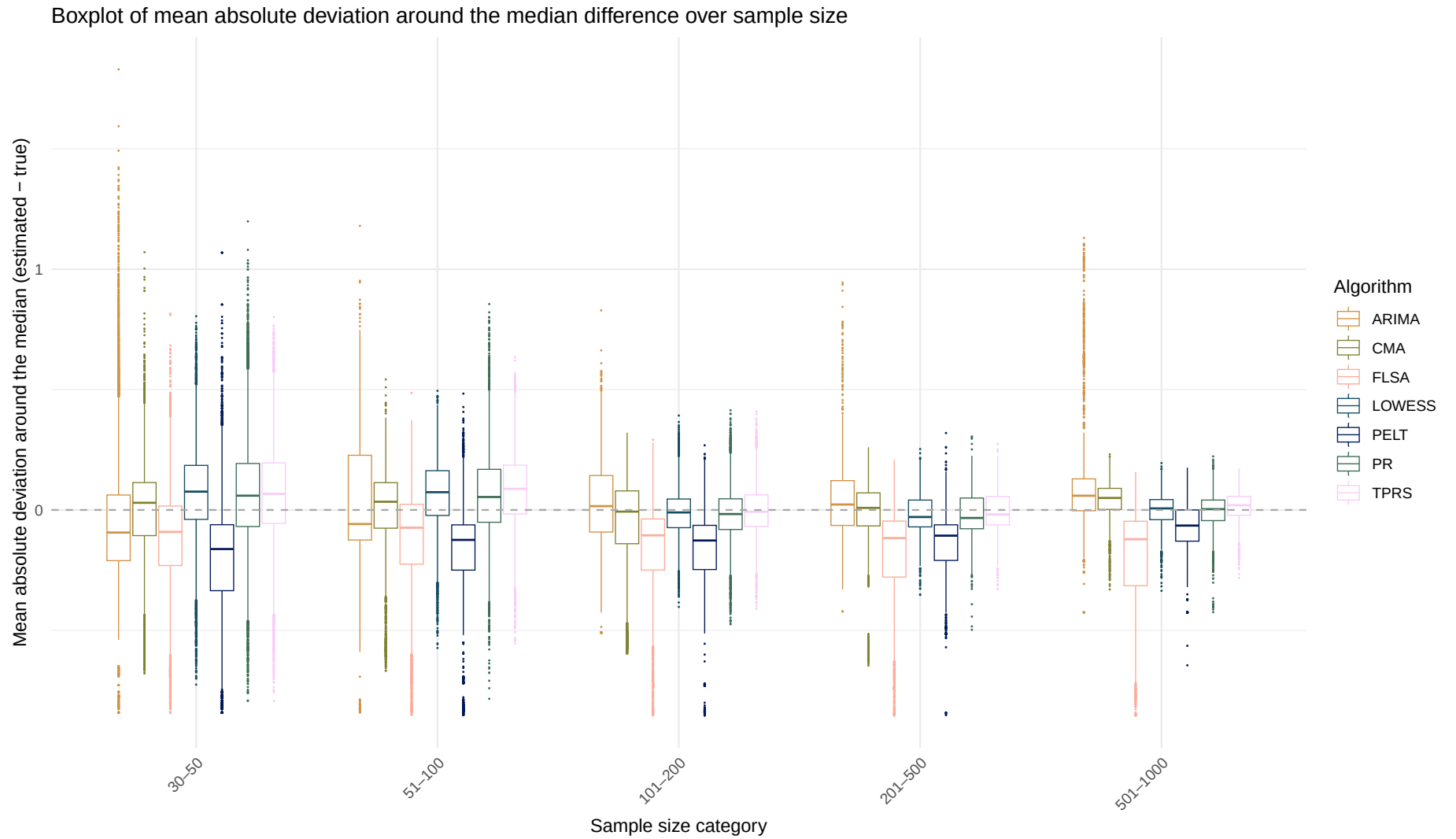
(Dashed line indicates the true mean absolute deviation around the median)



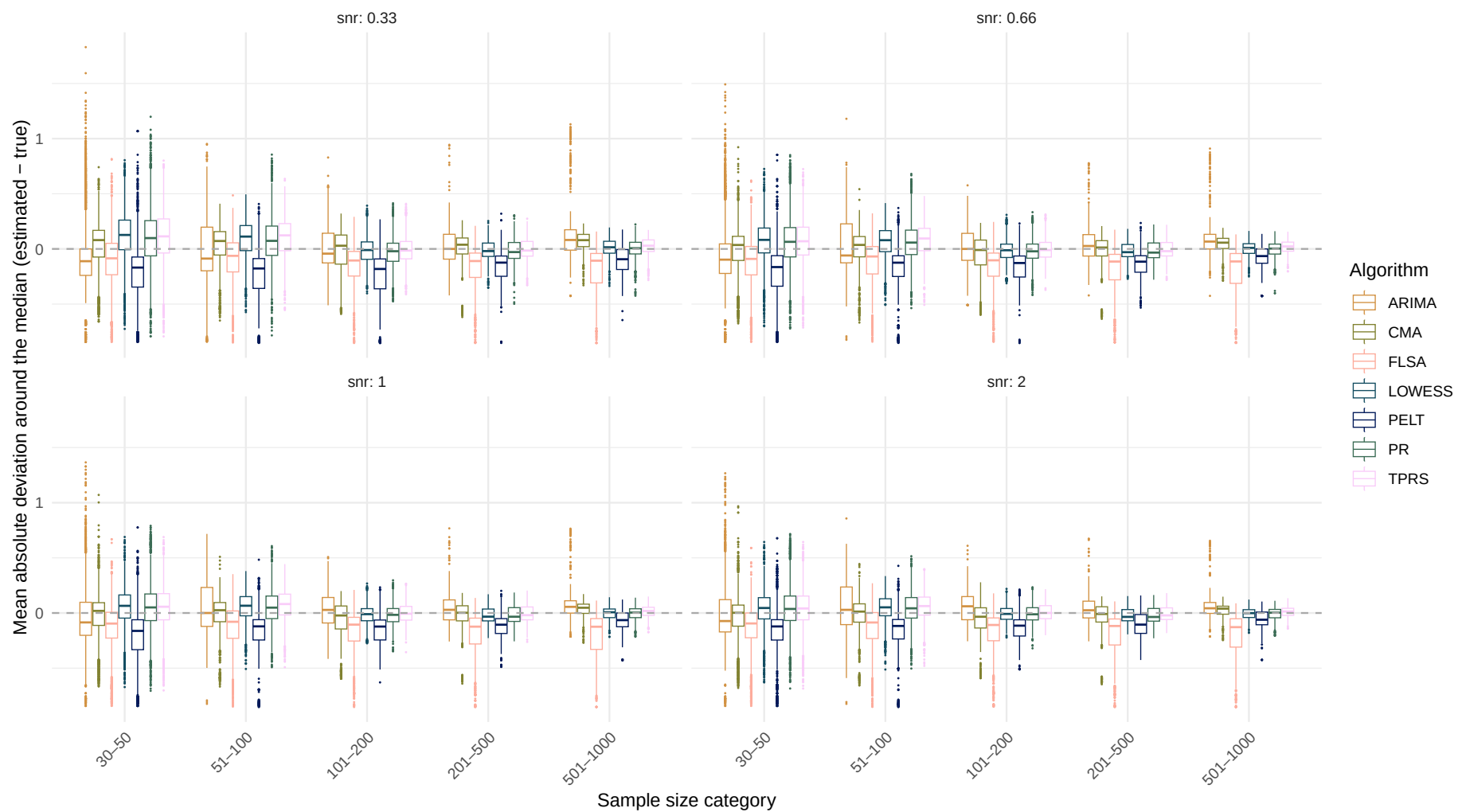
Boxplot of mean absolute deviation around the median difference over sample size



Systematic change pattern: All patterns combined



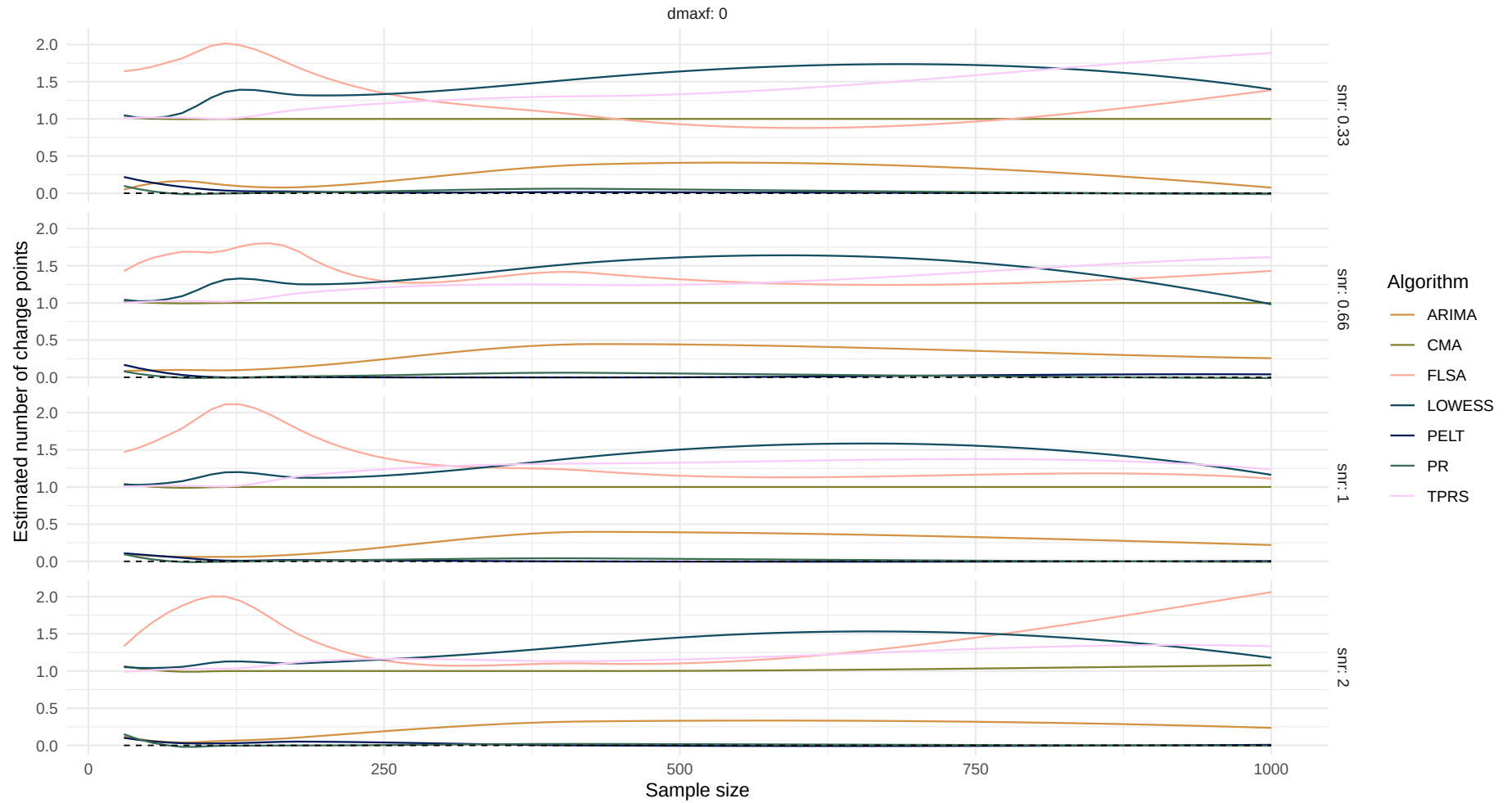
Boxplot of mean absolute deviation around the median difference over sample size



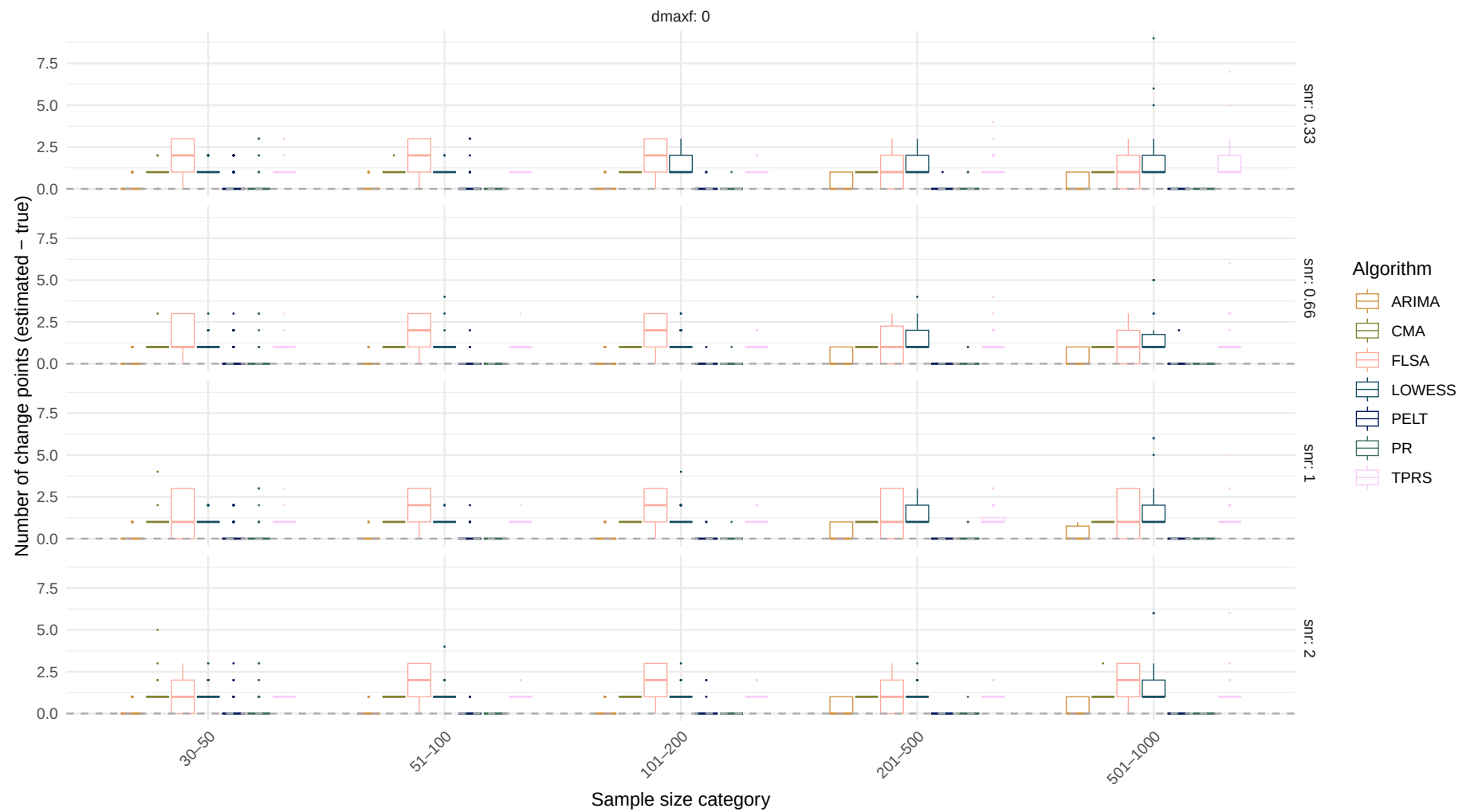
Estimand: Number of change points

Systematic change pattern: No change

Estimated number of change points over sample size
(Dashed line indicates the true number of change points)

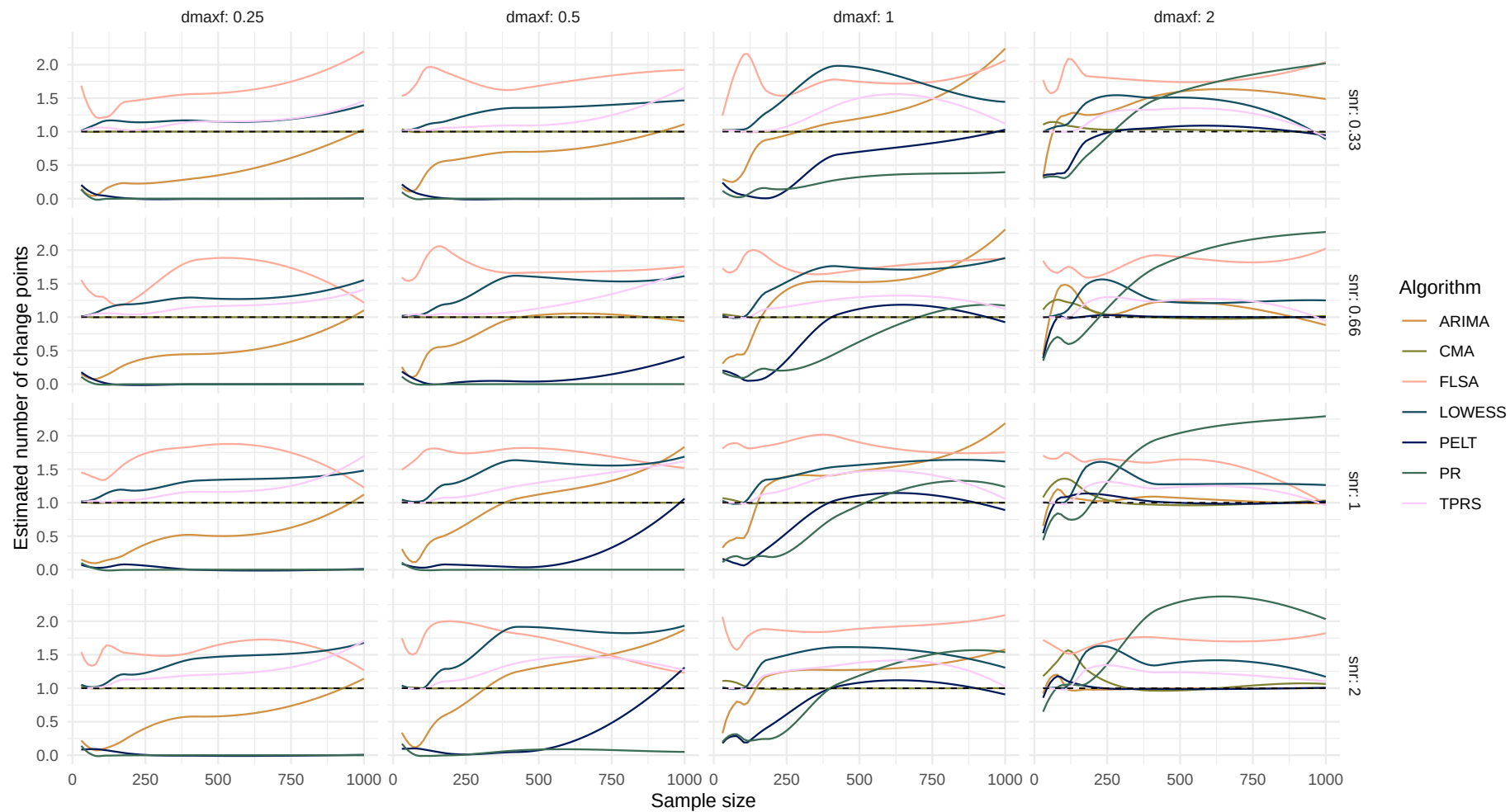


Boxplot of number of change points difference over sample size

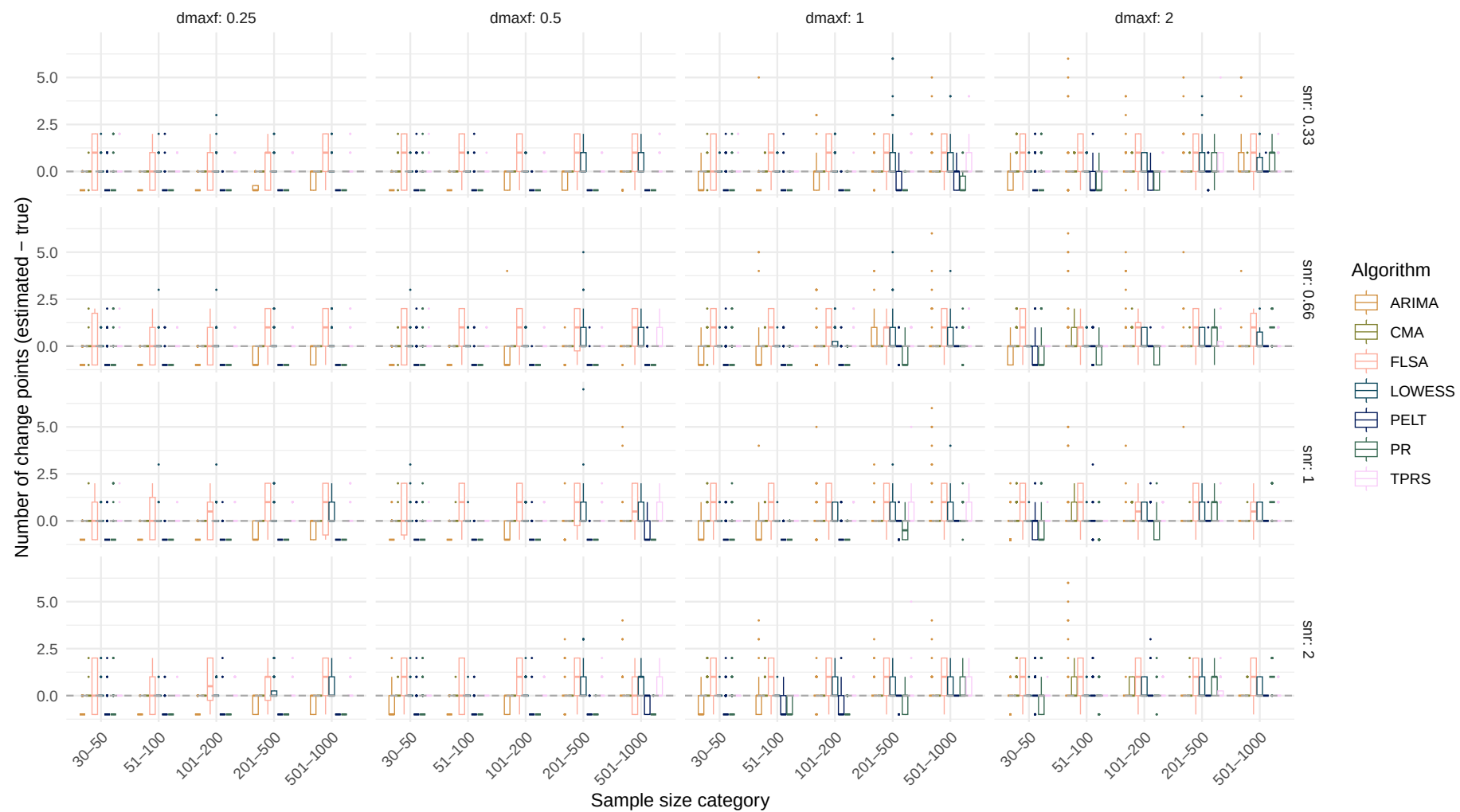


Systematic change pattern: Jump

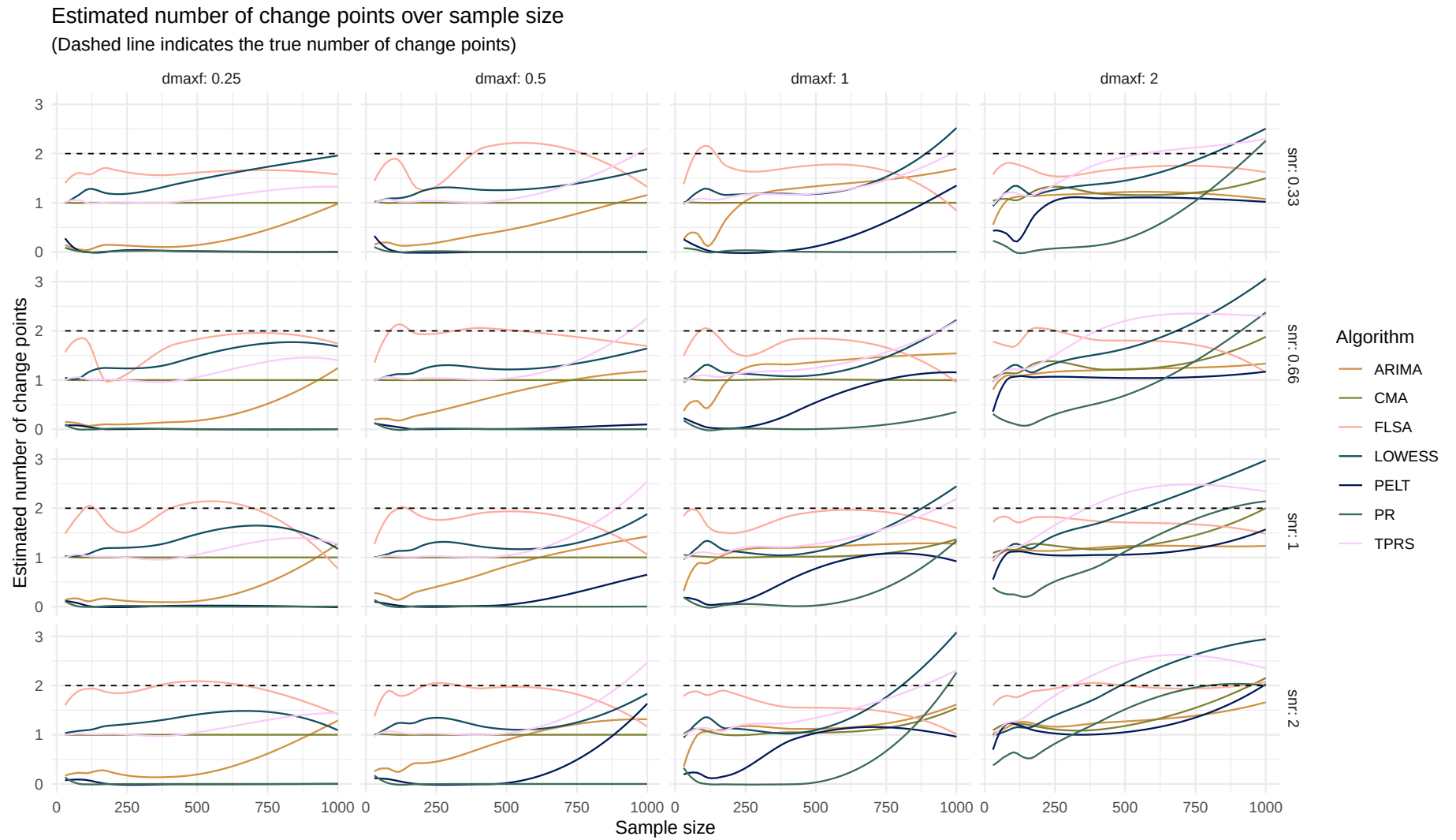
Estimated number of change points over sample size
(Dashed line indicates the true number of change points)



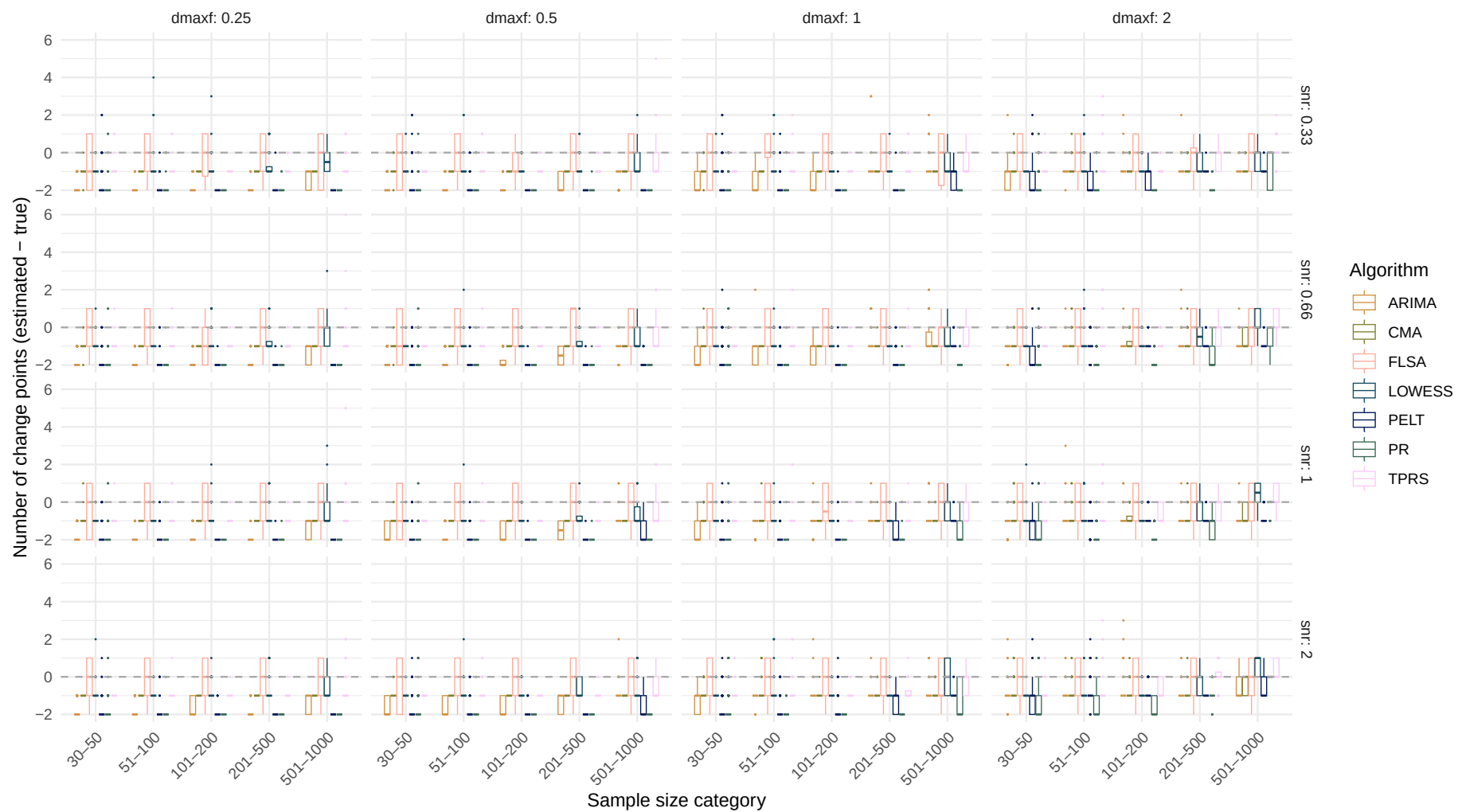
Boxplot of number of change points difference over sample size



Systematic change pattern: Linear

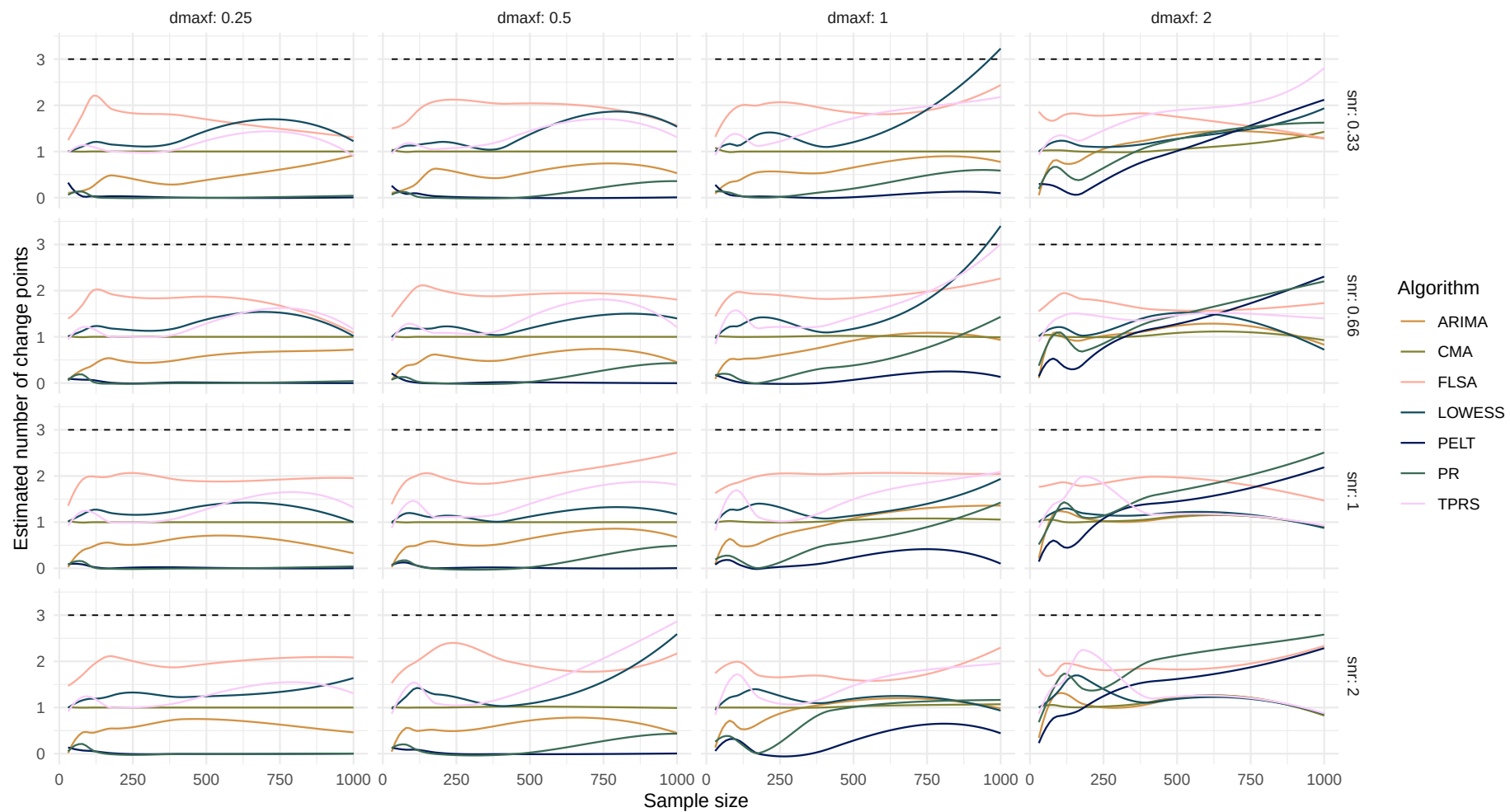


Boxplot of number of change points difference over sample size

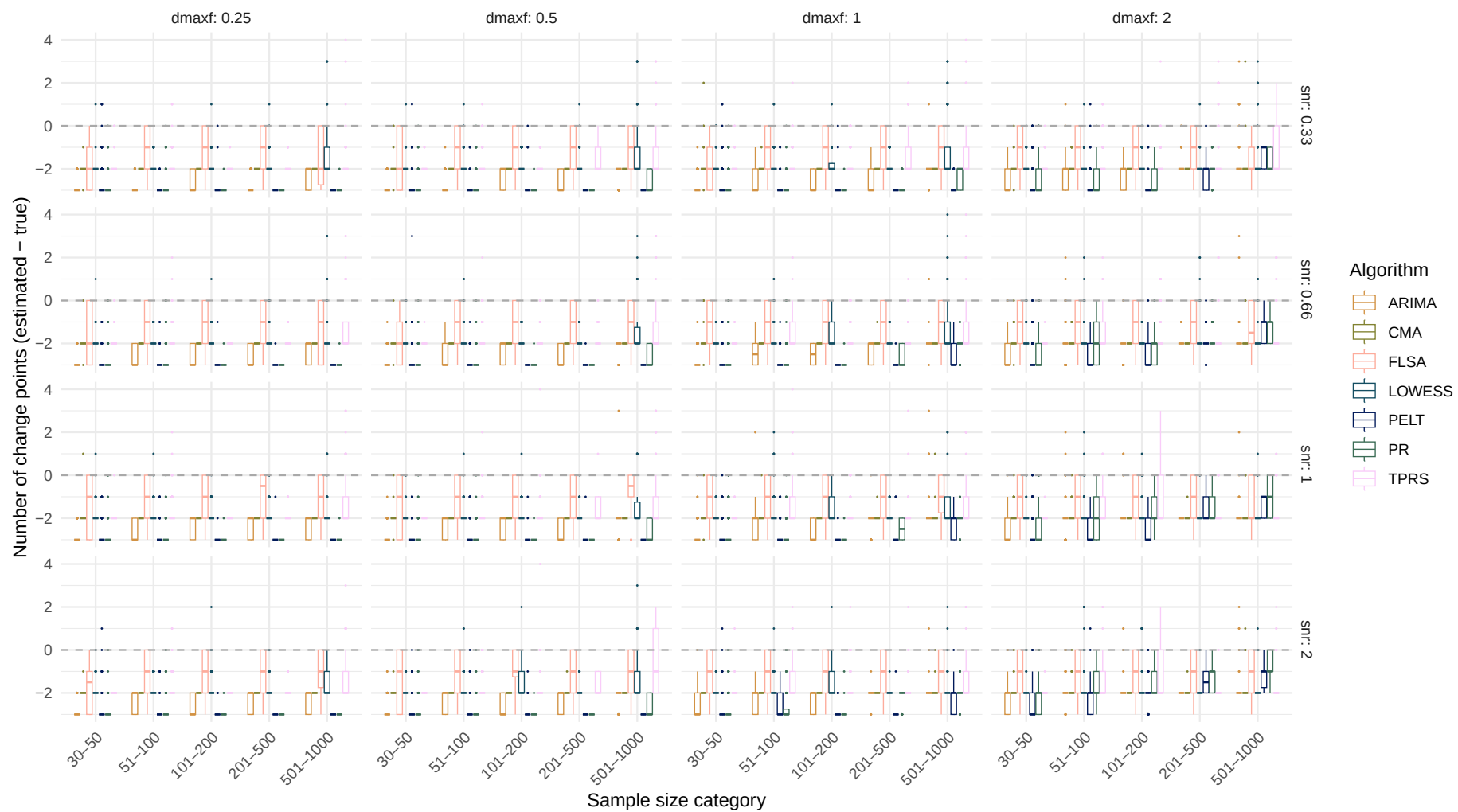


Systematic change pattern: Quadratic

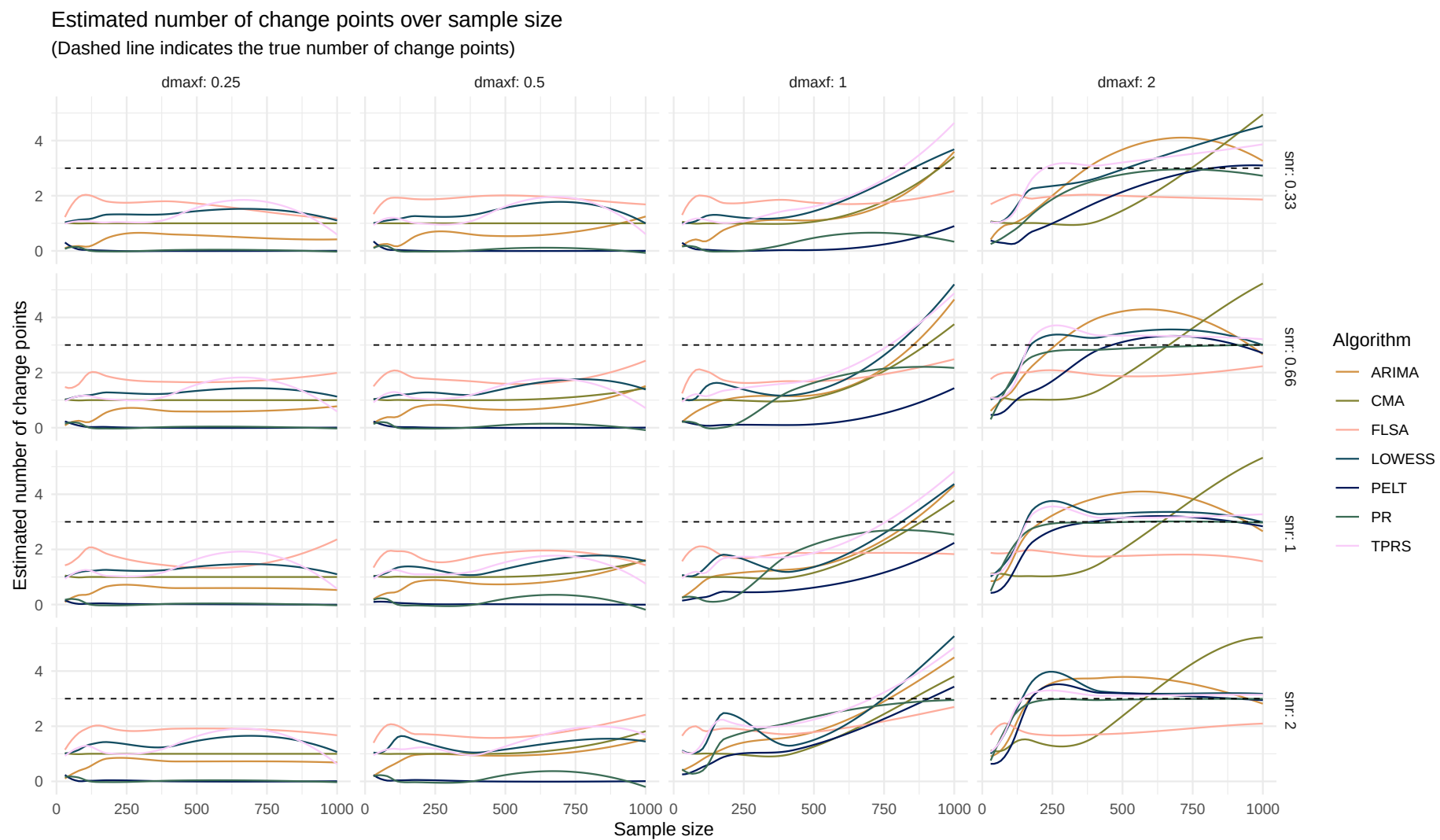
Estimated number of change points over sample size
(Dashed line indicates the true number of change points)



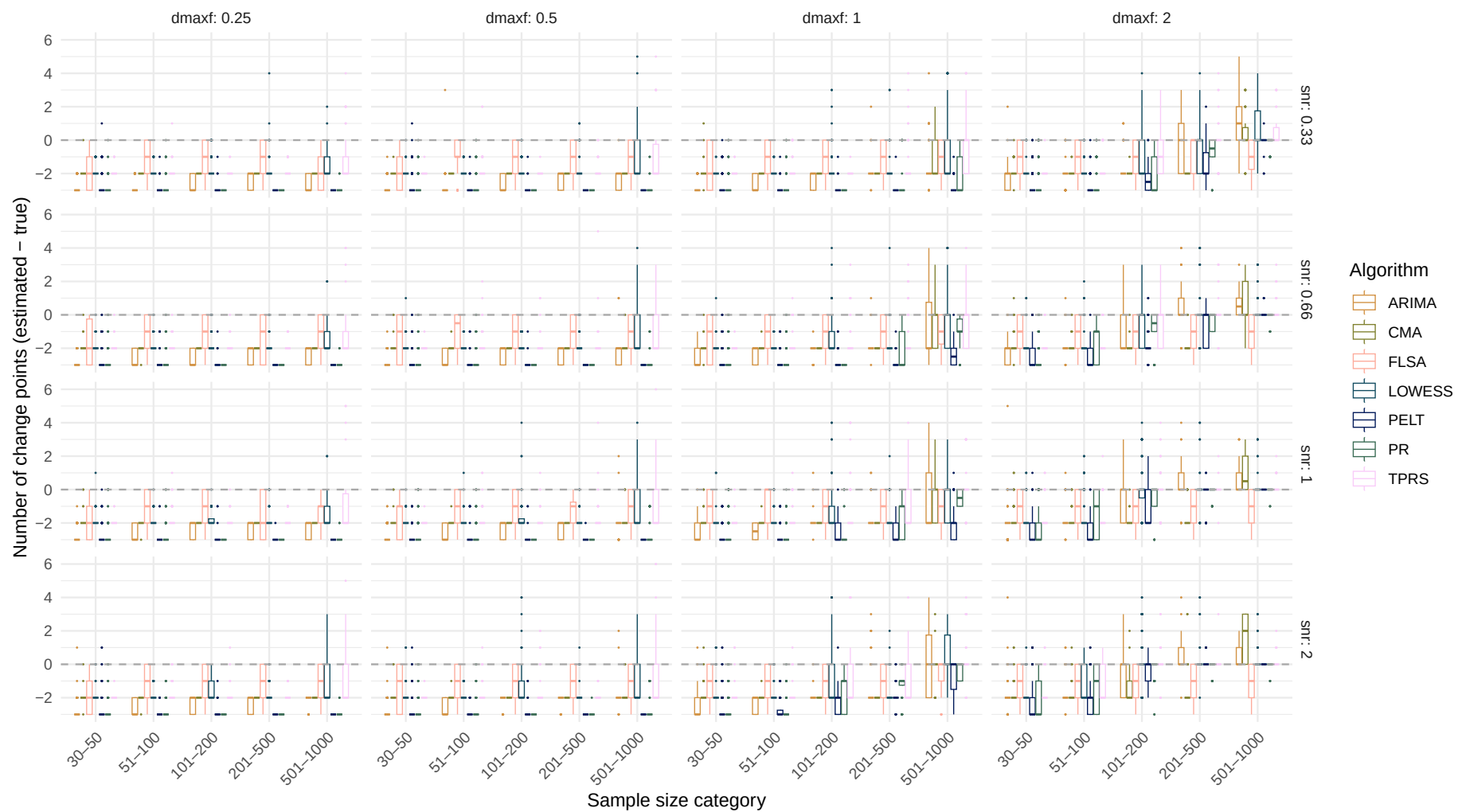
Boxplot of number of change points difference over sample size



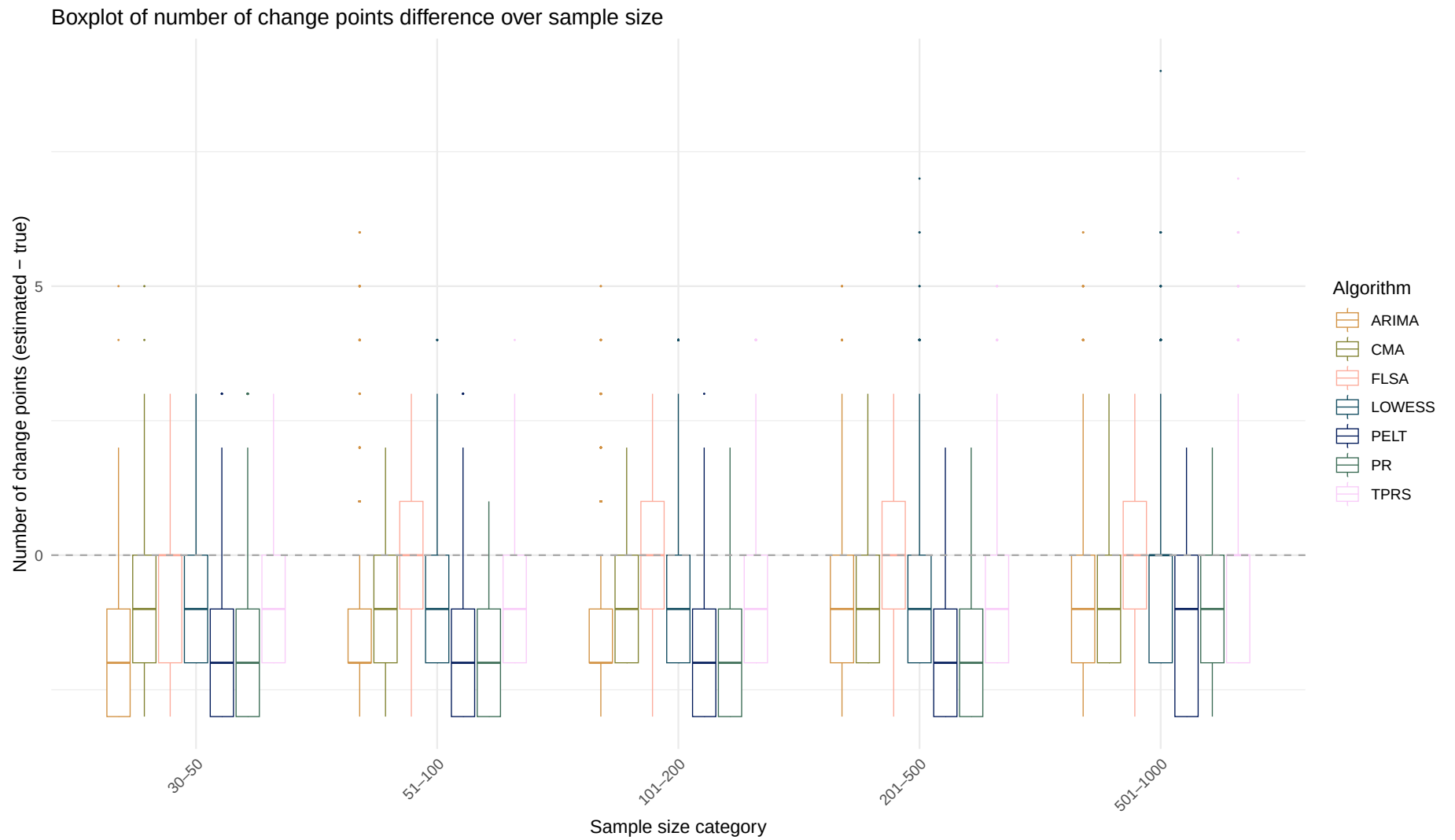
Systematic change pattern: Recurrent



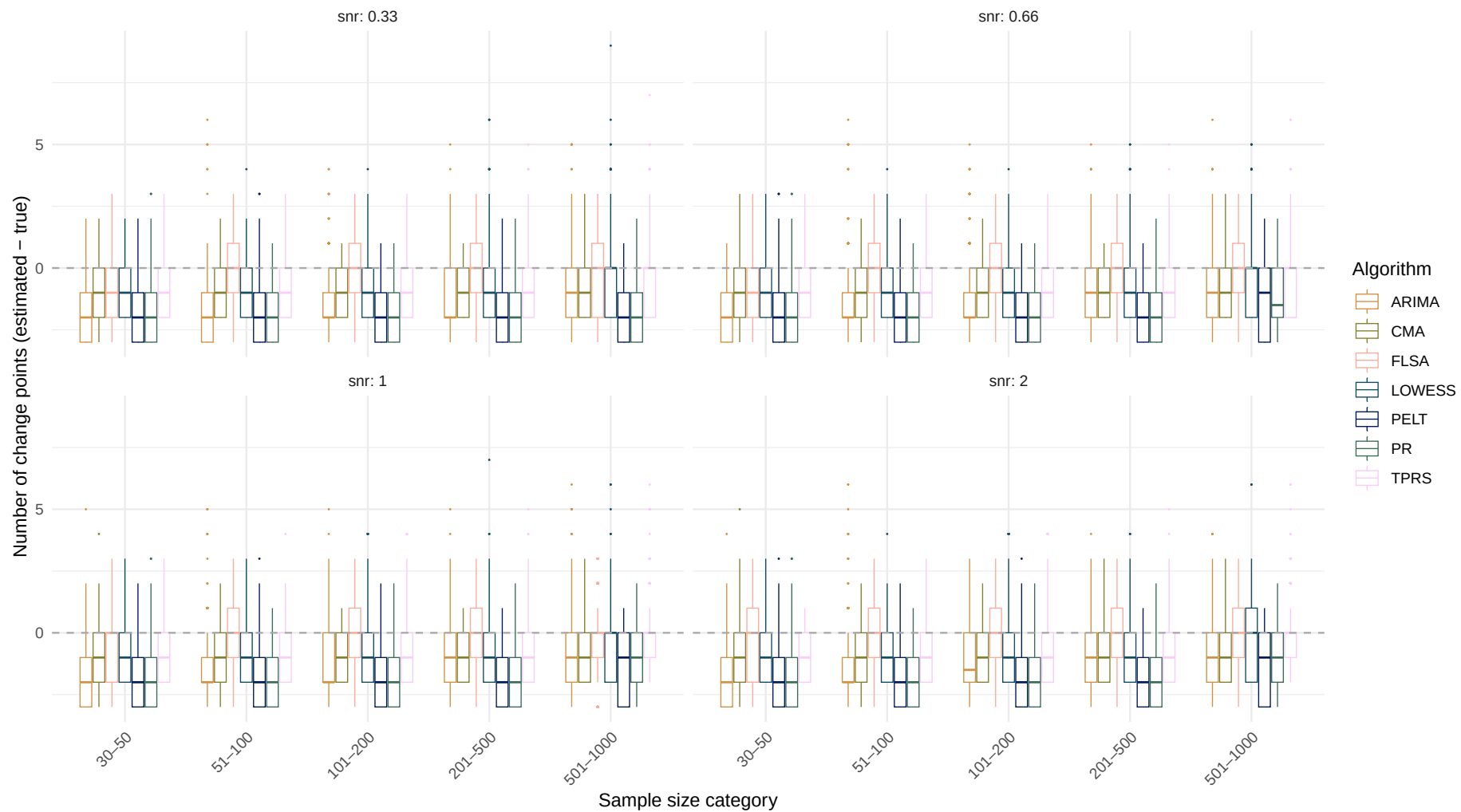
Boxplot of number of change points difference over sample size



Systematic change pattern: All patterns combined



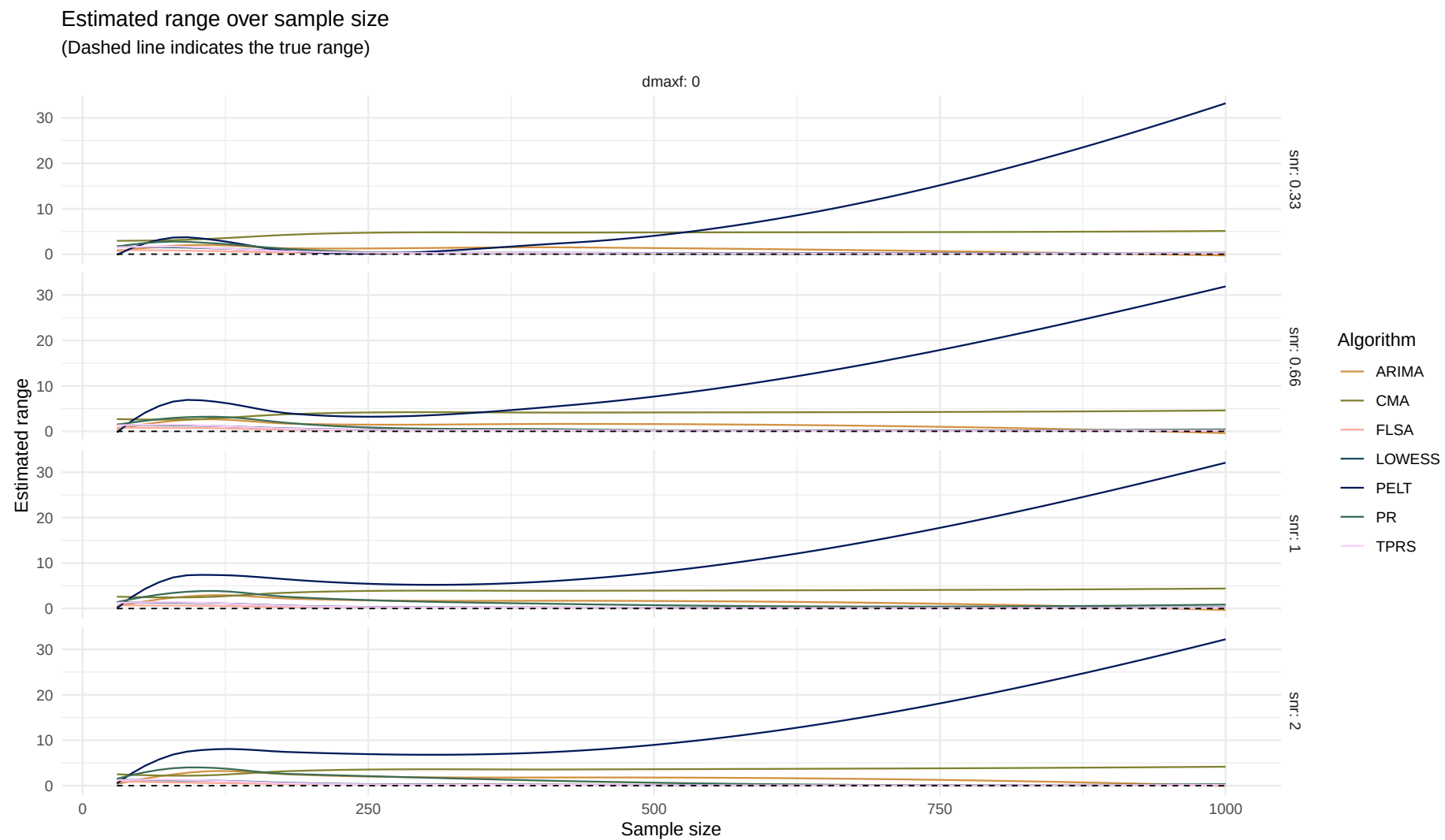
Boxplot of number of change points difference over sample size



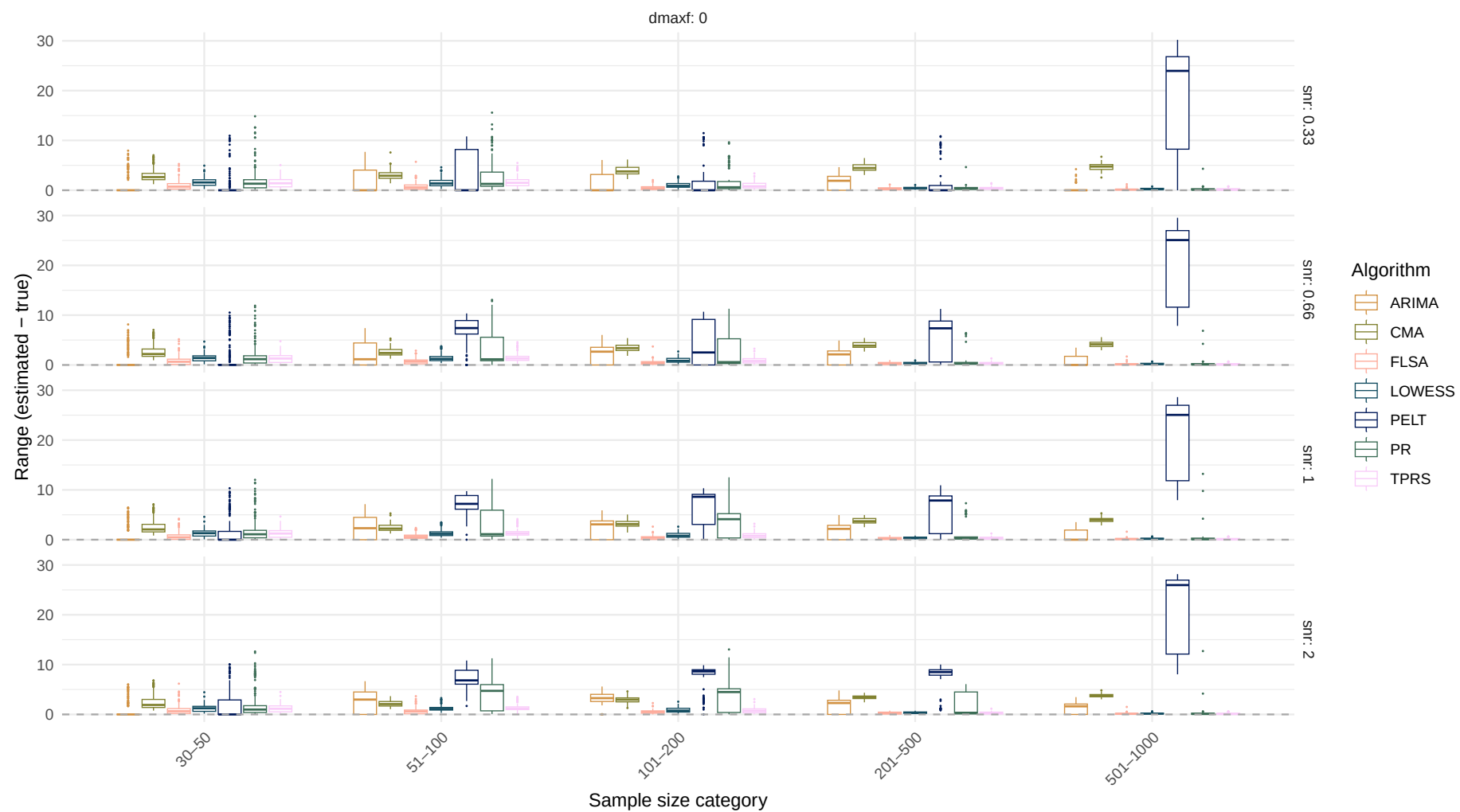
Log-normal distribution

Estimand: Range

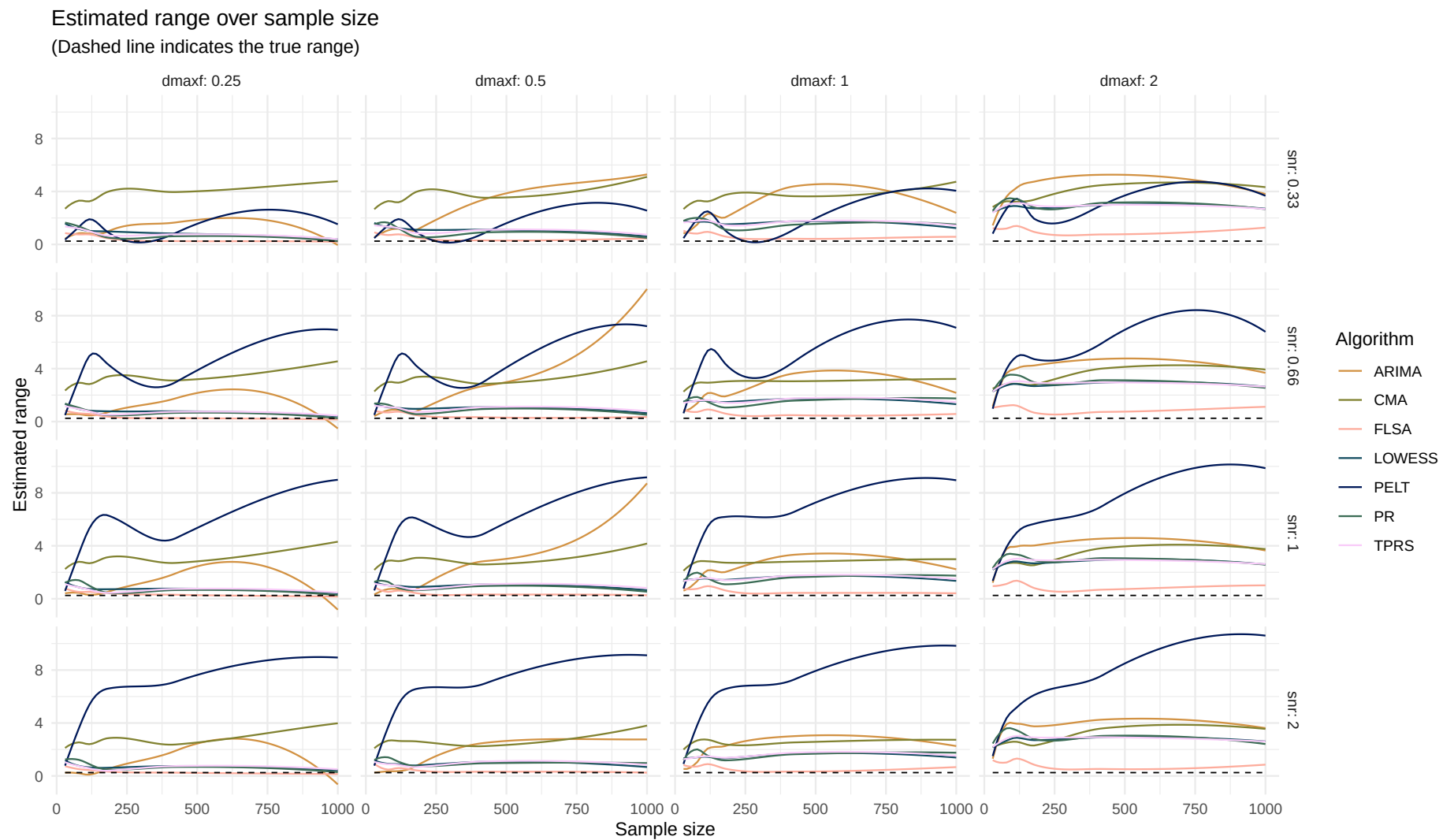
Systematic change pattern: No change



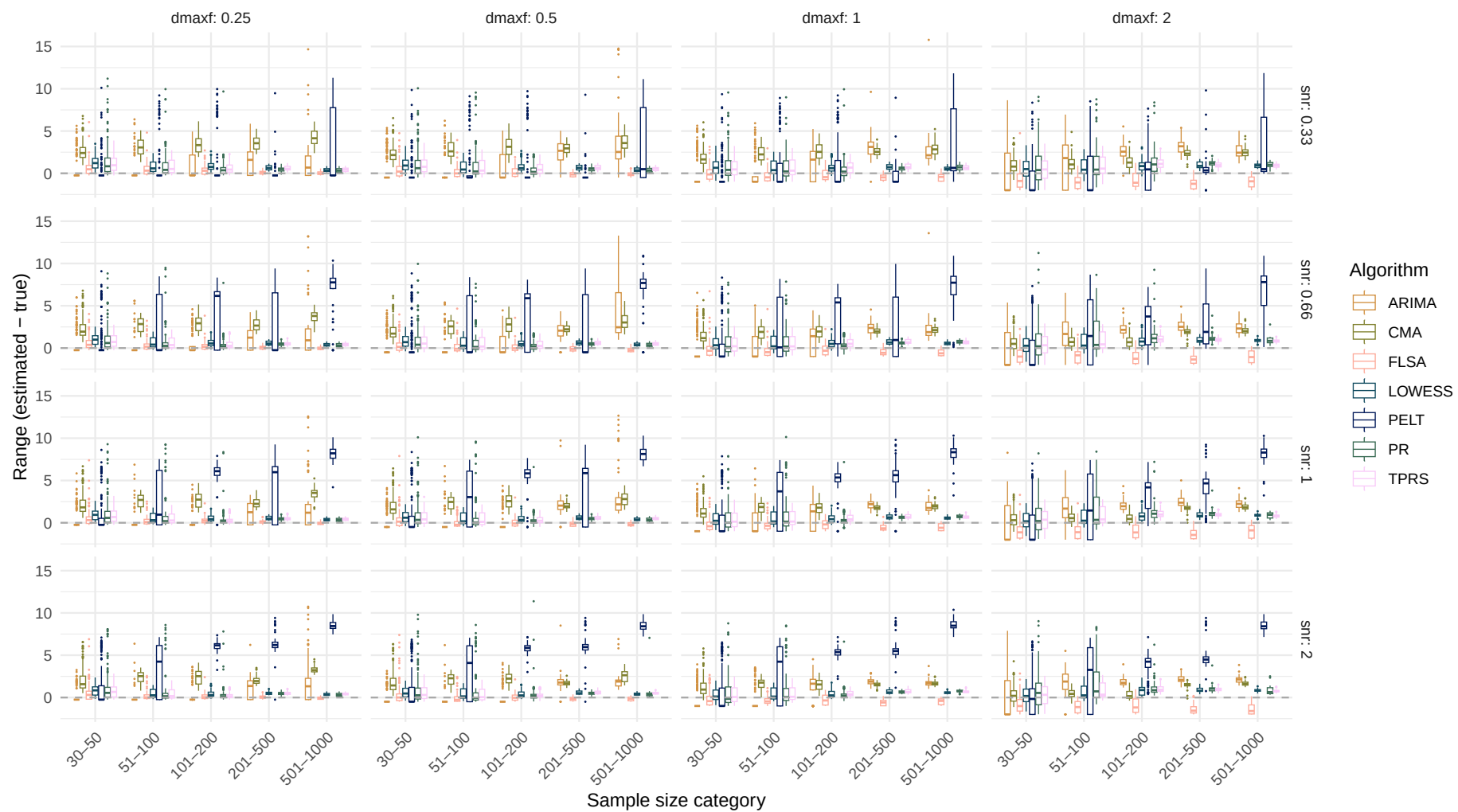
Boxplot of range difference over sample size



Systematic change pattern: Jump

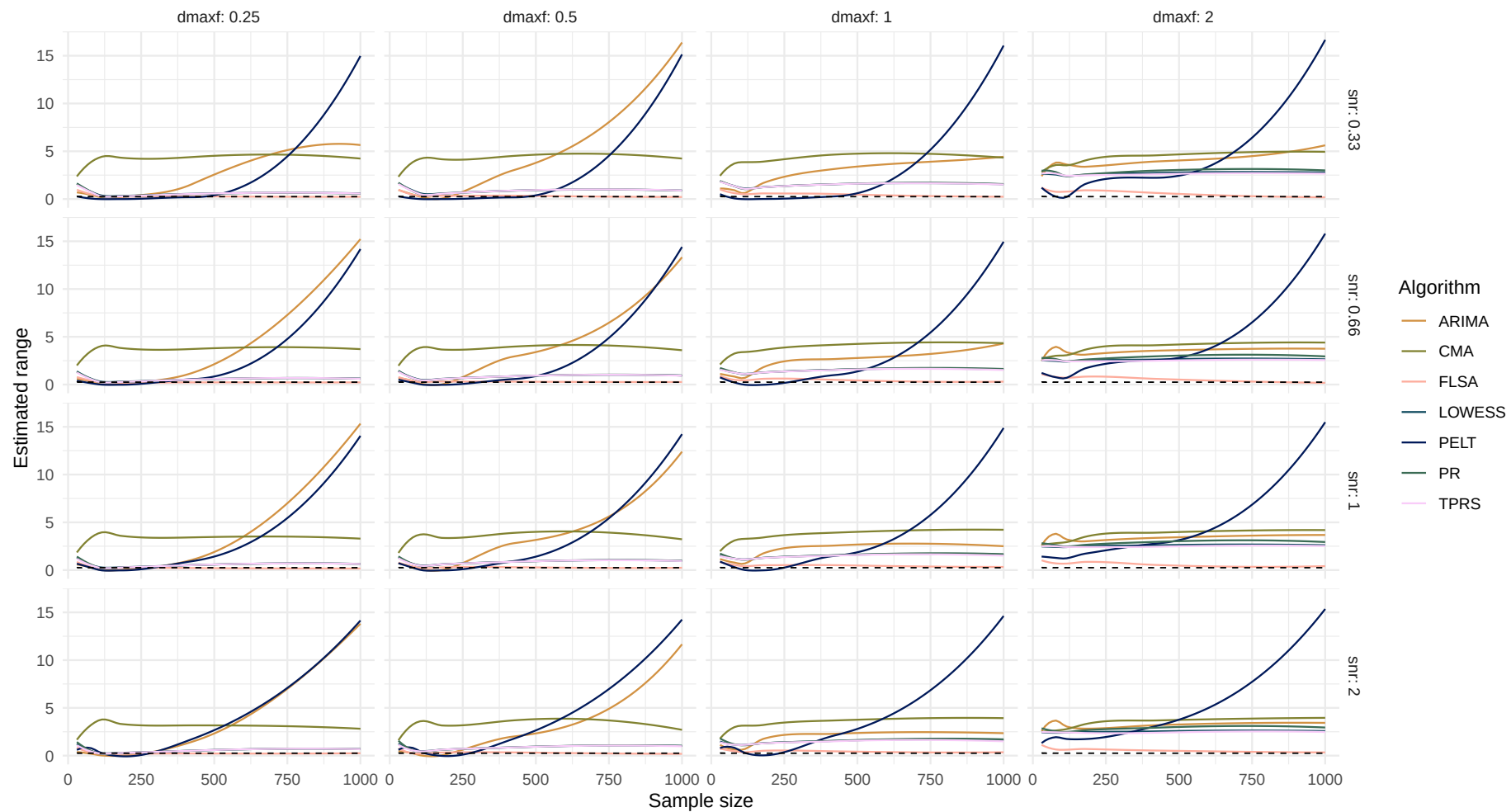


Boxplot of range difference over sample size

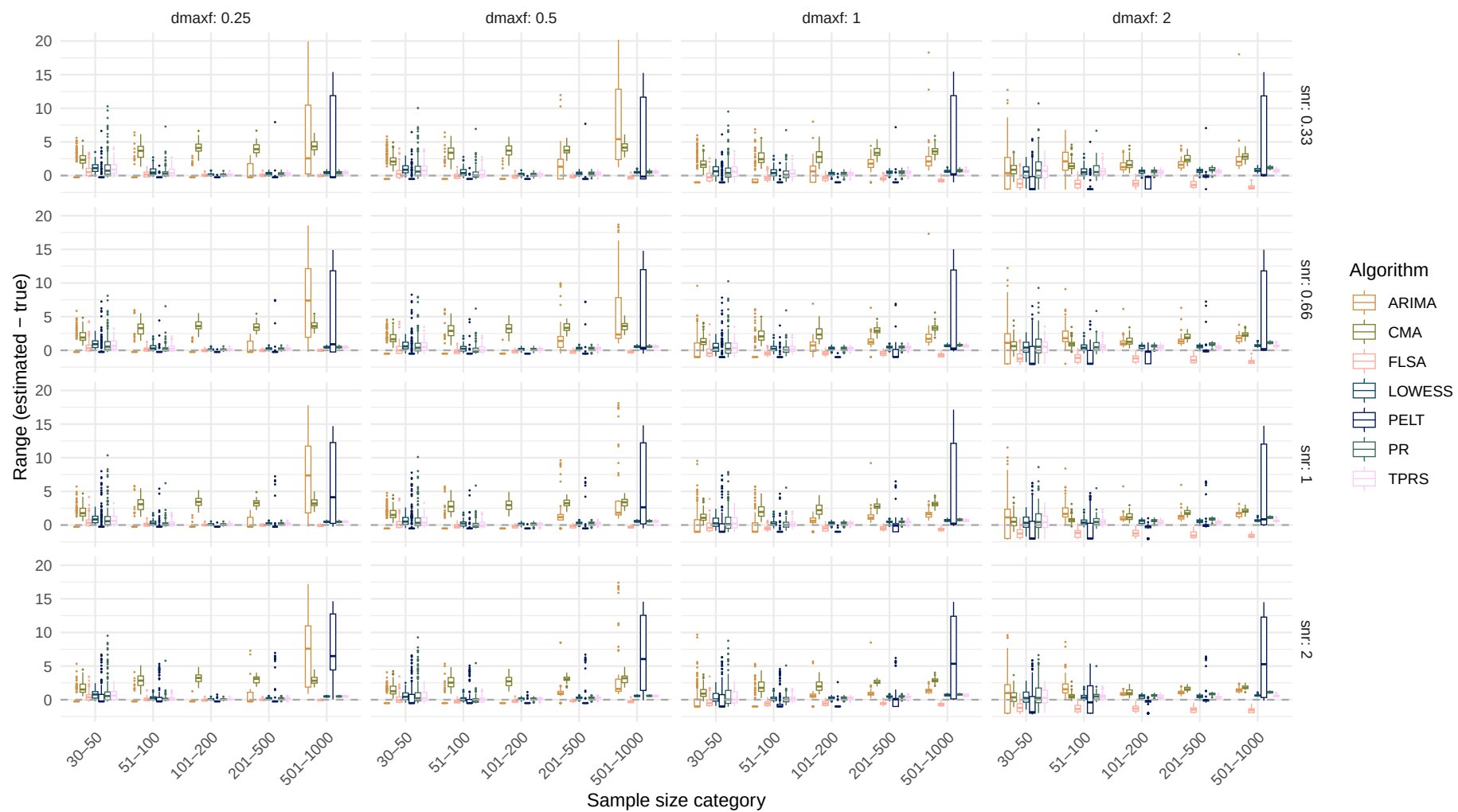


Systematic change pattern: Linear

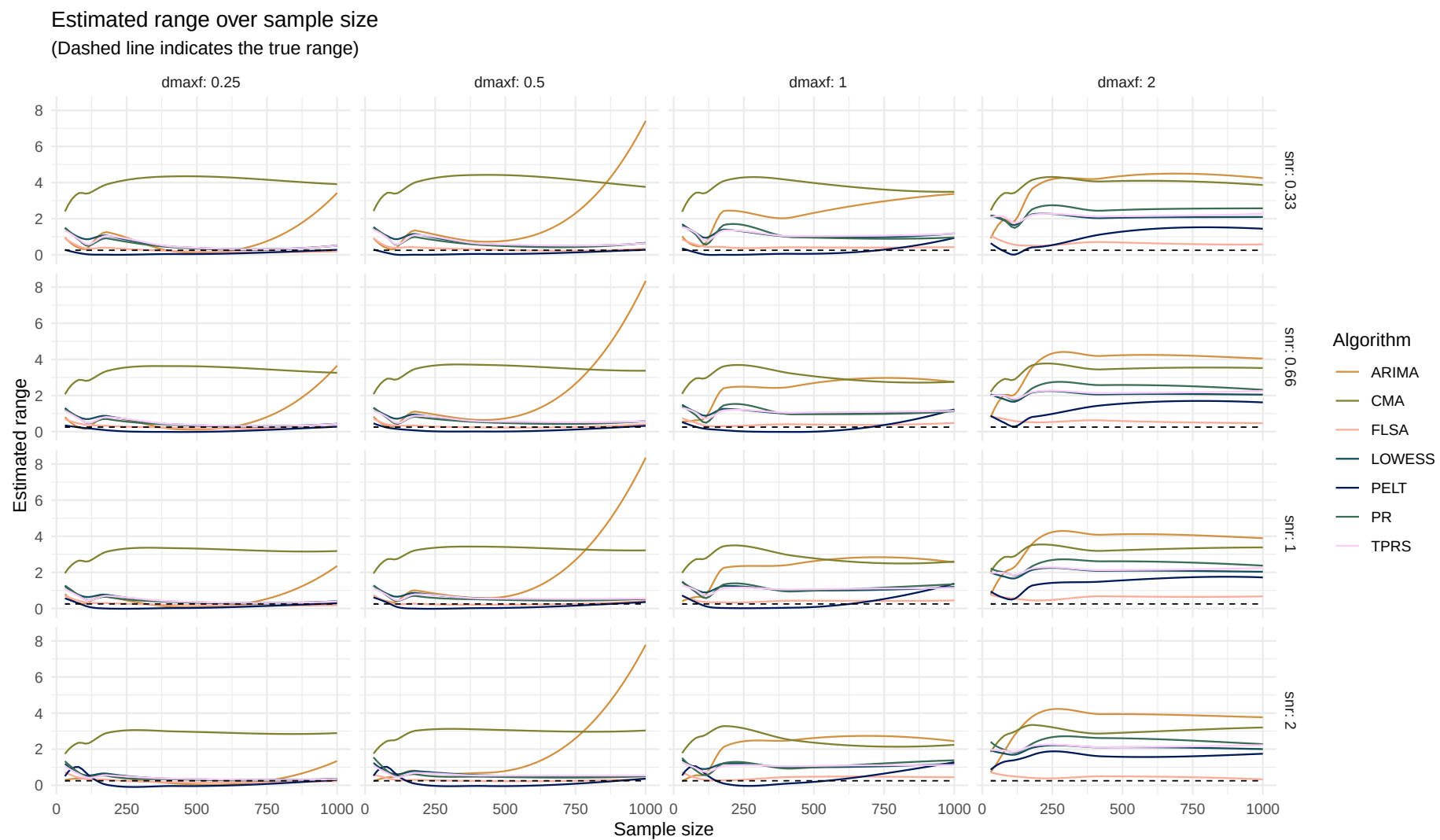
Estimated range over sample size
(Dashed line indicates the true range)



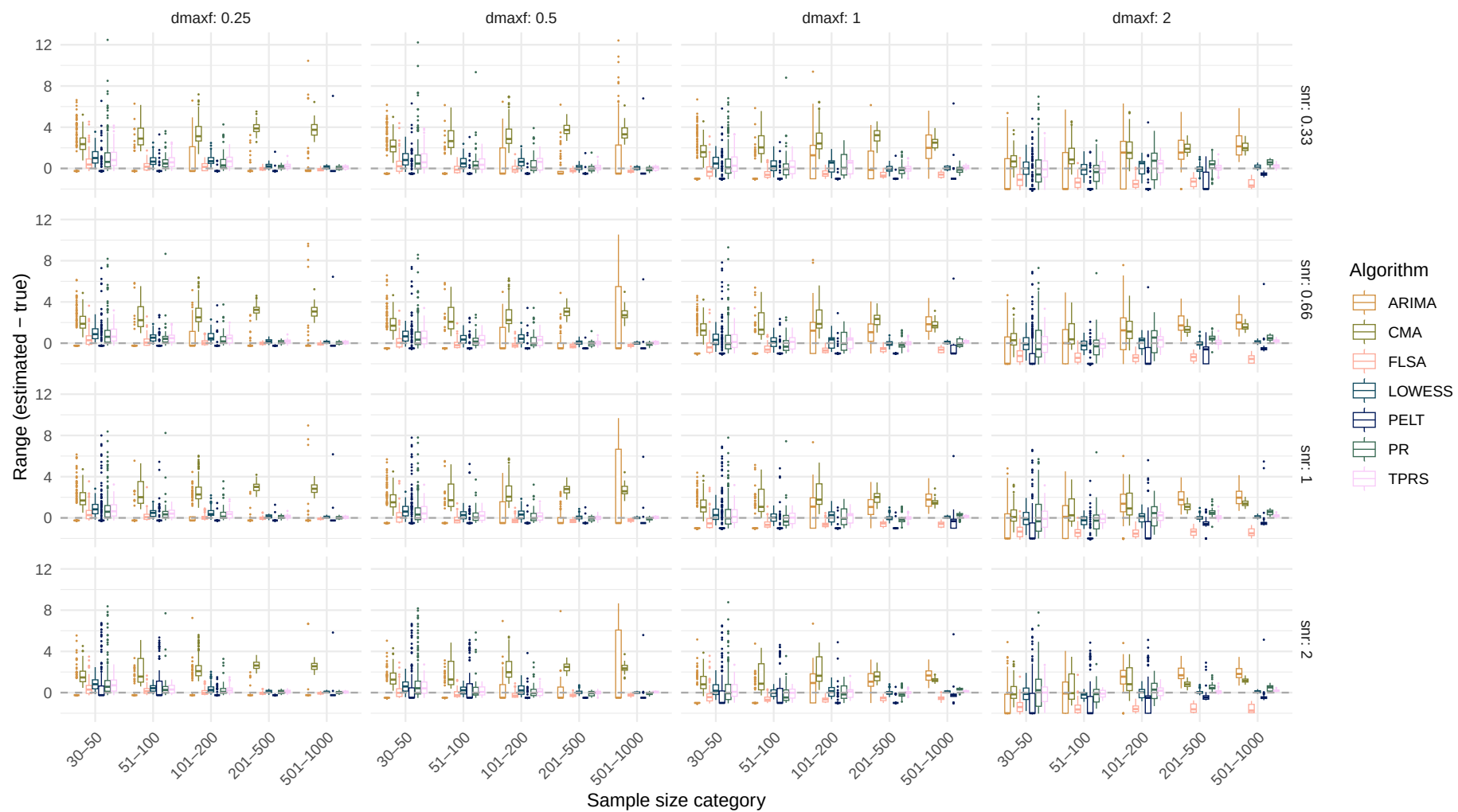
Boxplot of range difference over sample size



Systematic change pattern: Quadratic

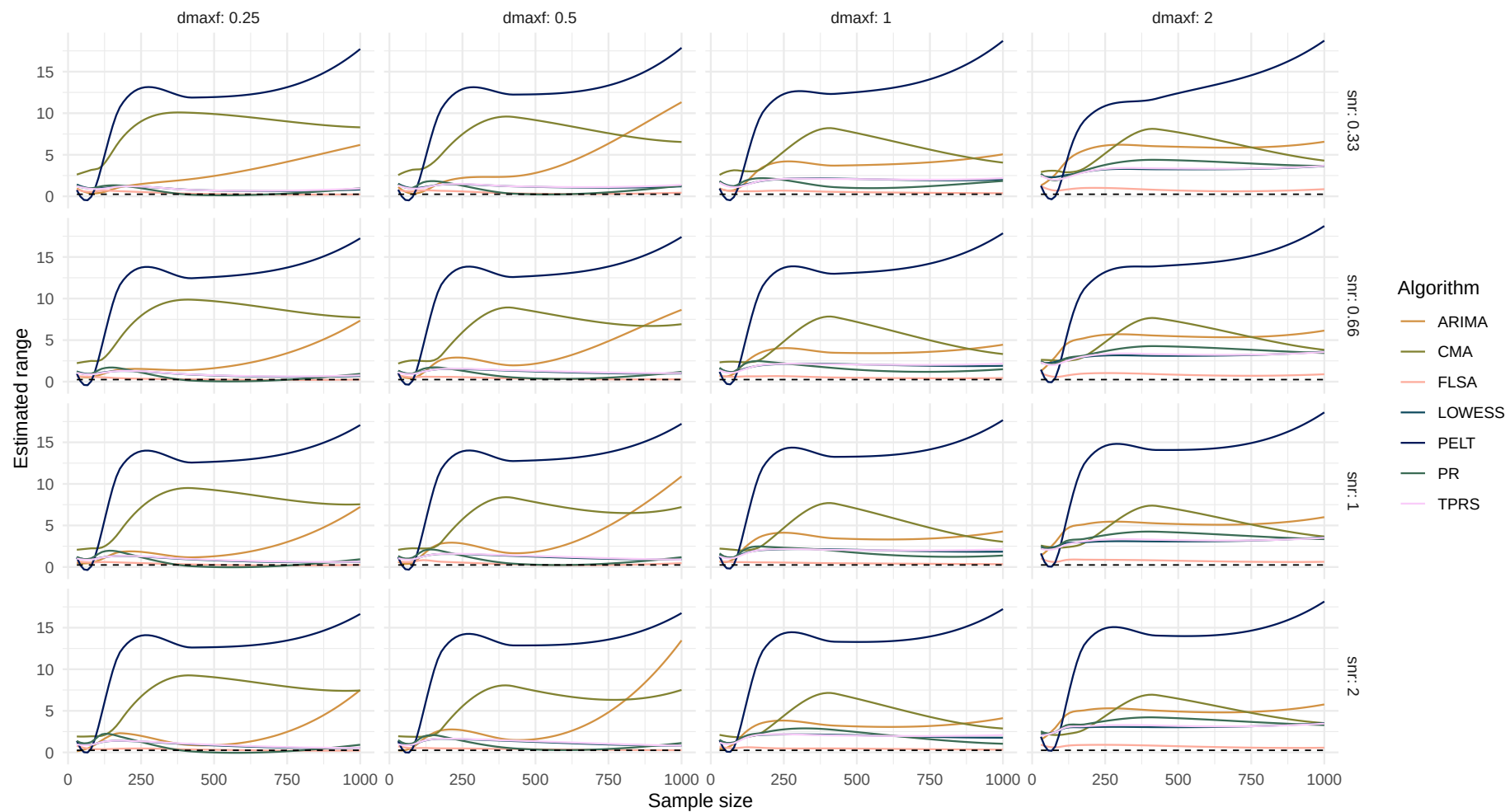


Boxplot of range difference over sample size

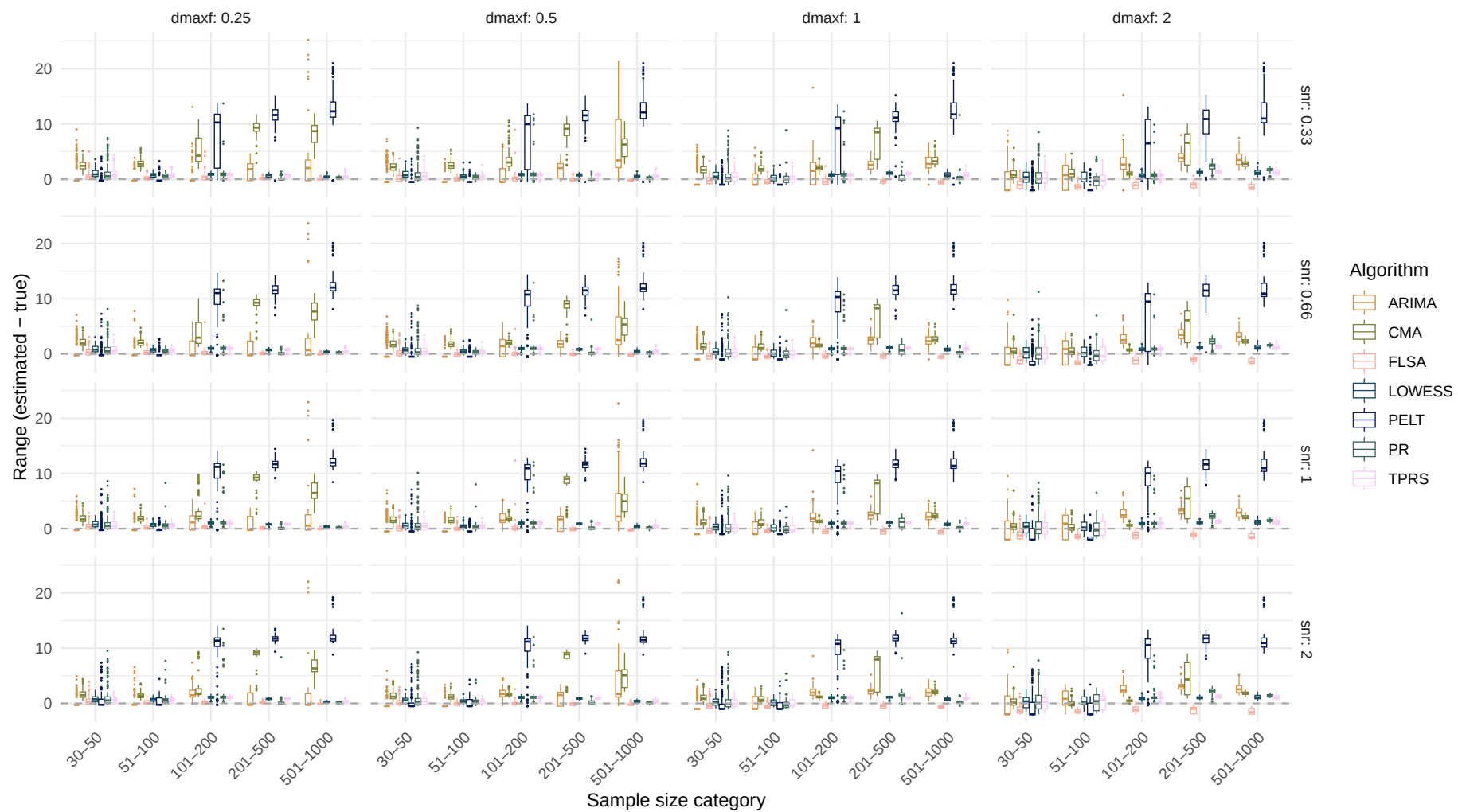


Systematic change pattern: Recurrent

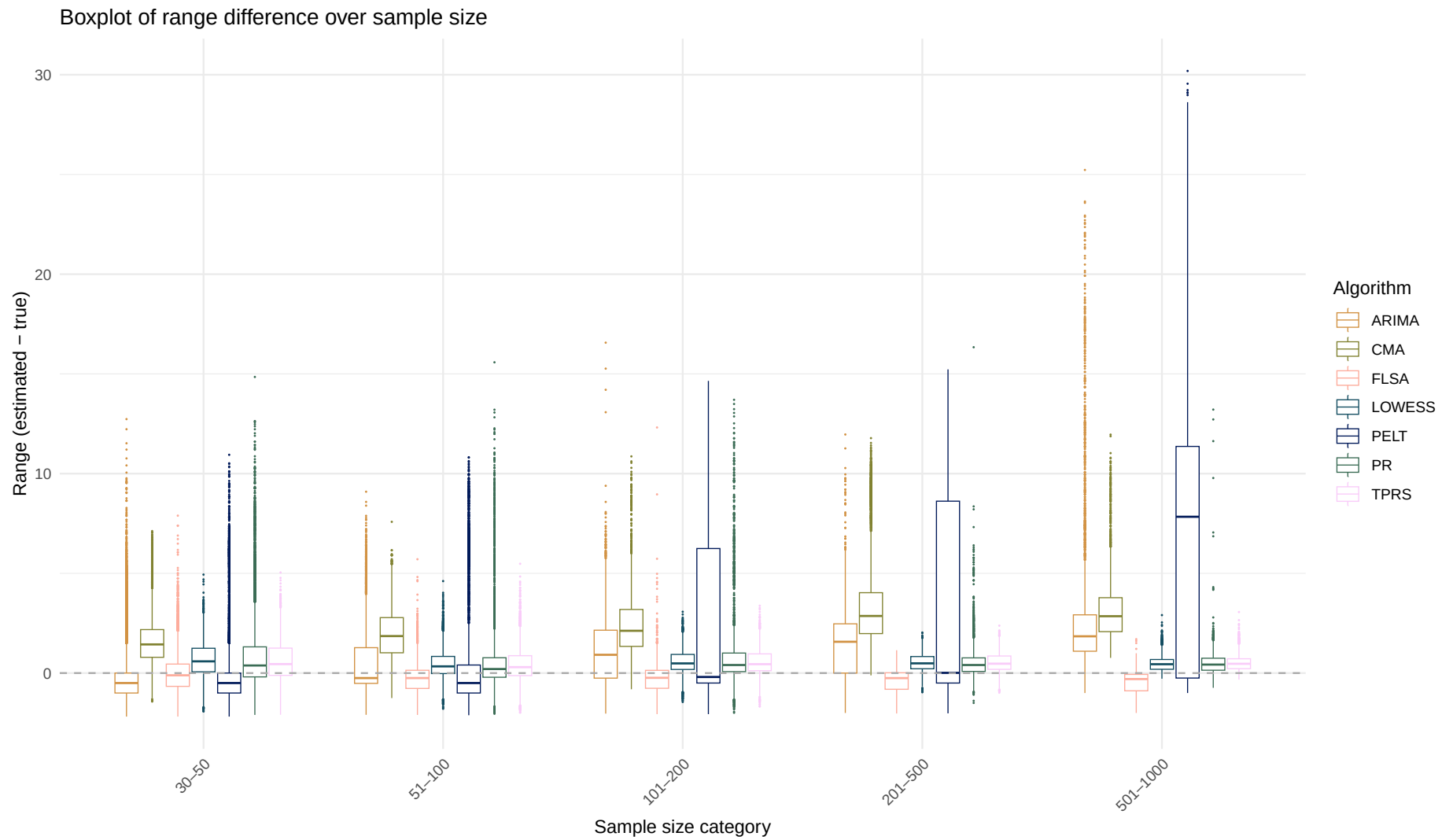
Estimated range over sample size
(Dashed line indicates the true range)



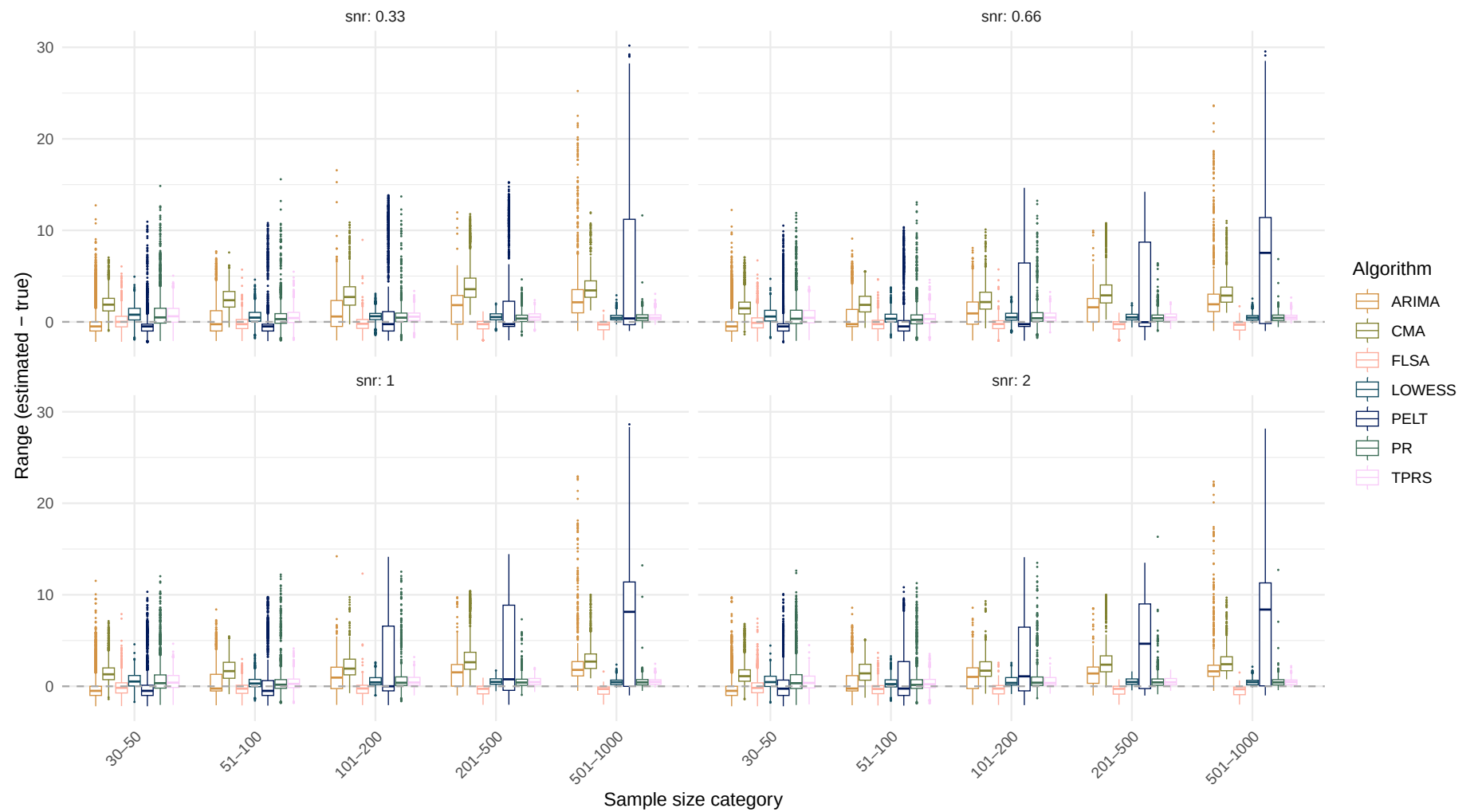
Boxplot of range difference over sample size



Systematic change pattern: All patterns combined

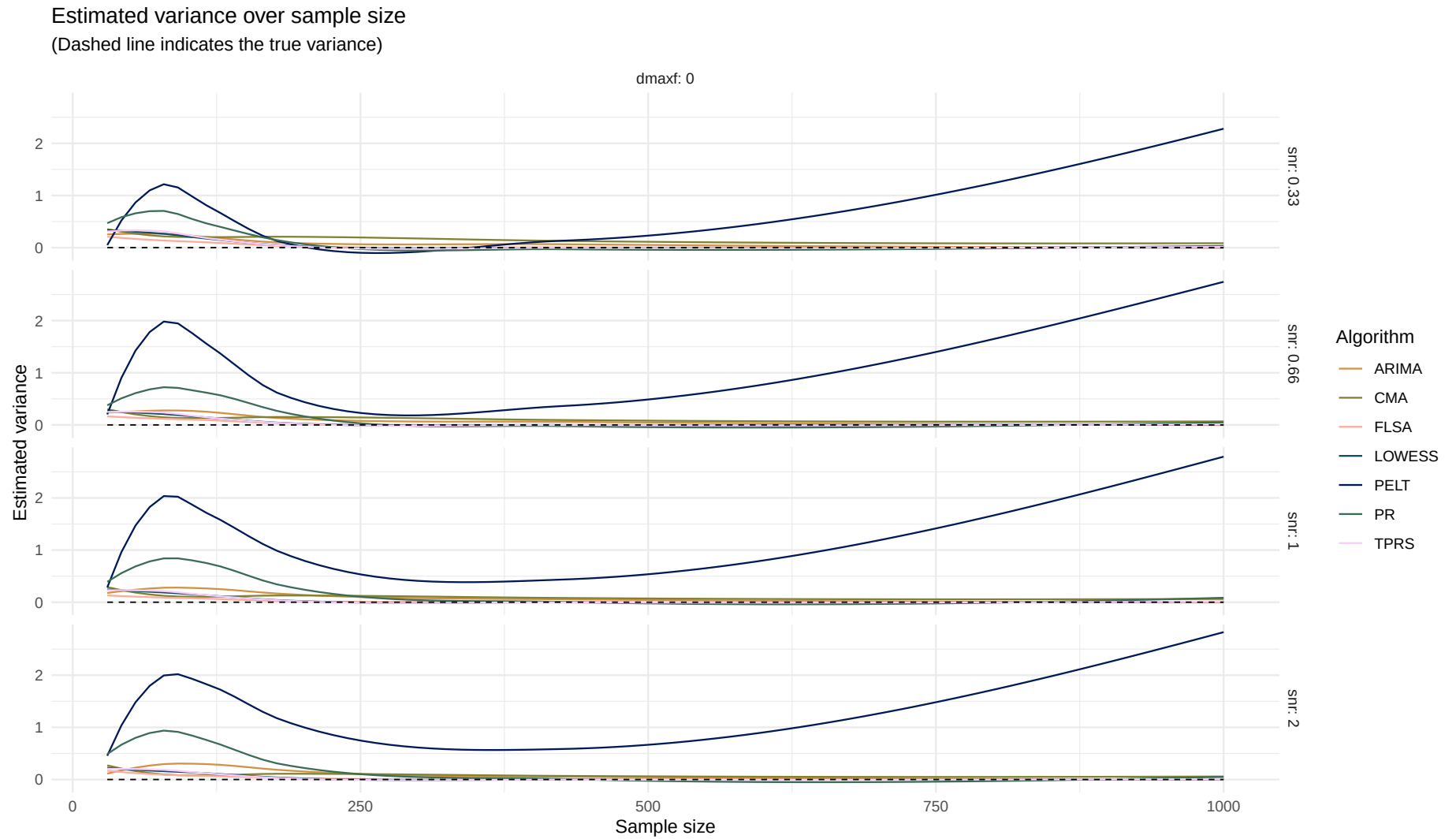


Boxplot of range difference over sample size

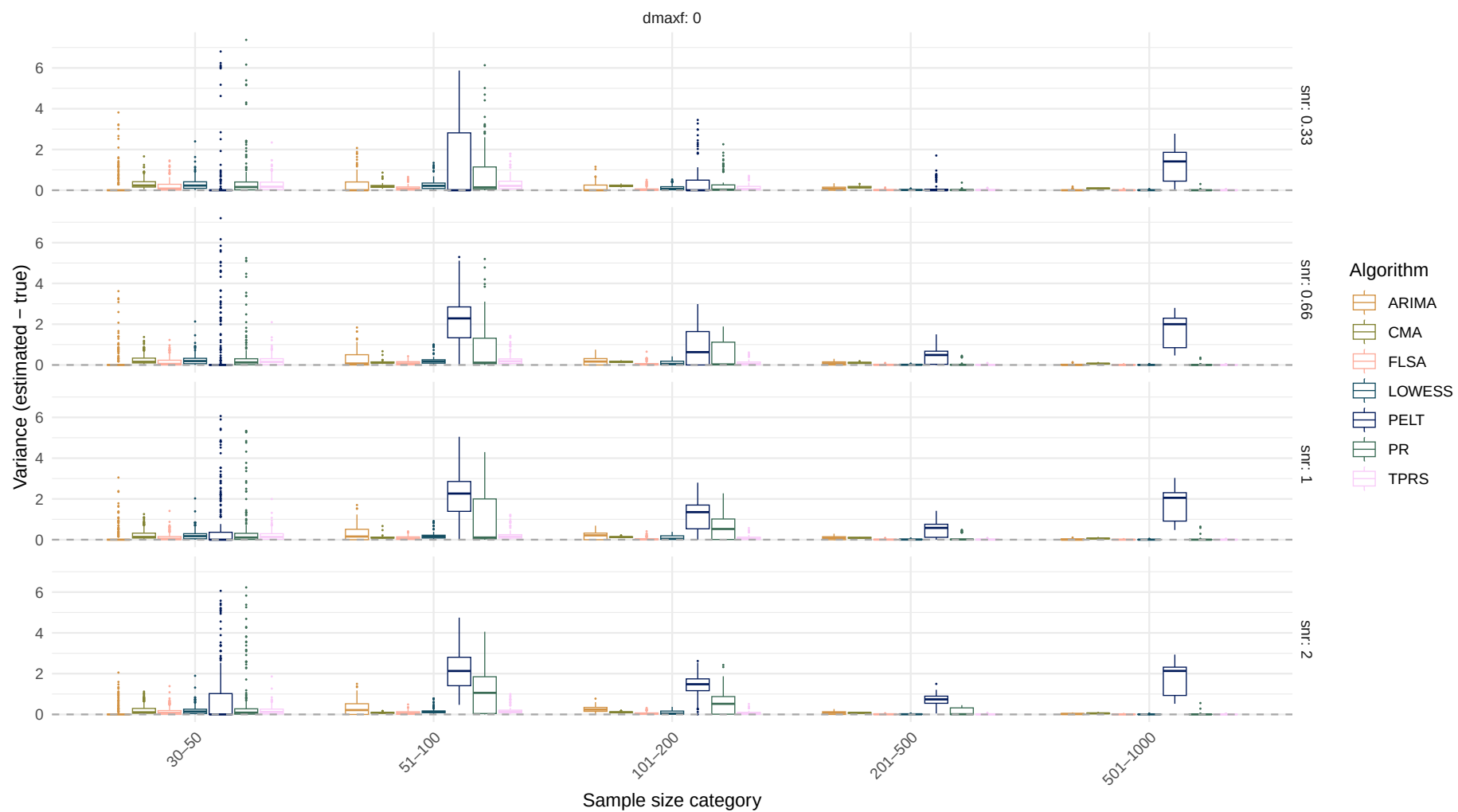


Estimand: Variance

Systematic change pattern: No change

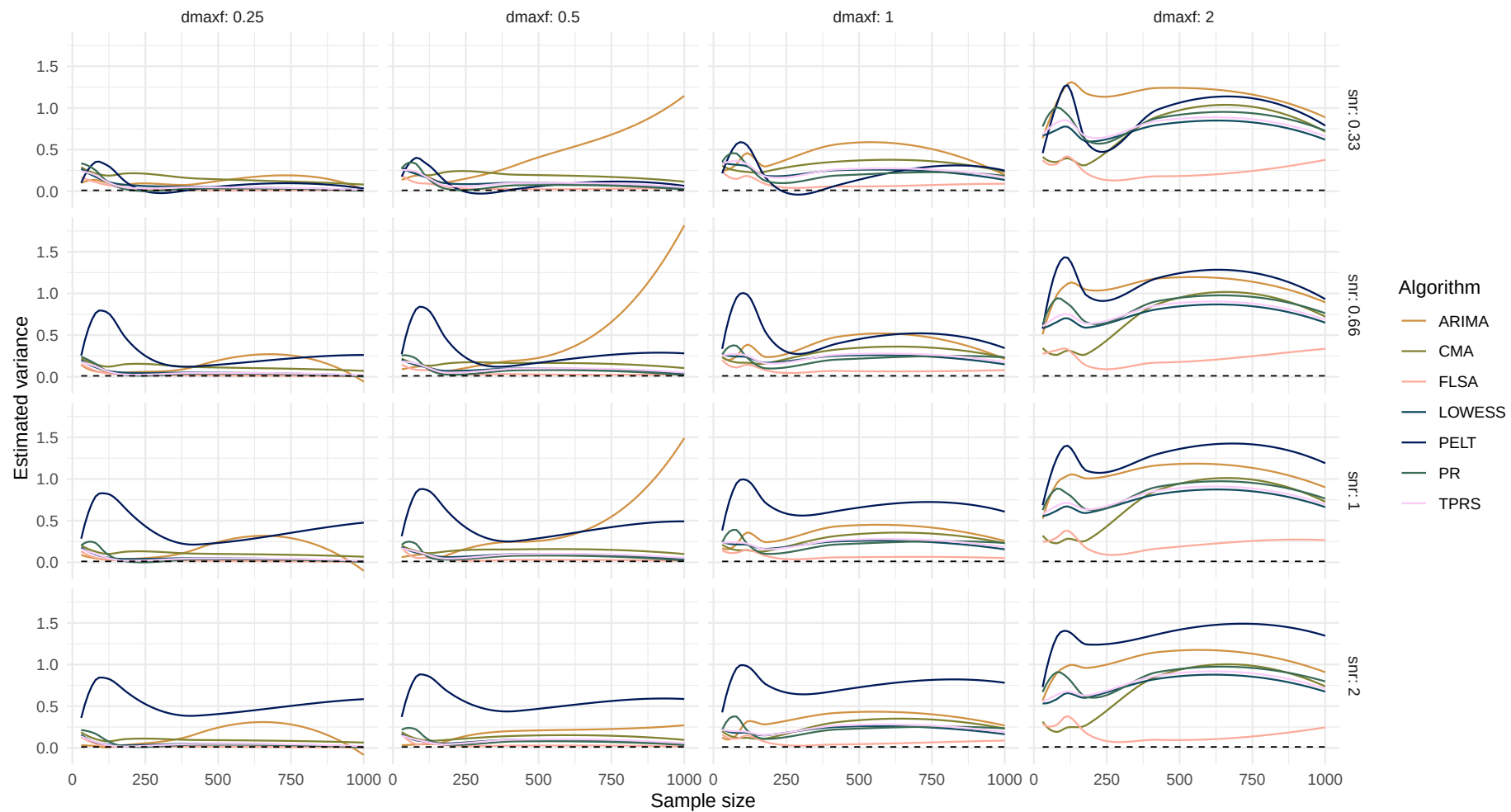


Boxplot of variance difference over sample size

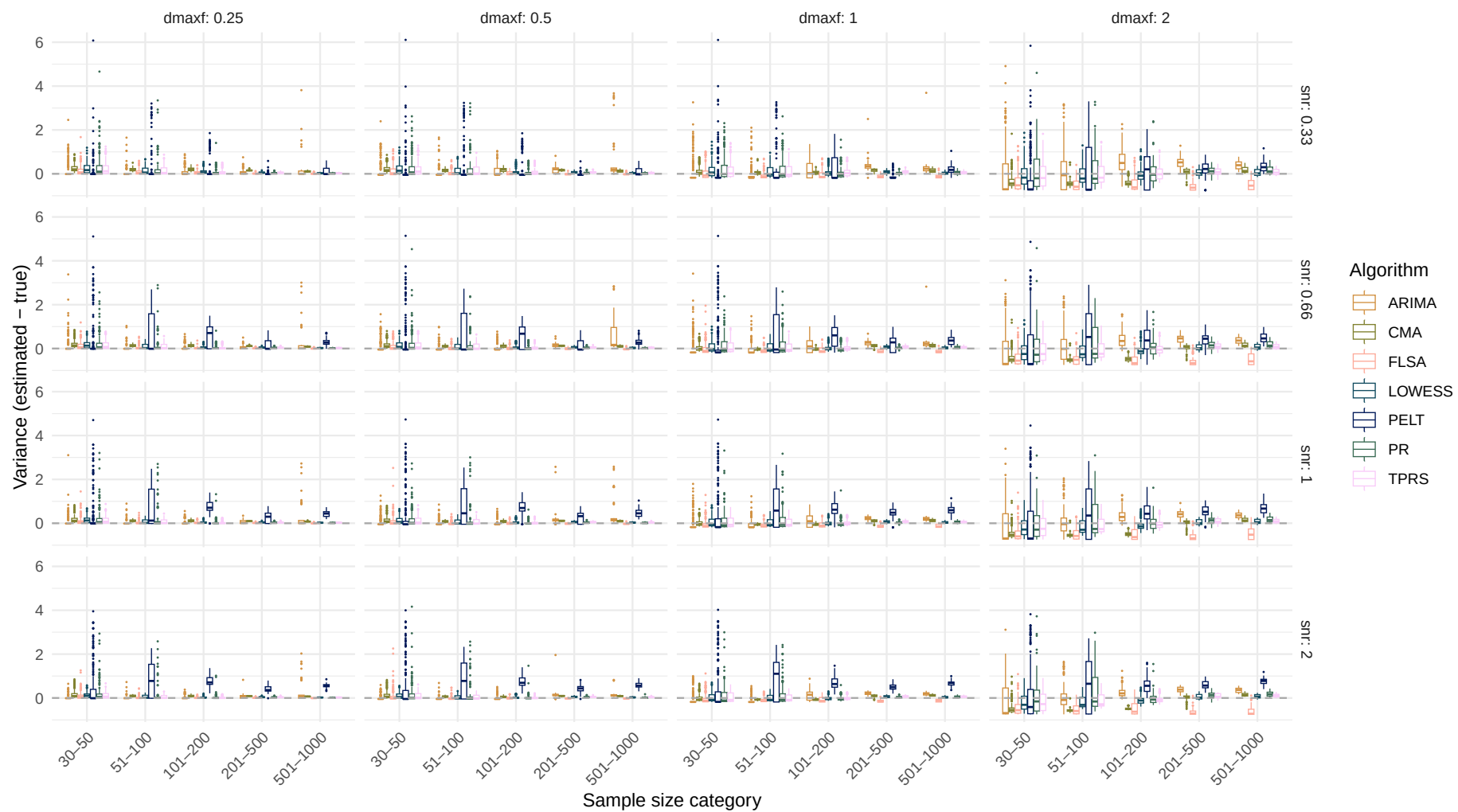


Systematic change pattern: Jump

Estimated variance over sample size
(Dashed line indicates the true variance)

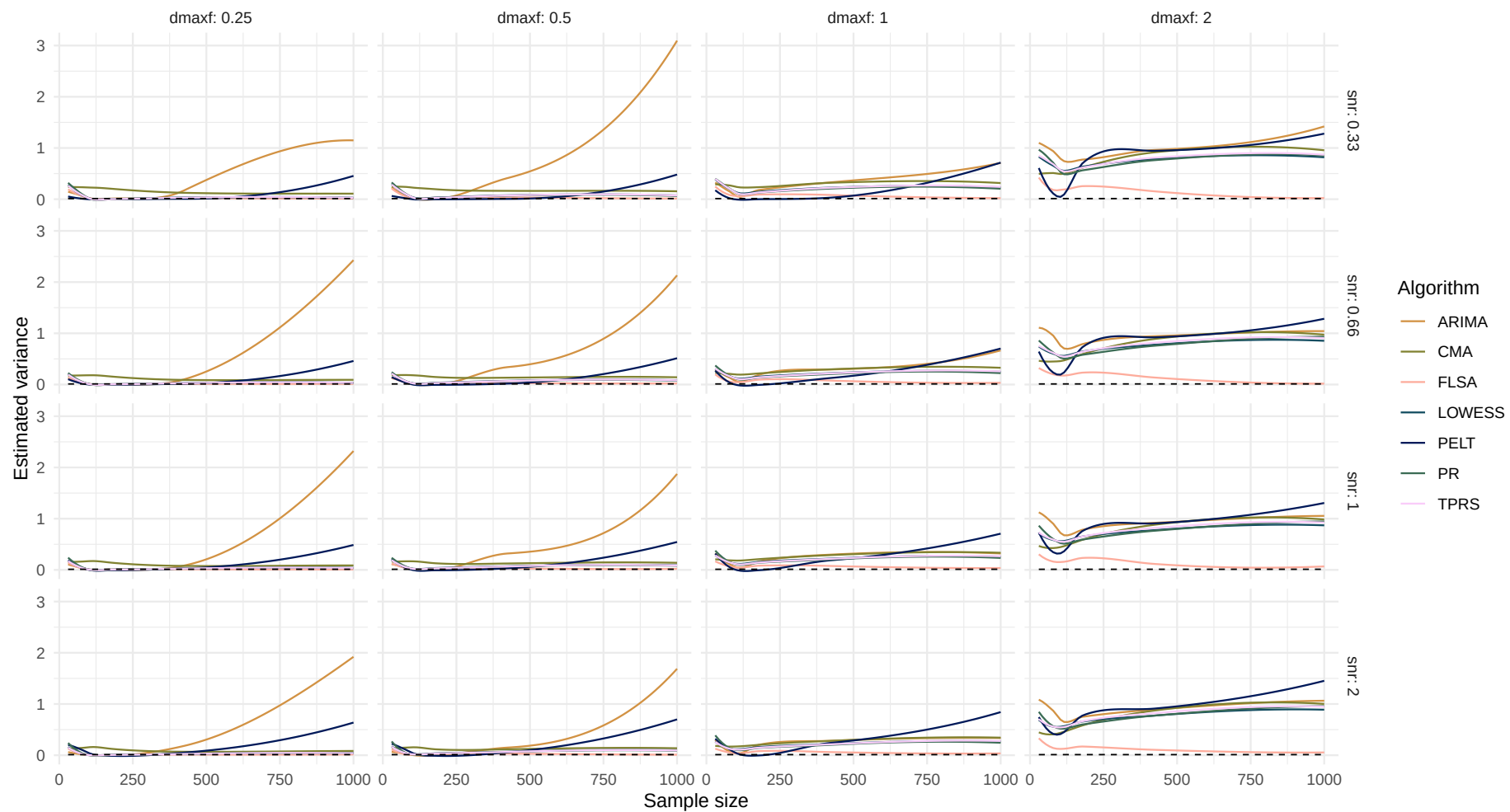


Boxplot of variance difference over sample size

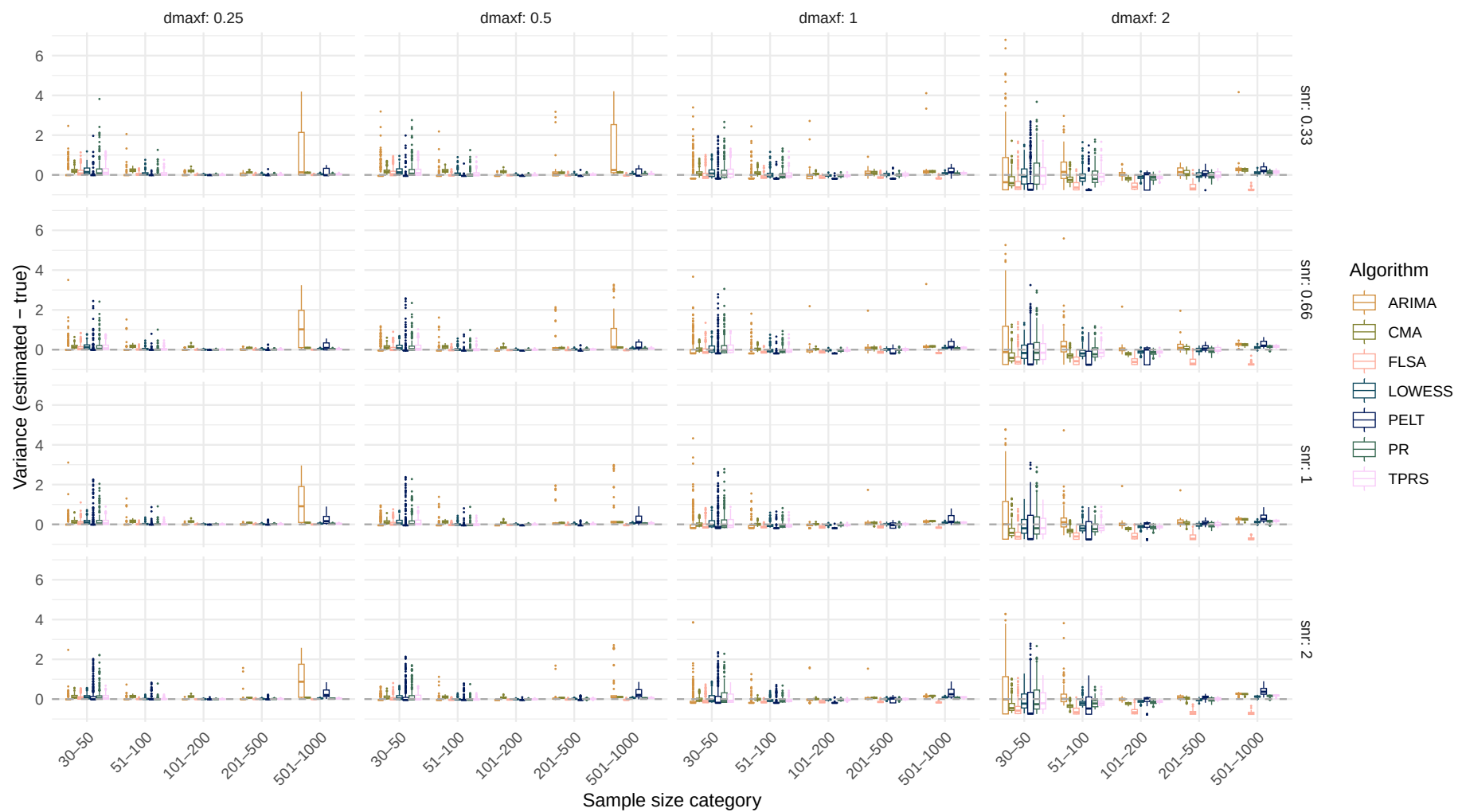


Systematic change pattern: Linear

Estimated variance over sample size
(Dashed line indicates the true variance)

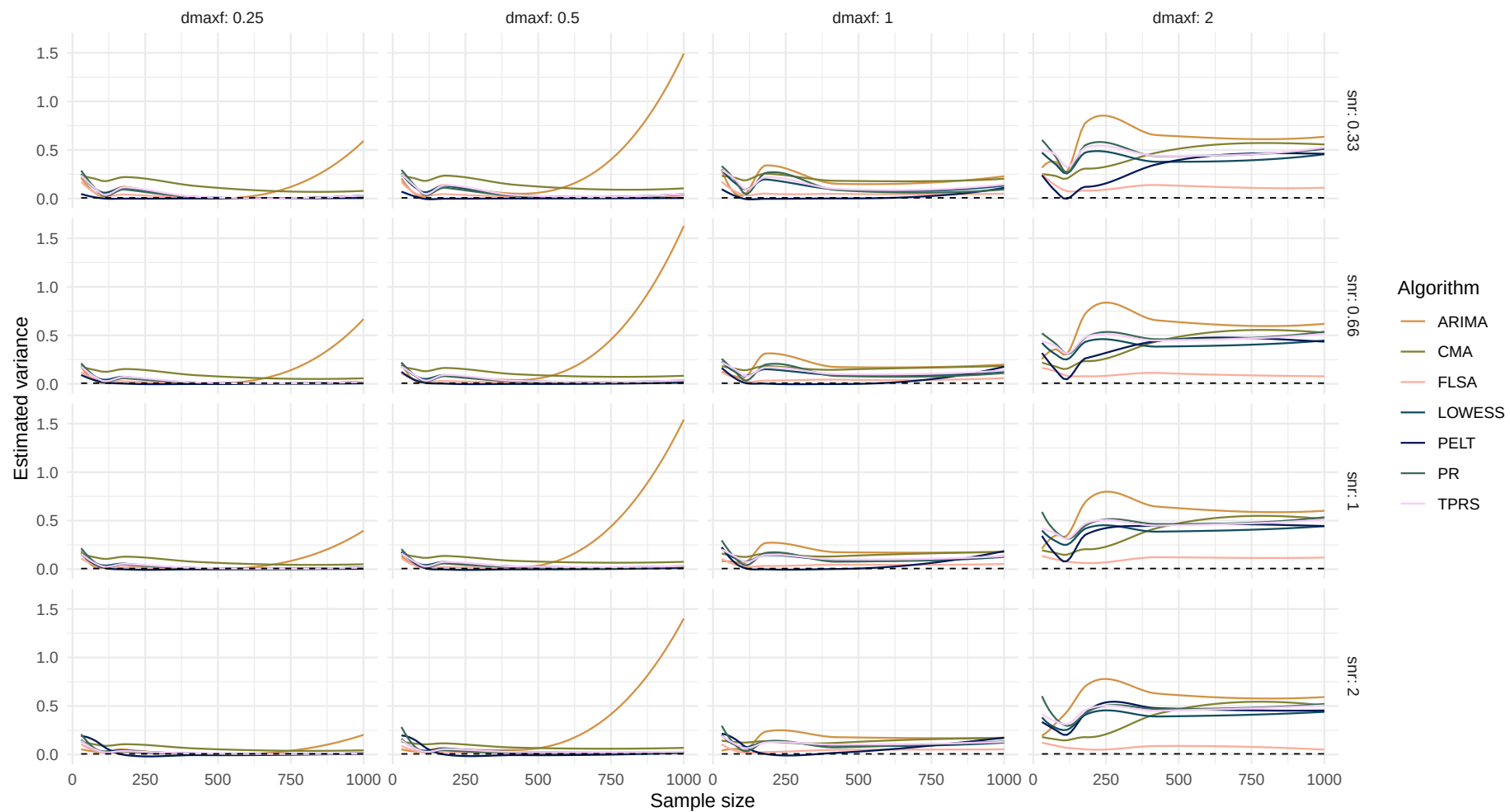


Boxplot of variance difference over sample size

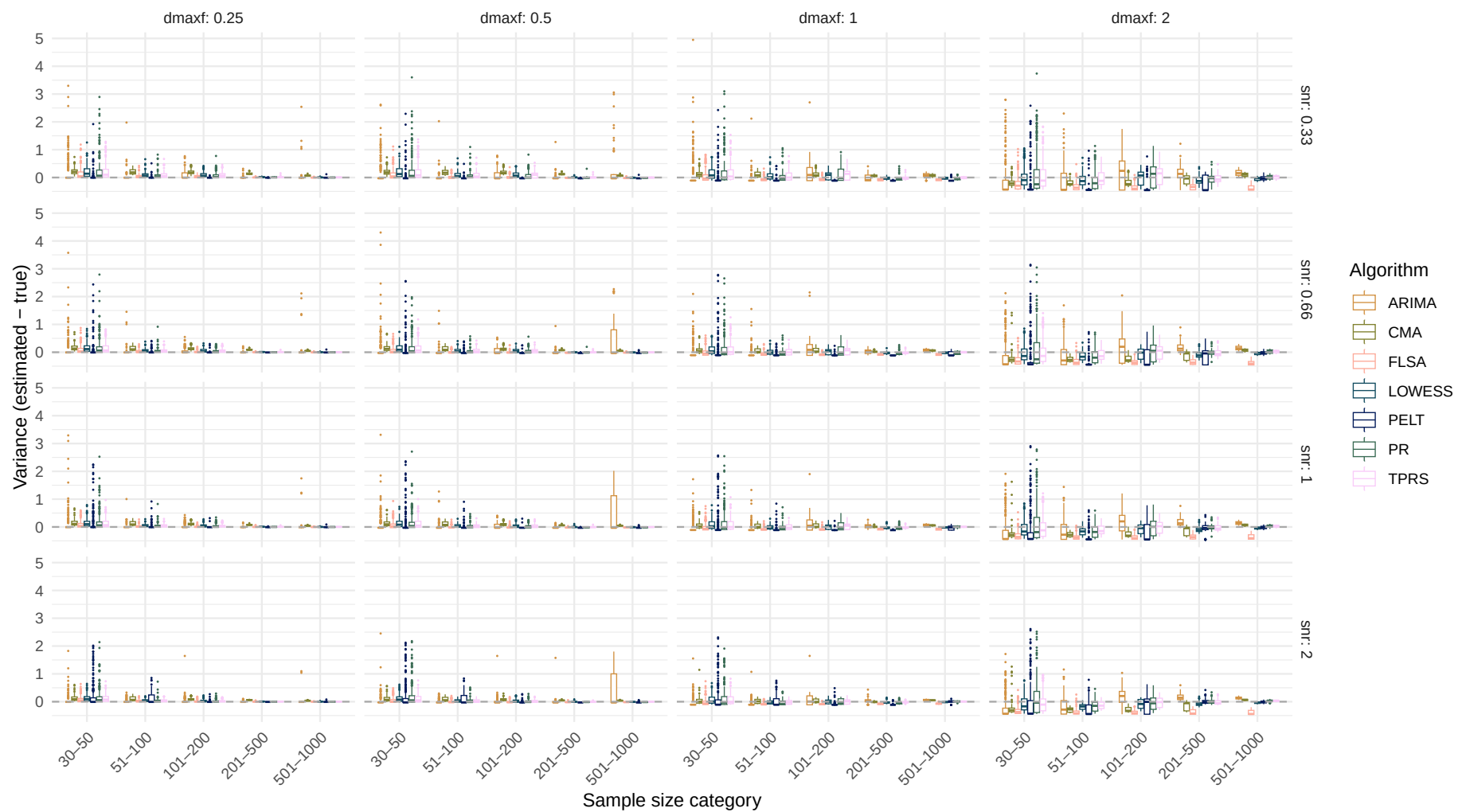


Systematic change pattern: Quadratic

Estimated variance over sample size
(Dashed line indicates the true variance)

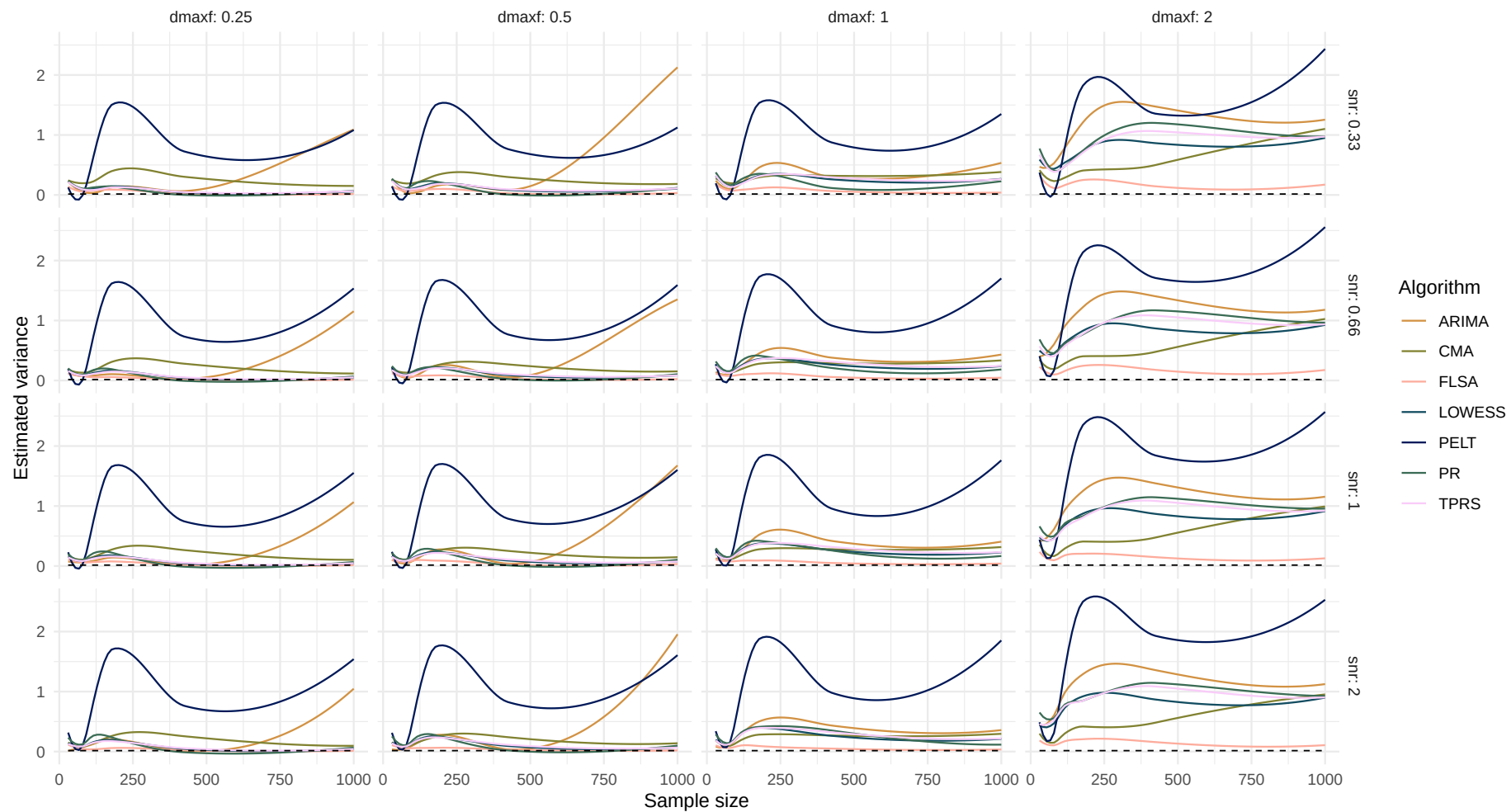


Boxplot of variance difference over sample size

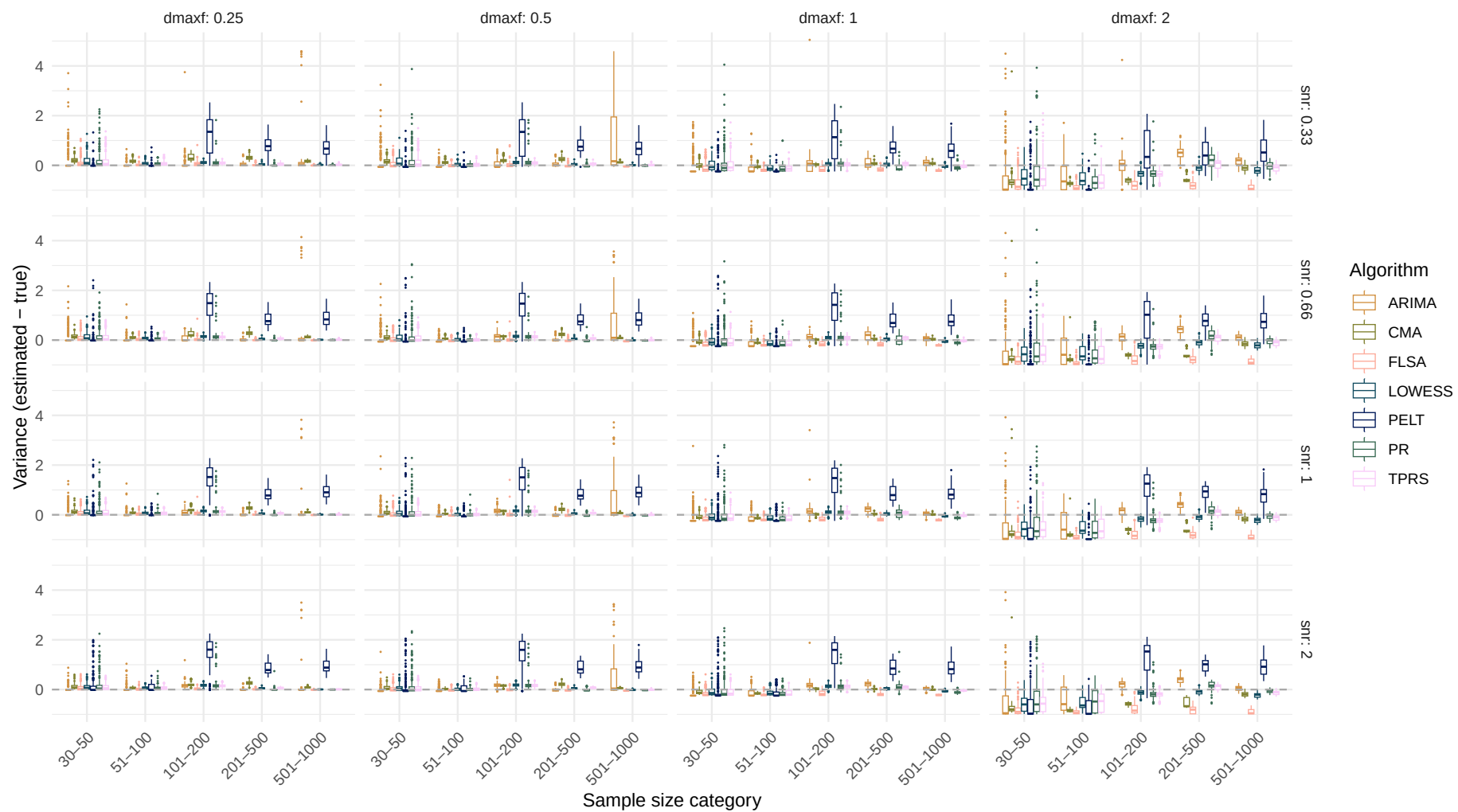


Systematic change pattern: Recurrent

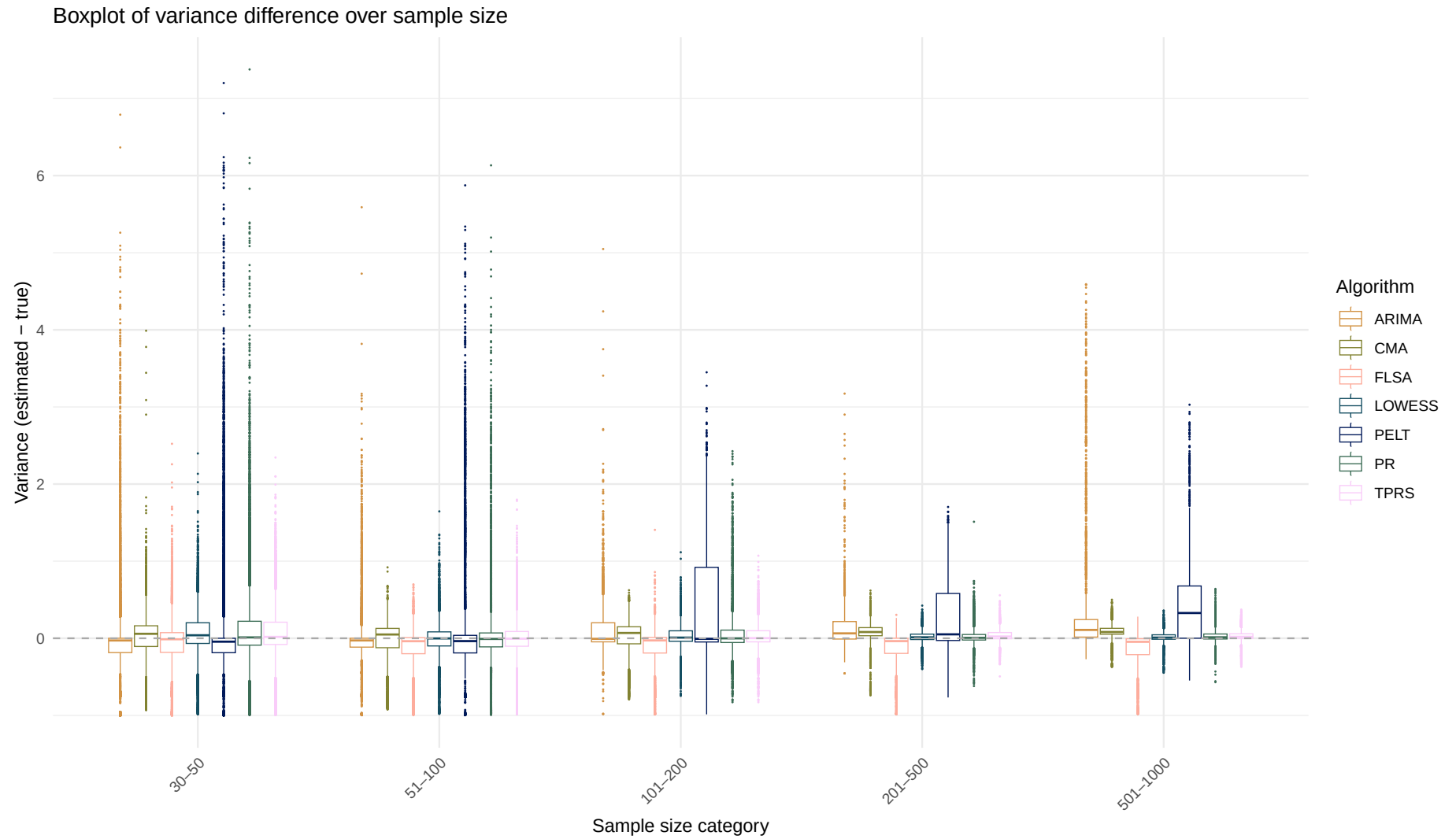
Estimated variance over sample size
(Dashed line indicates the true variance)



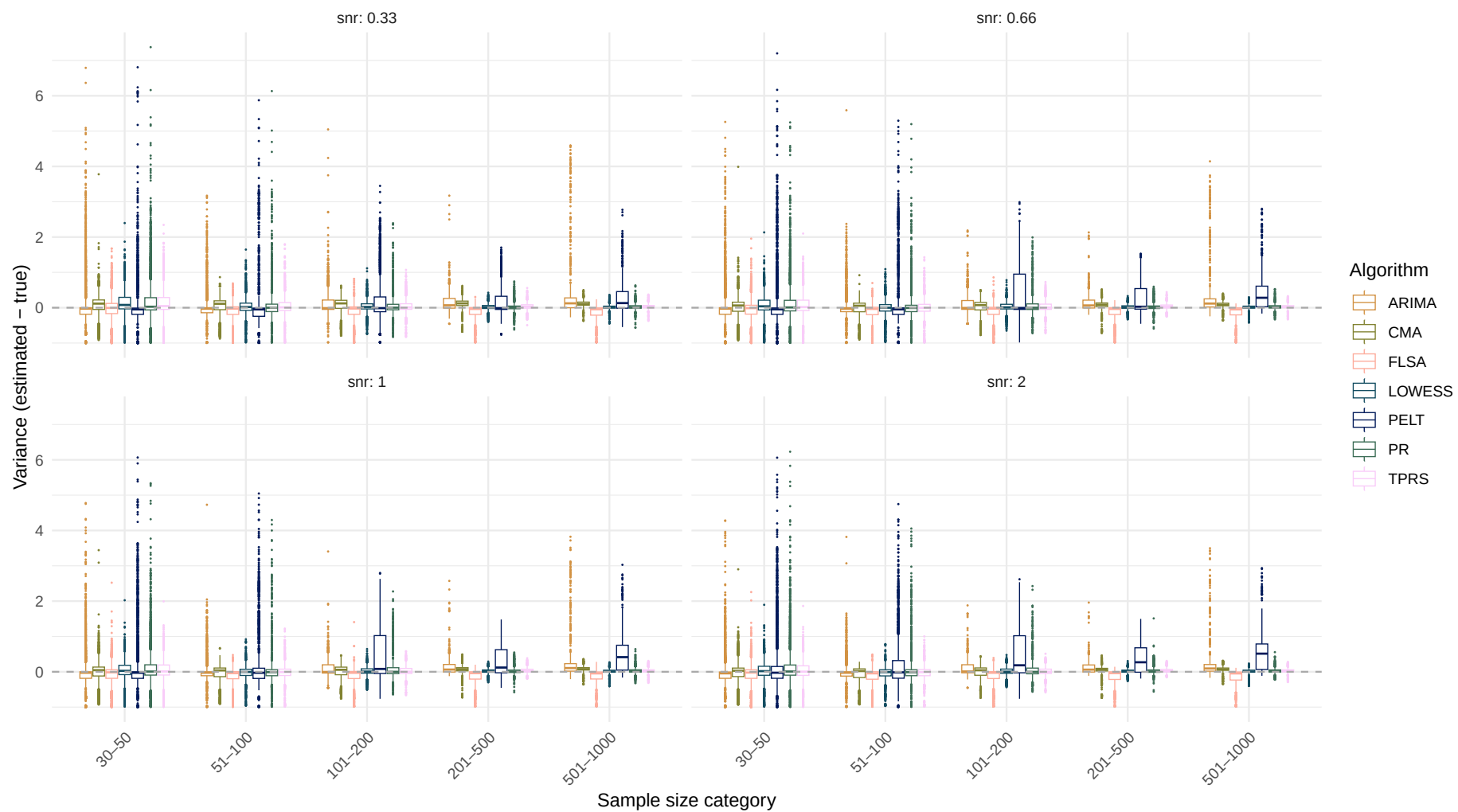
Boxplot of variance difference over sample size



Systematic change pattern: All patterns combined



Boxplot of variance difference over sample size

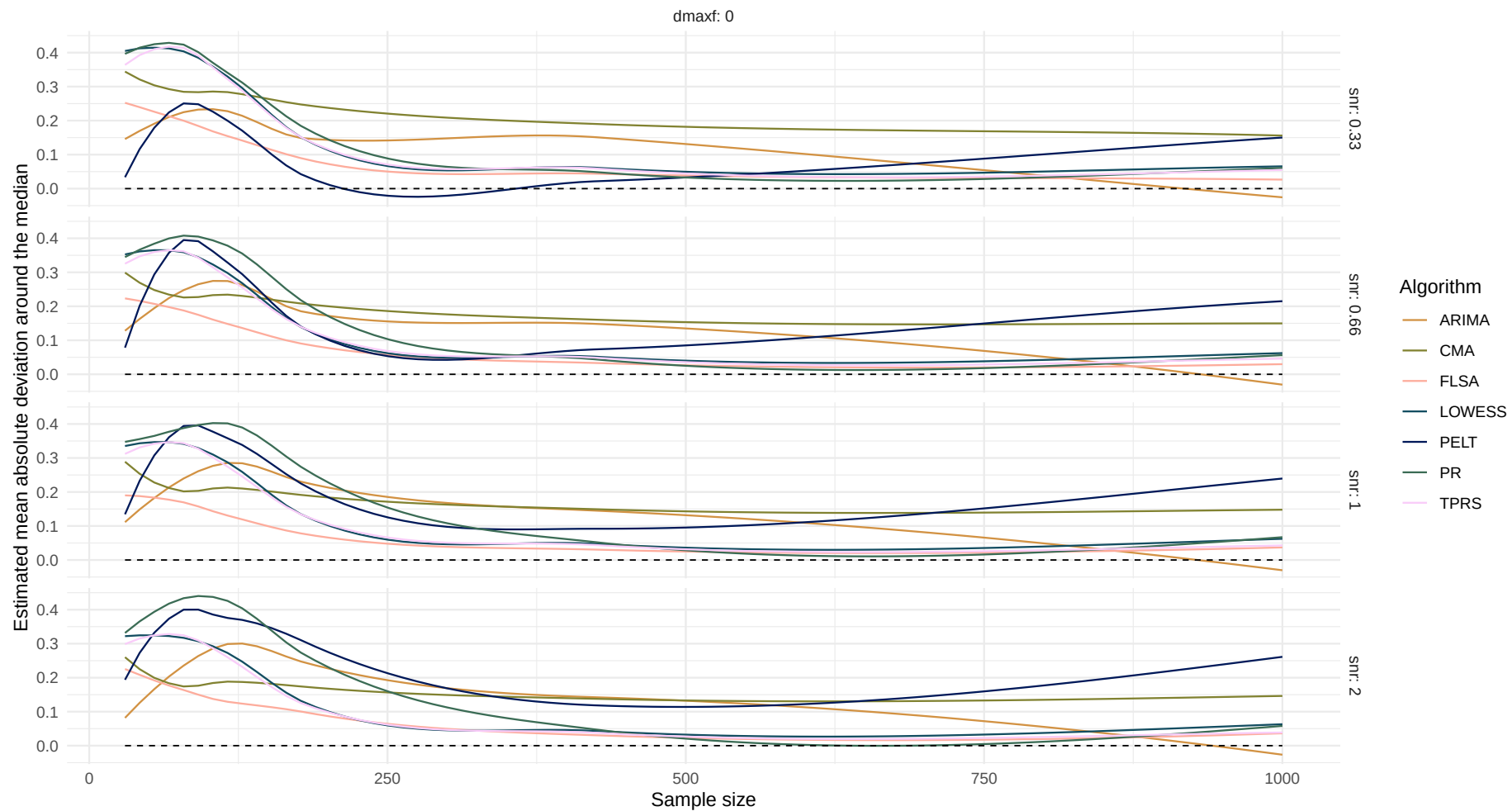


Estimand: Mean absolute deviation around the median

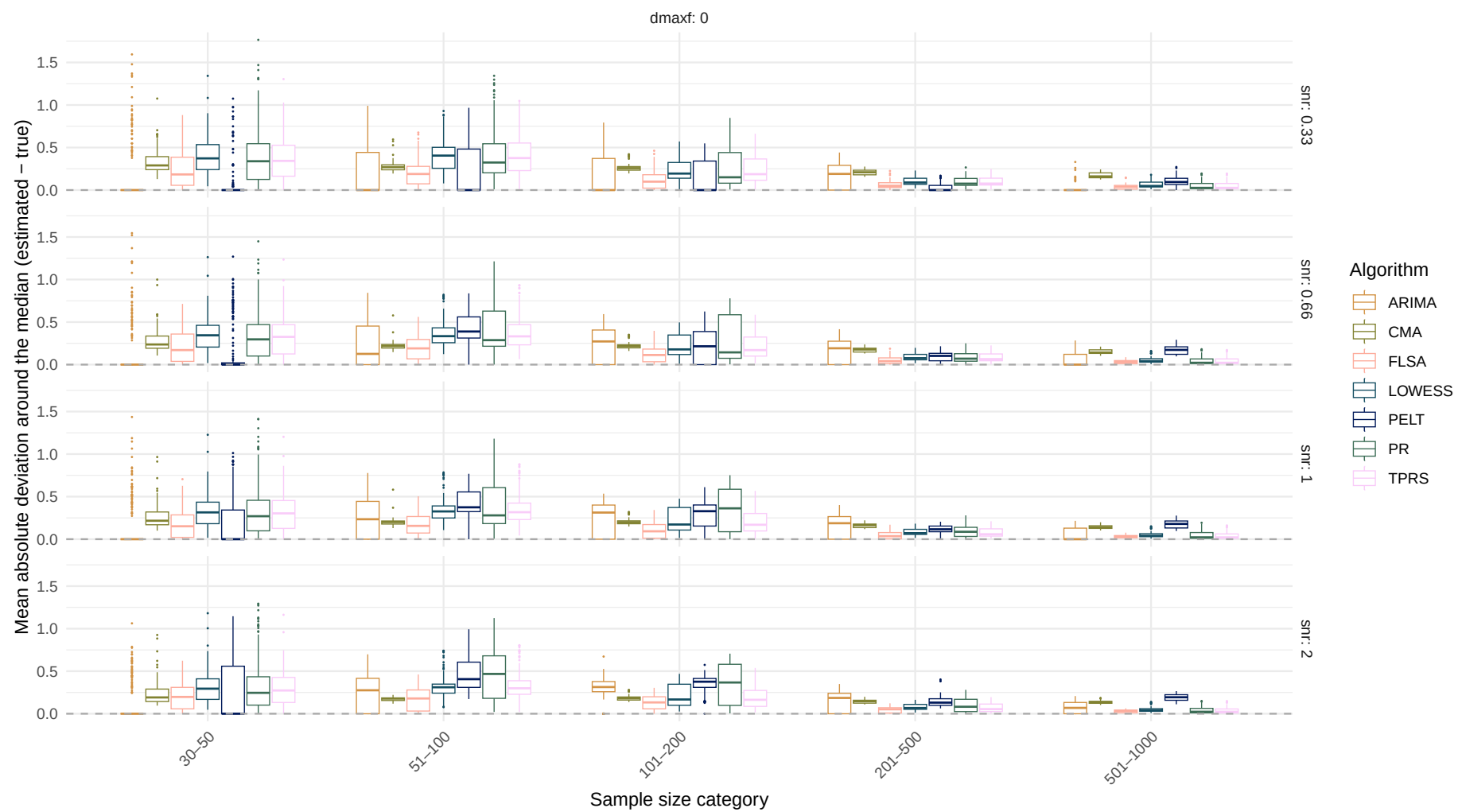
Systematic change pattern: No change

Estimated mean absolute deviation around the median over sample size

(Dashed line indicates the true mean absolute deviation around the median)



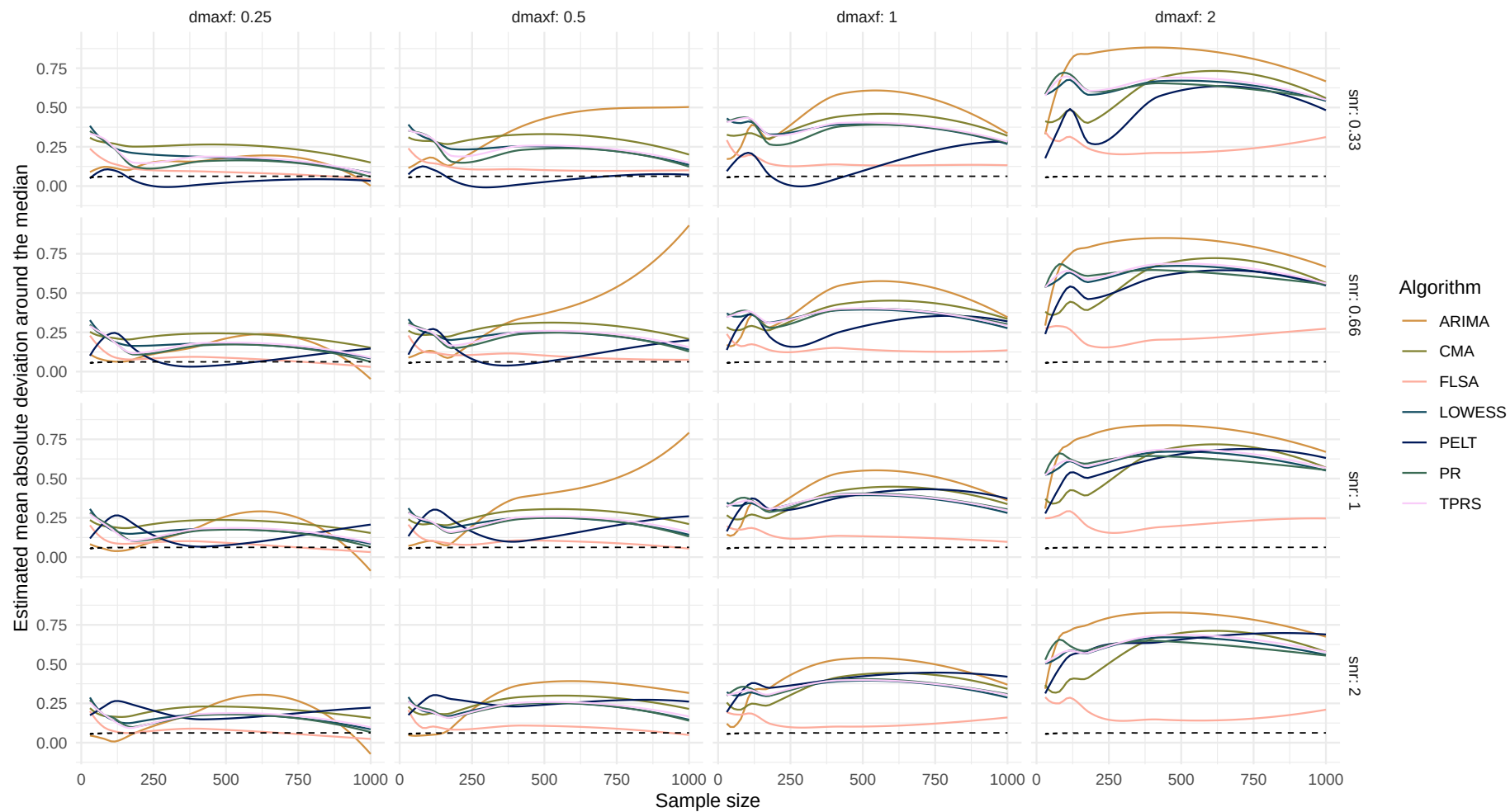
Boxplot of mean absolute deviation around the median difference over sample size



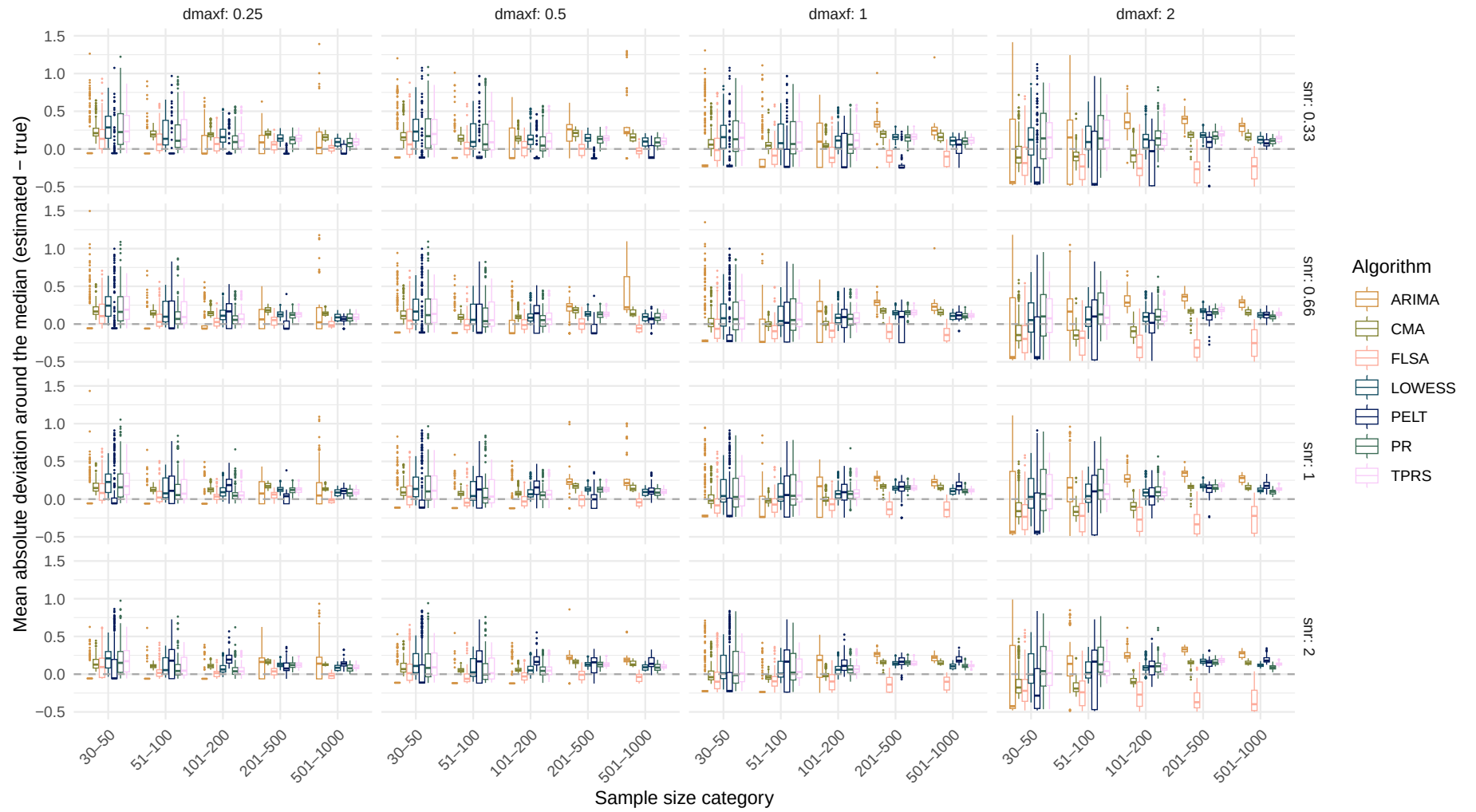
Systematic change pattern: Jump

Estimated mean absolute deviation around the median over sample size

(Dashed line indicates the true mean absolute deviation around the median)

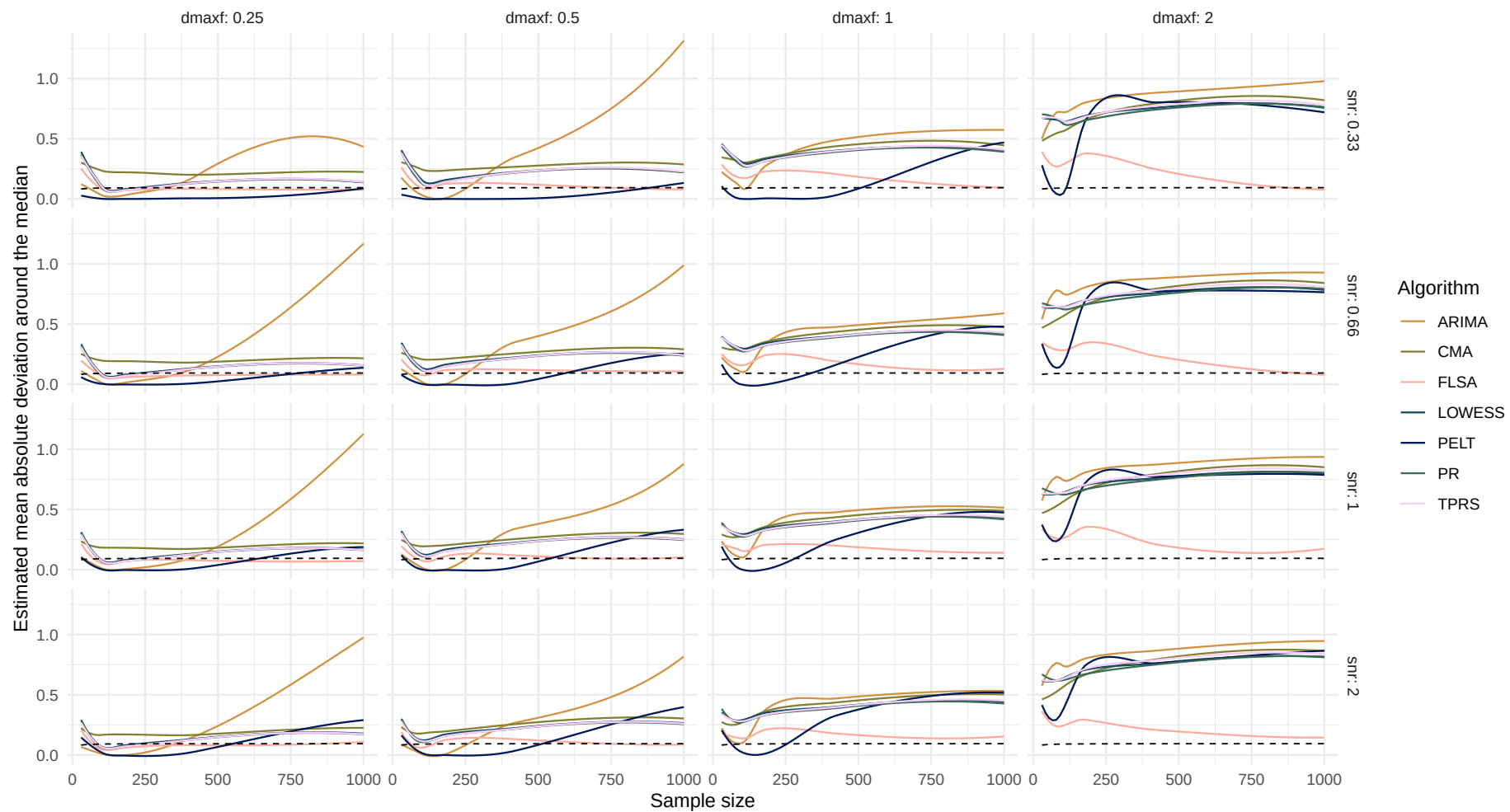


Boxplot of mean absolute deviation around the median difference over sample size

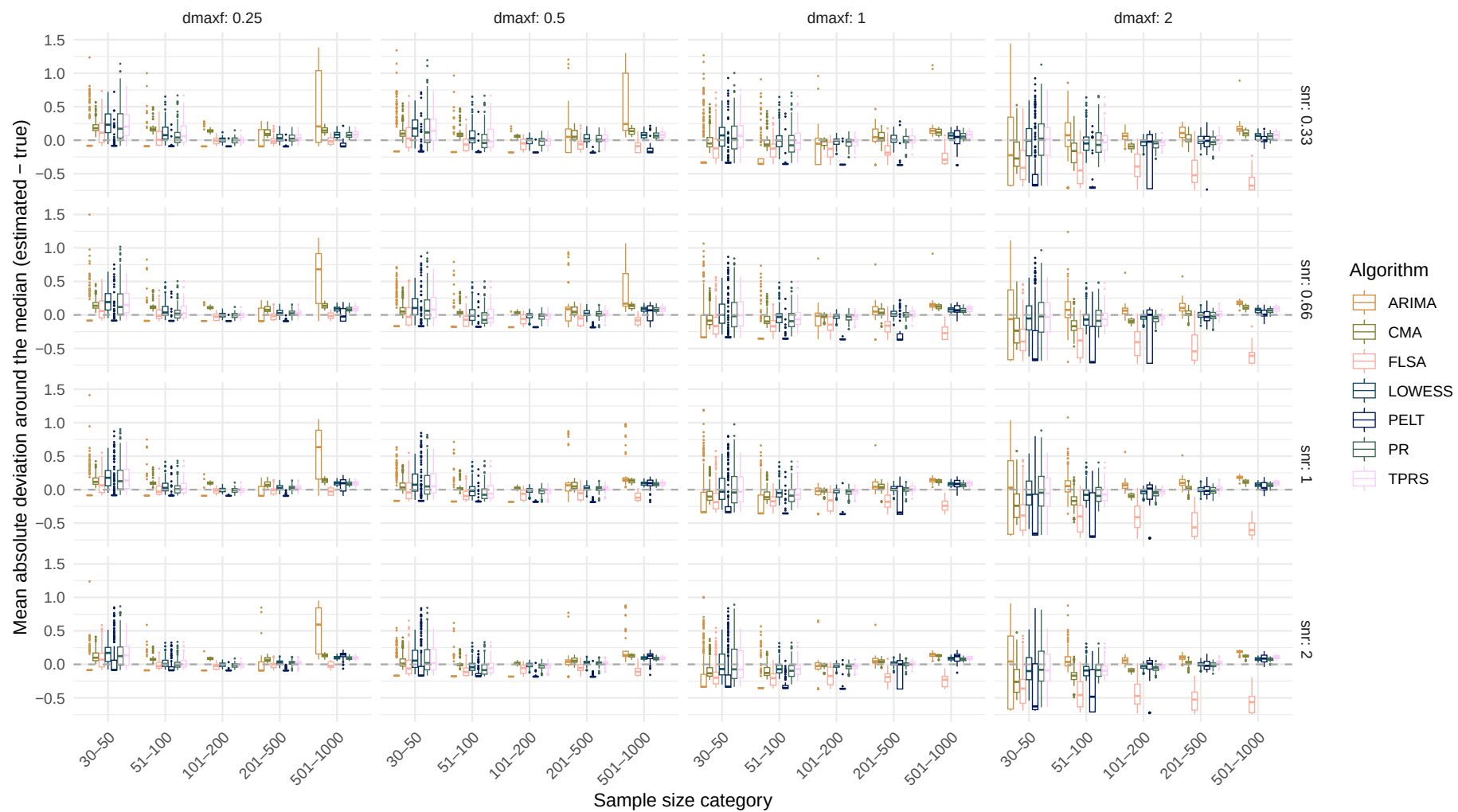


Systematic change pattern: Linear

Estimated mean absolute deviation around the median over sample size
(Dashed line indicates the true mean absolute deviation around the median)



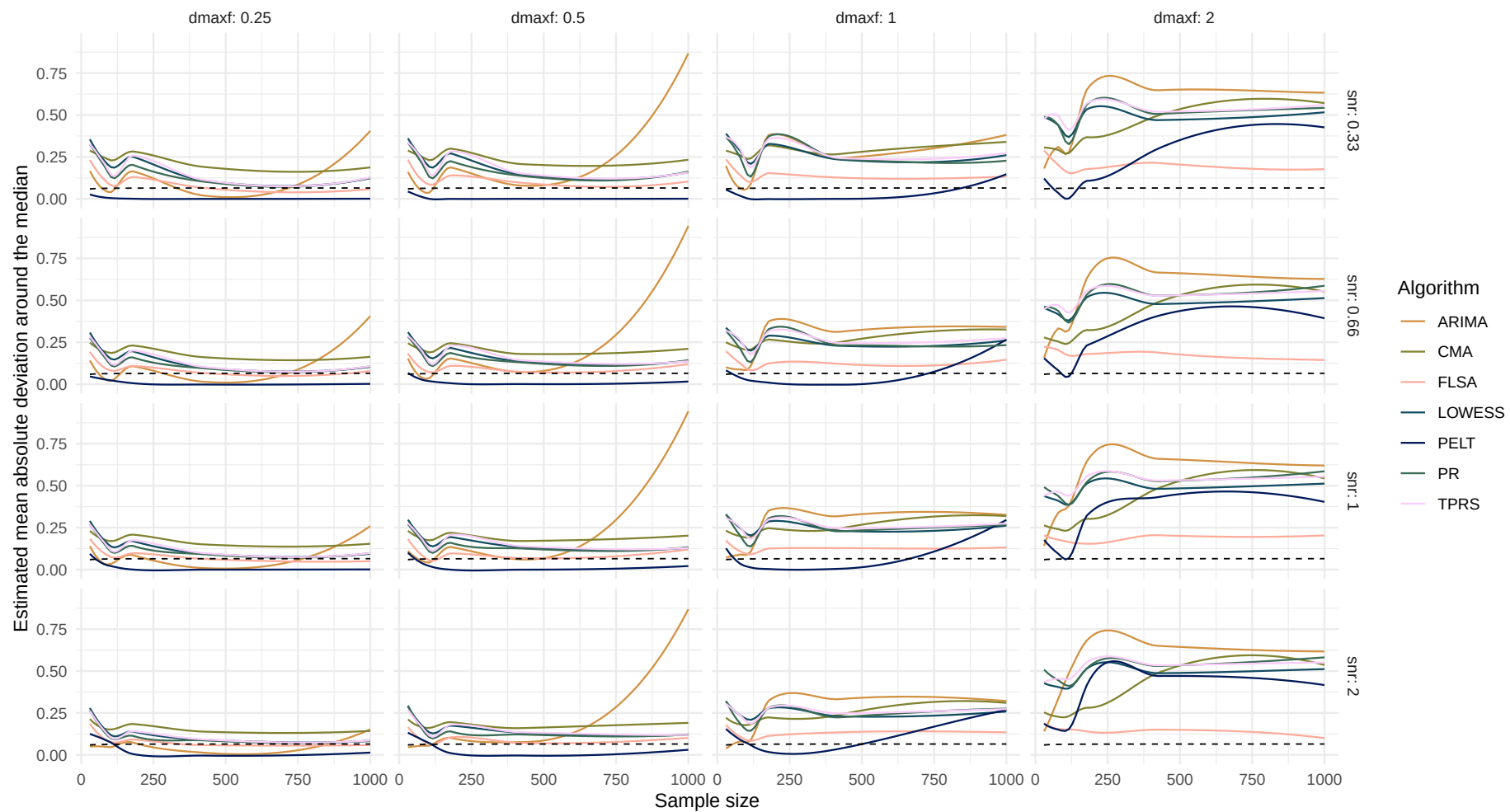
Boxplot of mean absolute deviation around the median difference over sample size



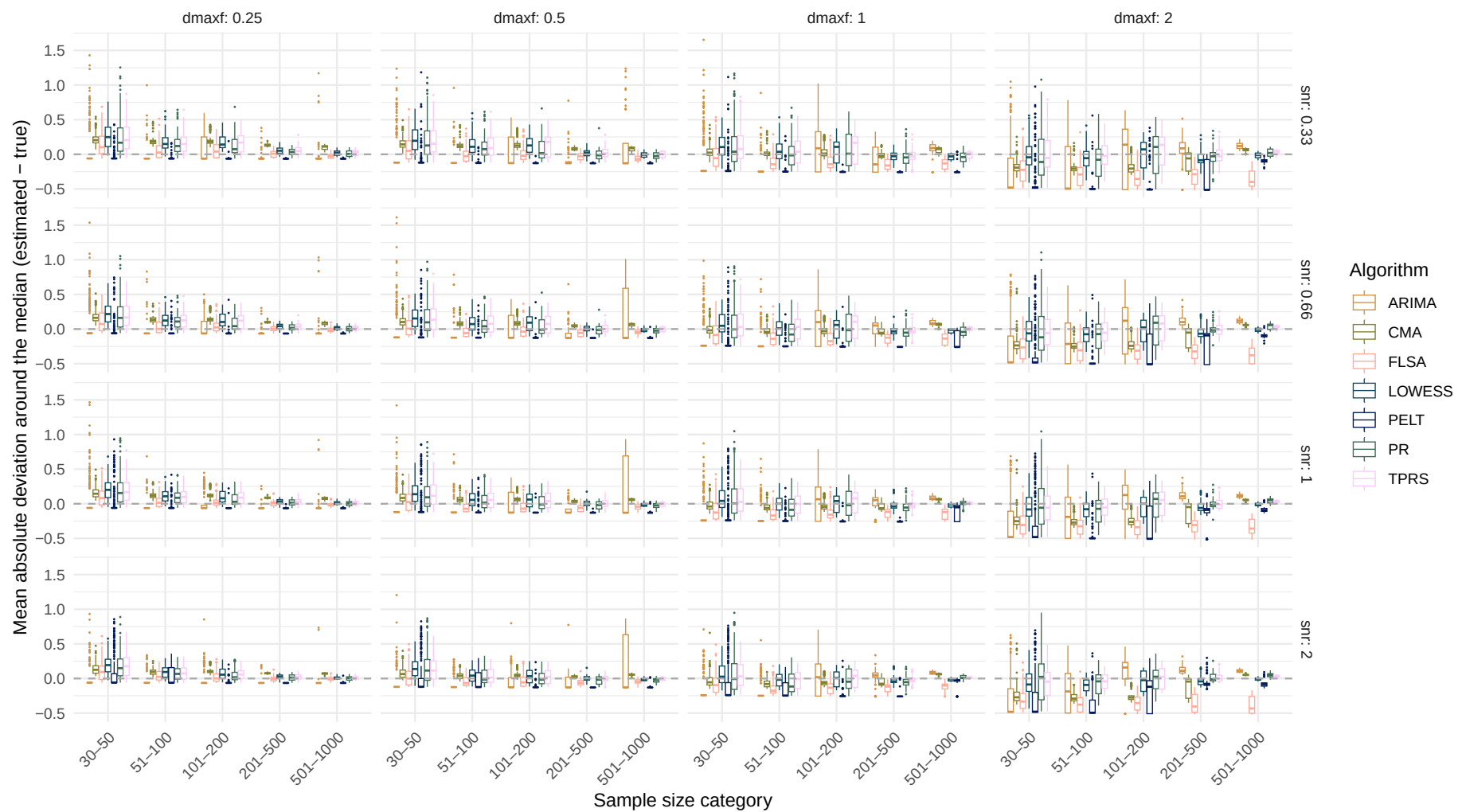
Systematic change pattern: Quadratic

Estimated mean absolute deviation around the median over sample size

(Dashed line indicates the true mean absolute deviation around the median)

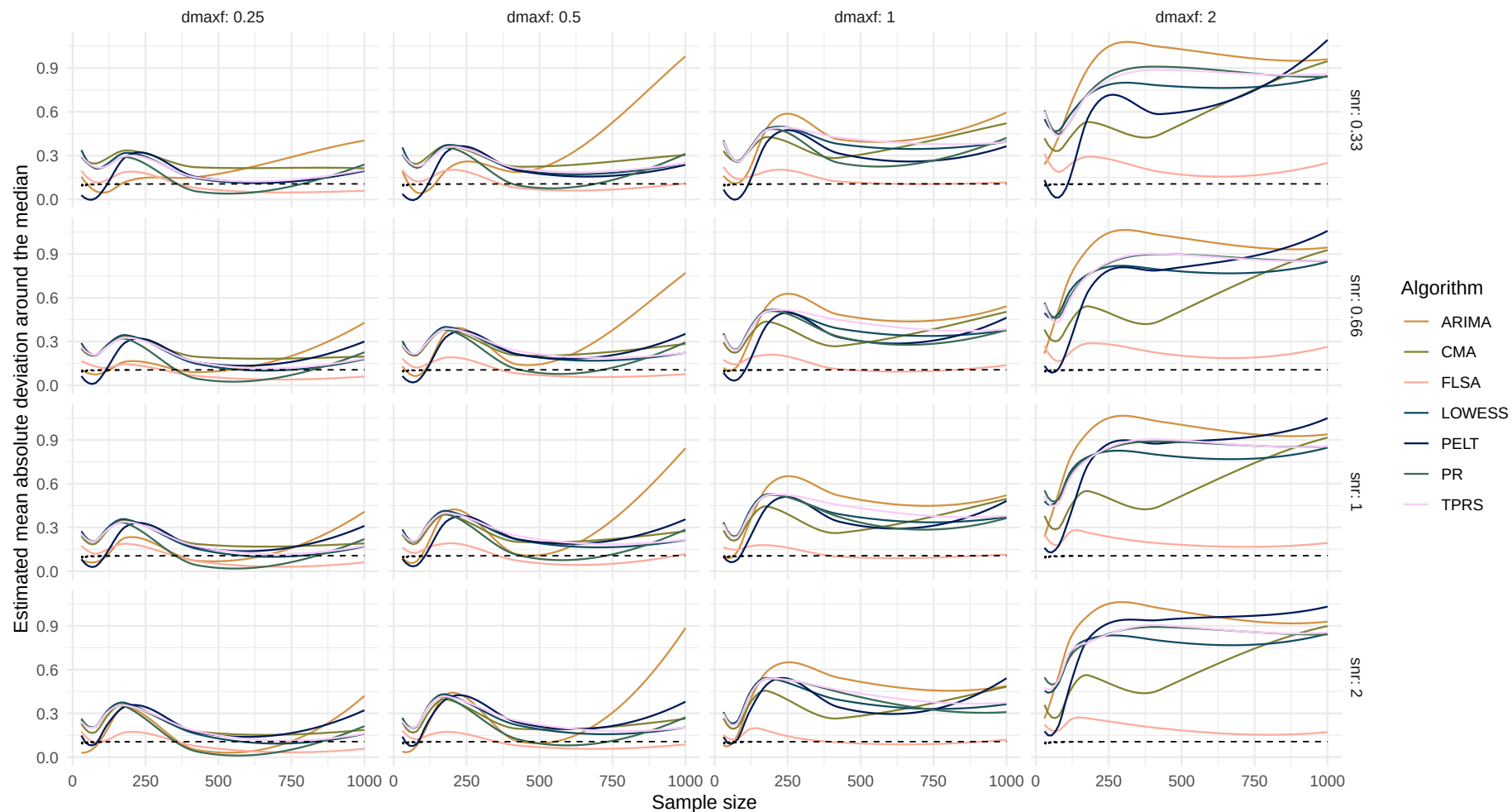


Boxplot of mean absolute deviation around the median difference over sample size

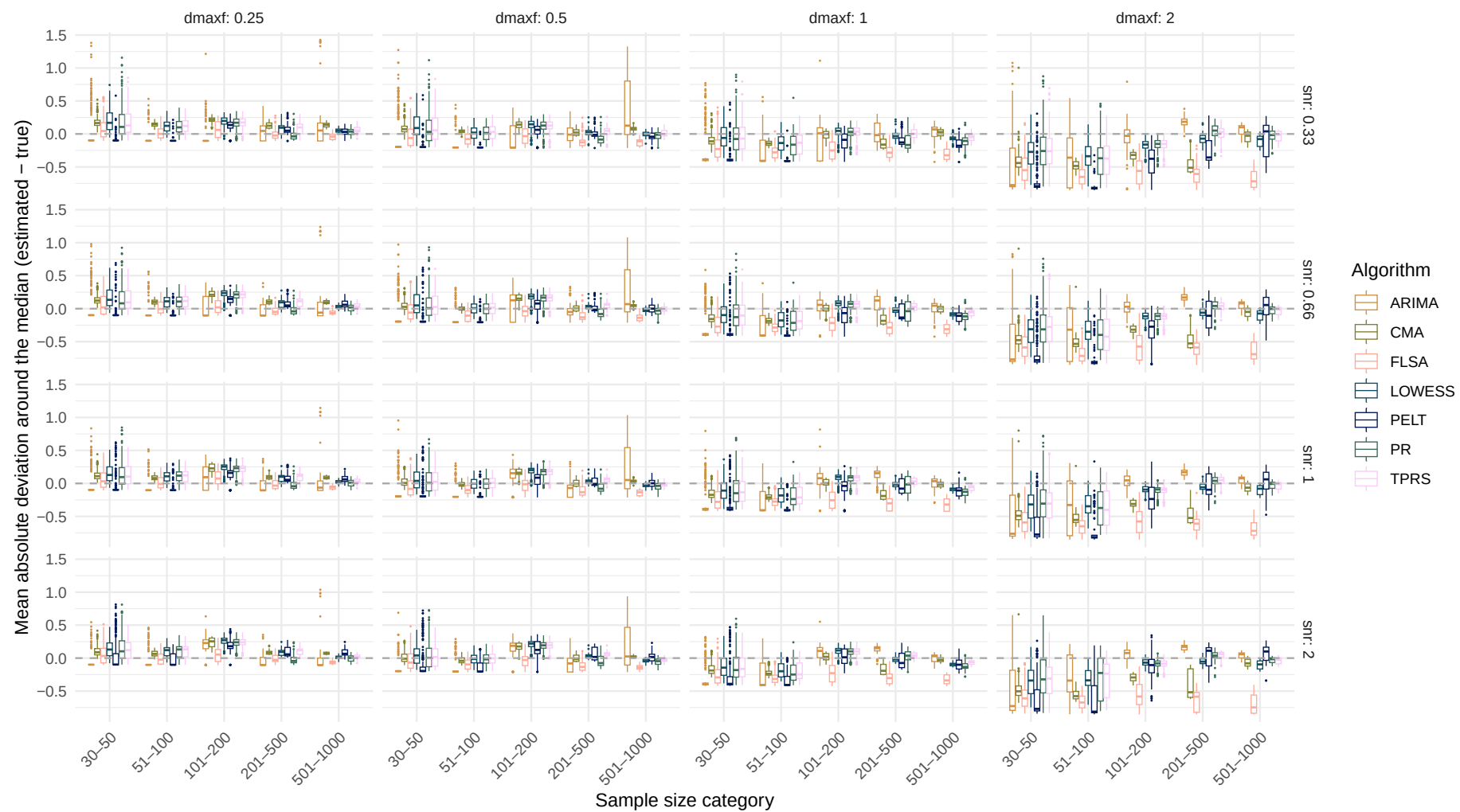


Systematic change pattern: Recurrent

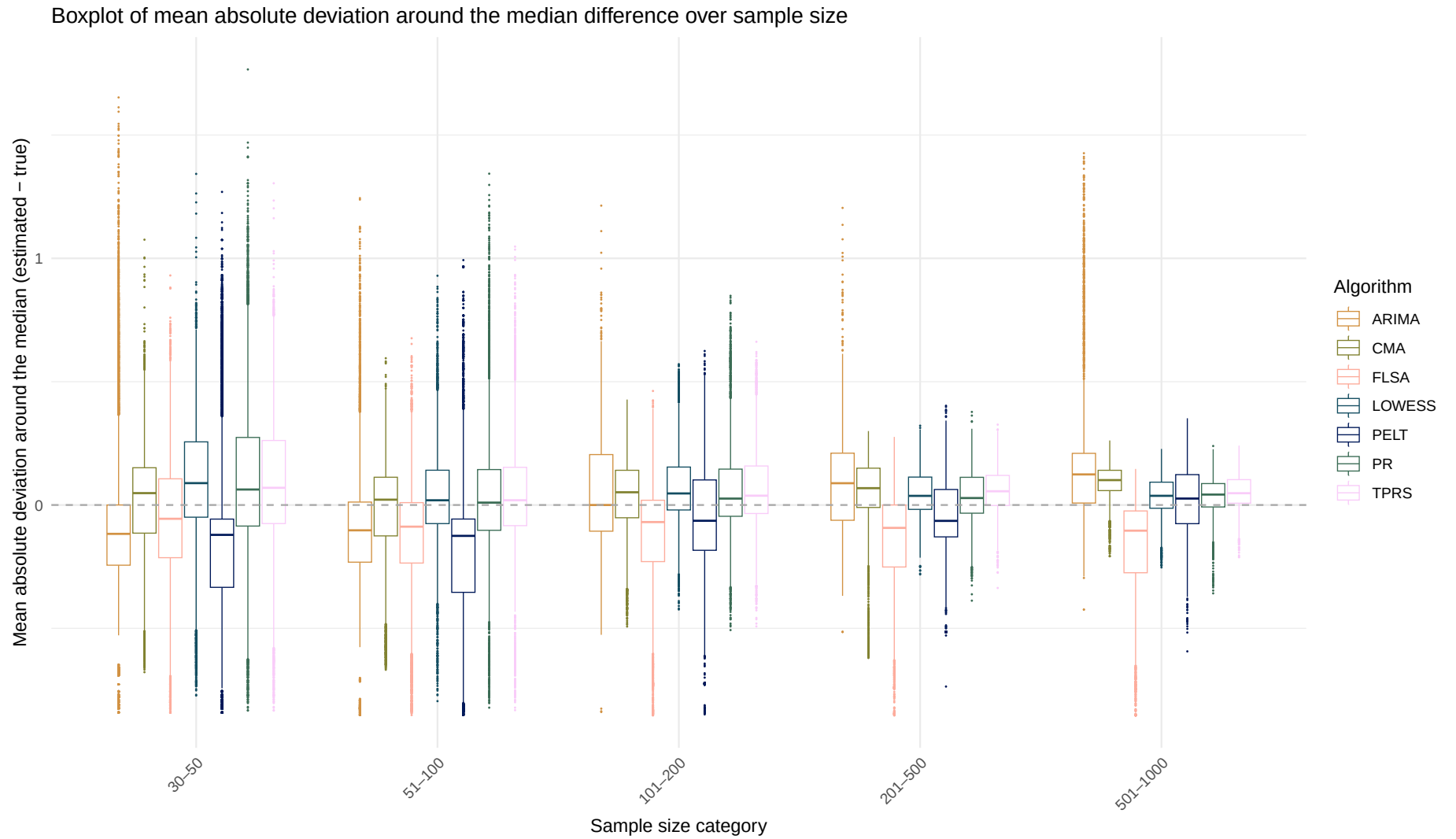
Estimated mean absolute deviation around the median over sample size
(Dashed line indicates the true mean absolute deviation around the median)



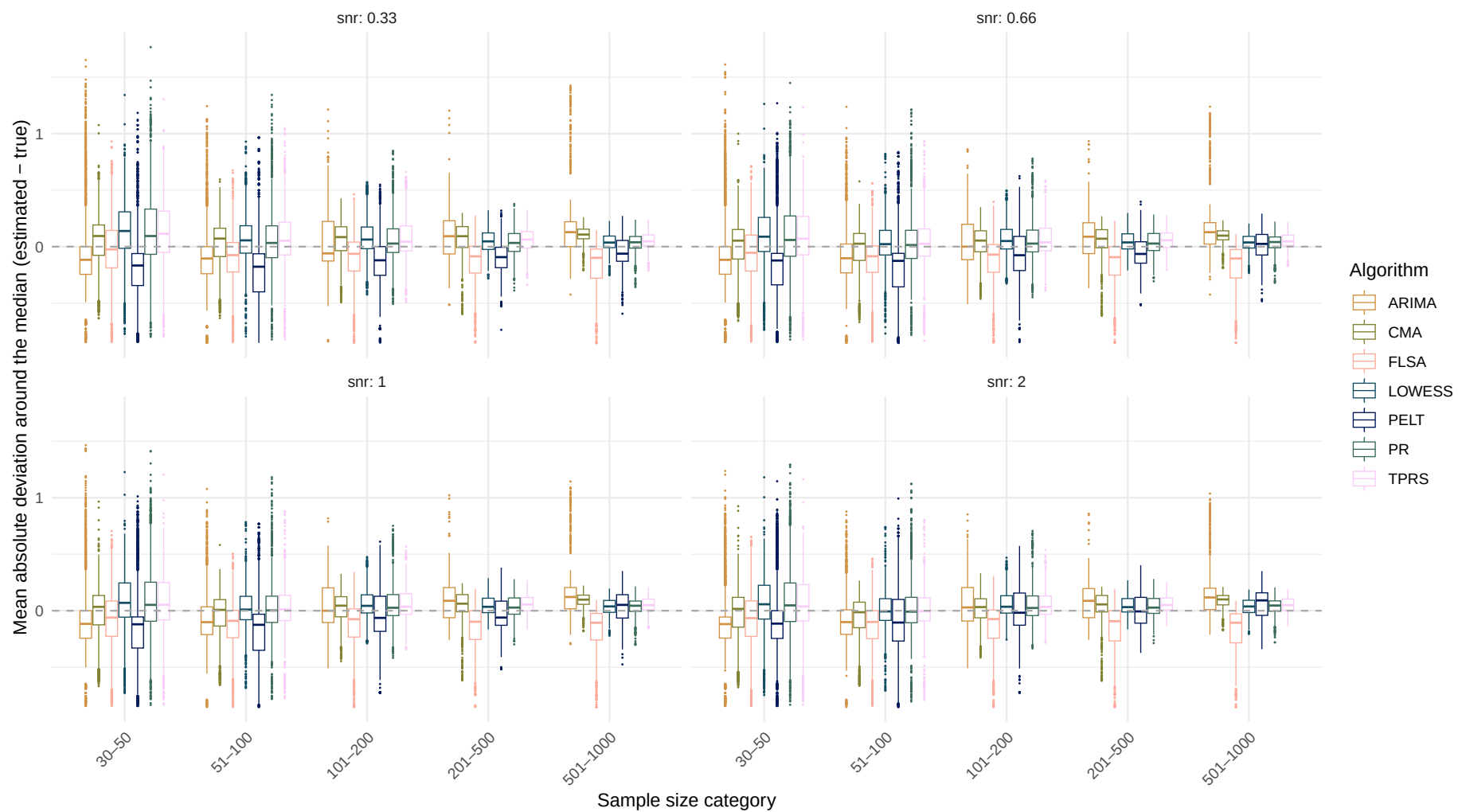
Boxplot of mean absolute deviation around the median difference over sample size



Systematic change pattern: All patterns combined



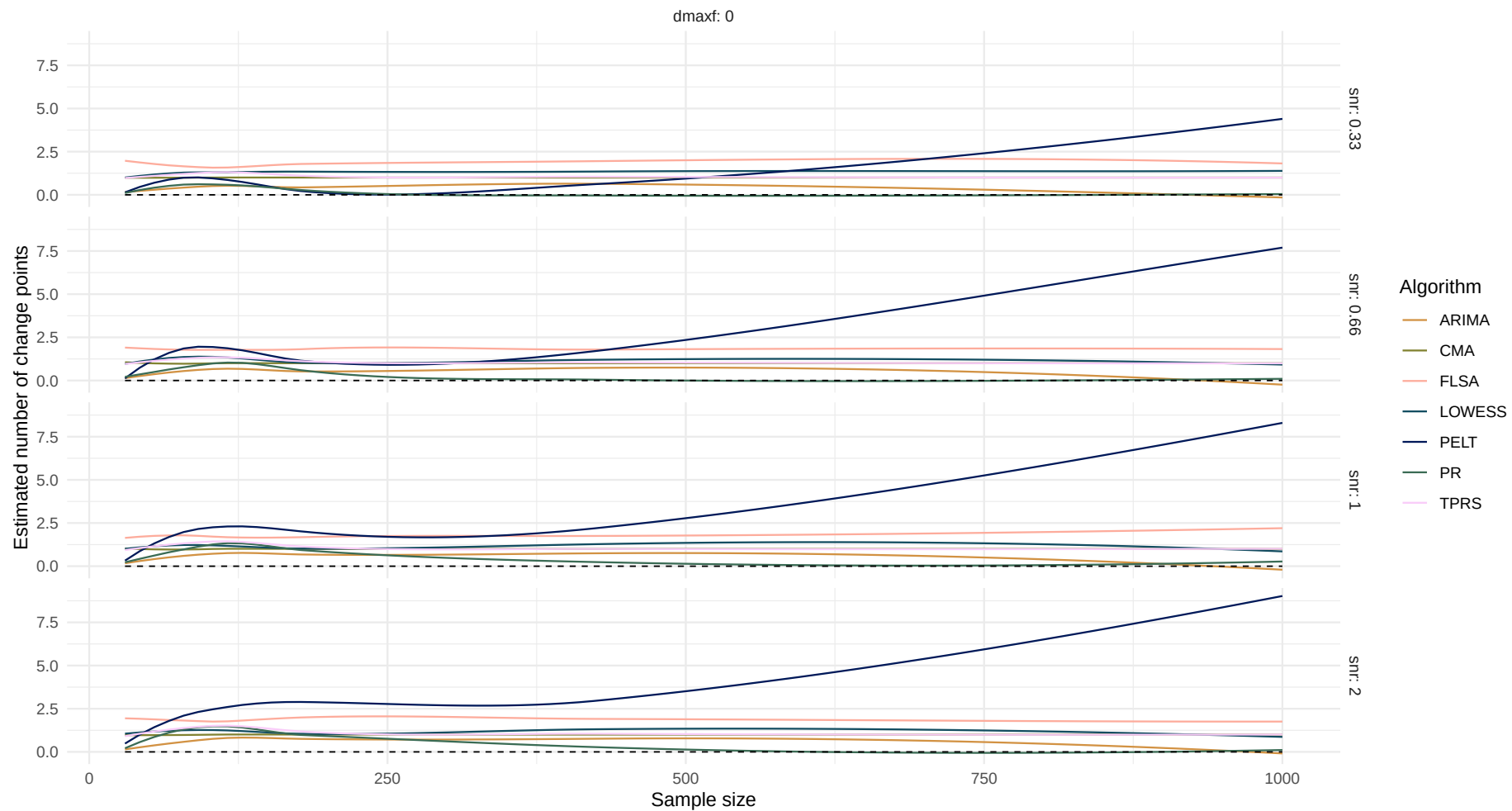
Boxplot of mean absolute deviation around the median difference over sample size



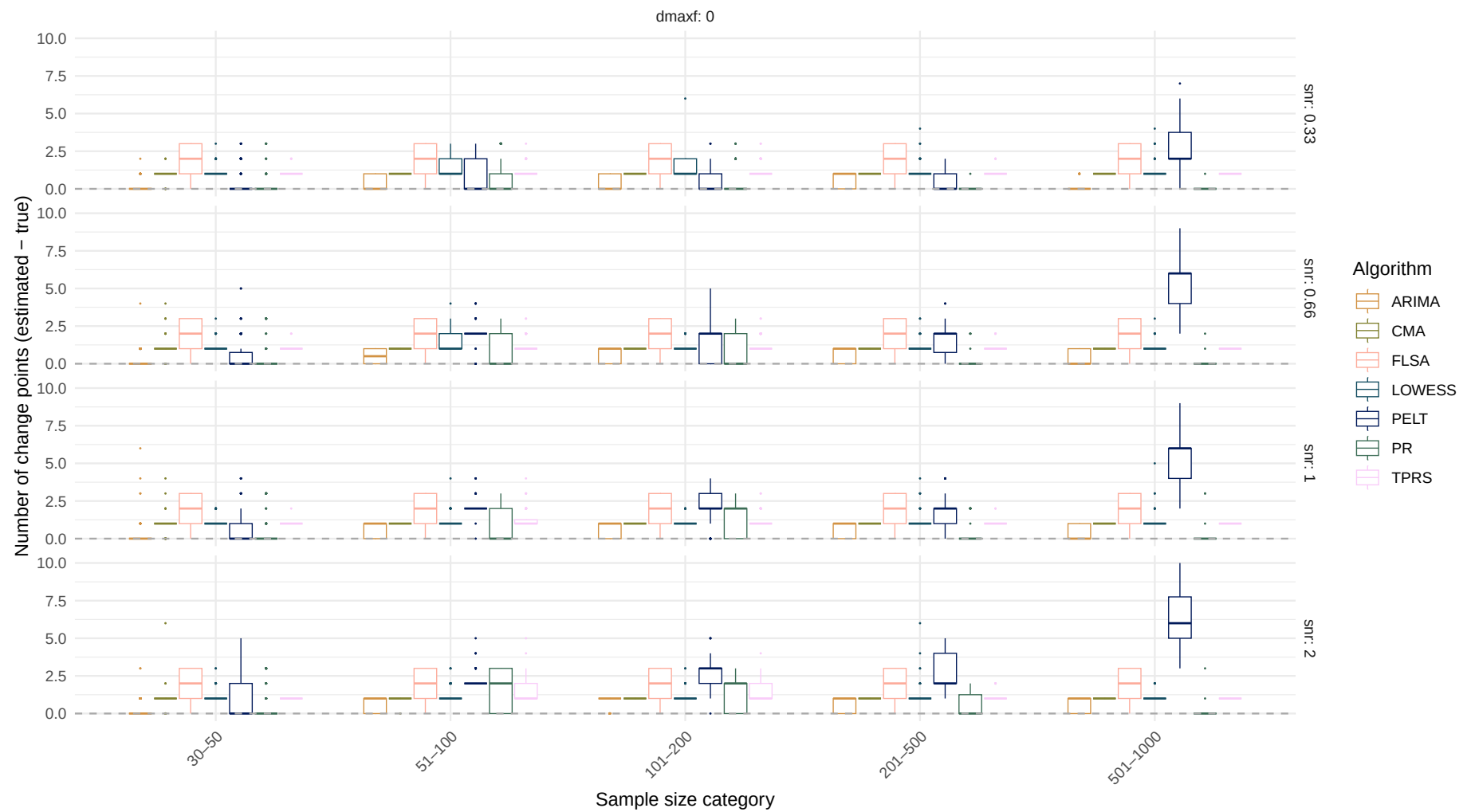
Estimand: Number of change points

Systematic change pattern: No change

Estimated number of change points over sample size
(Dashed line indicates the true number of change points)

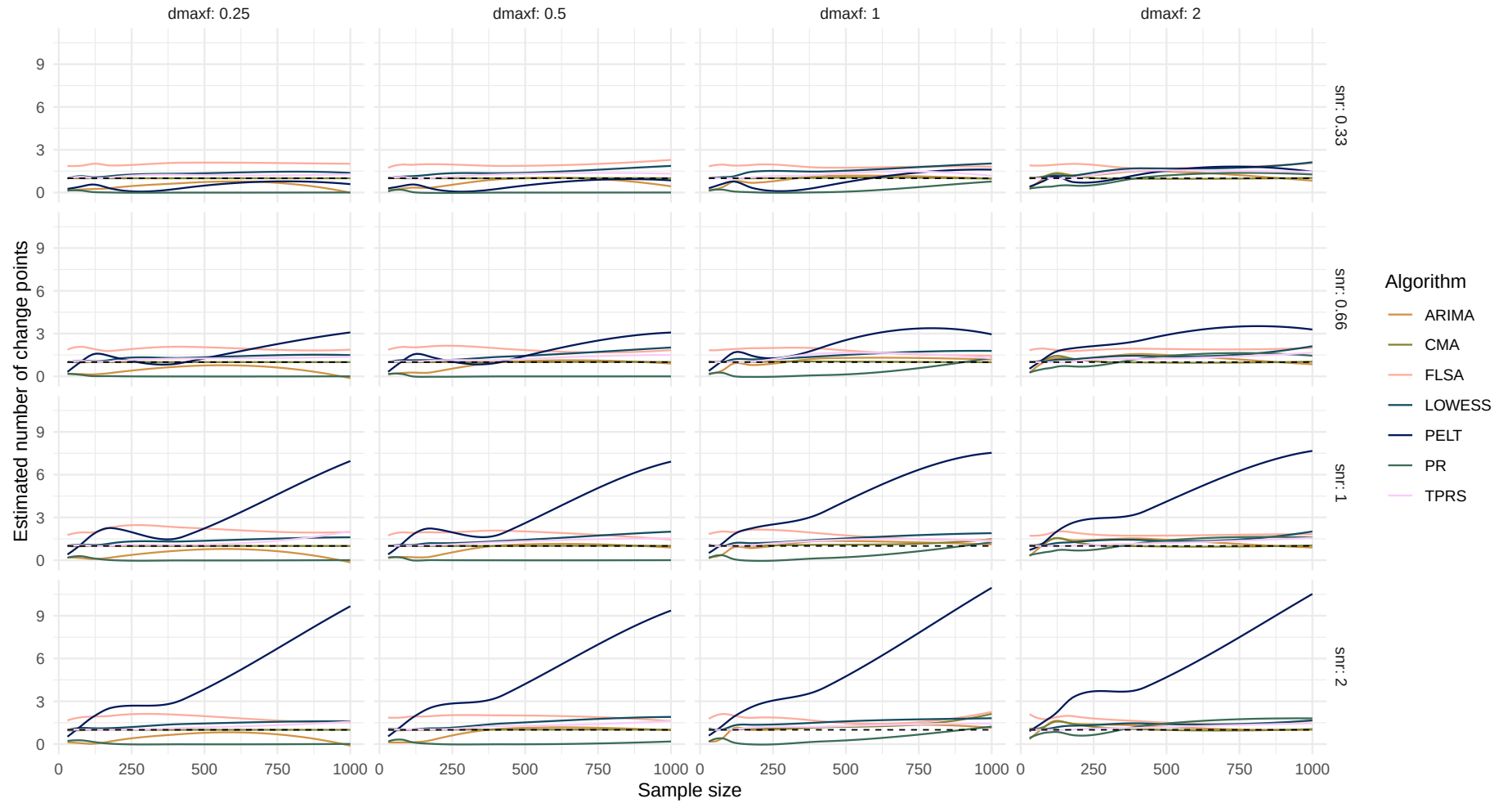


Boxplot of number of change points difference over sample size

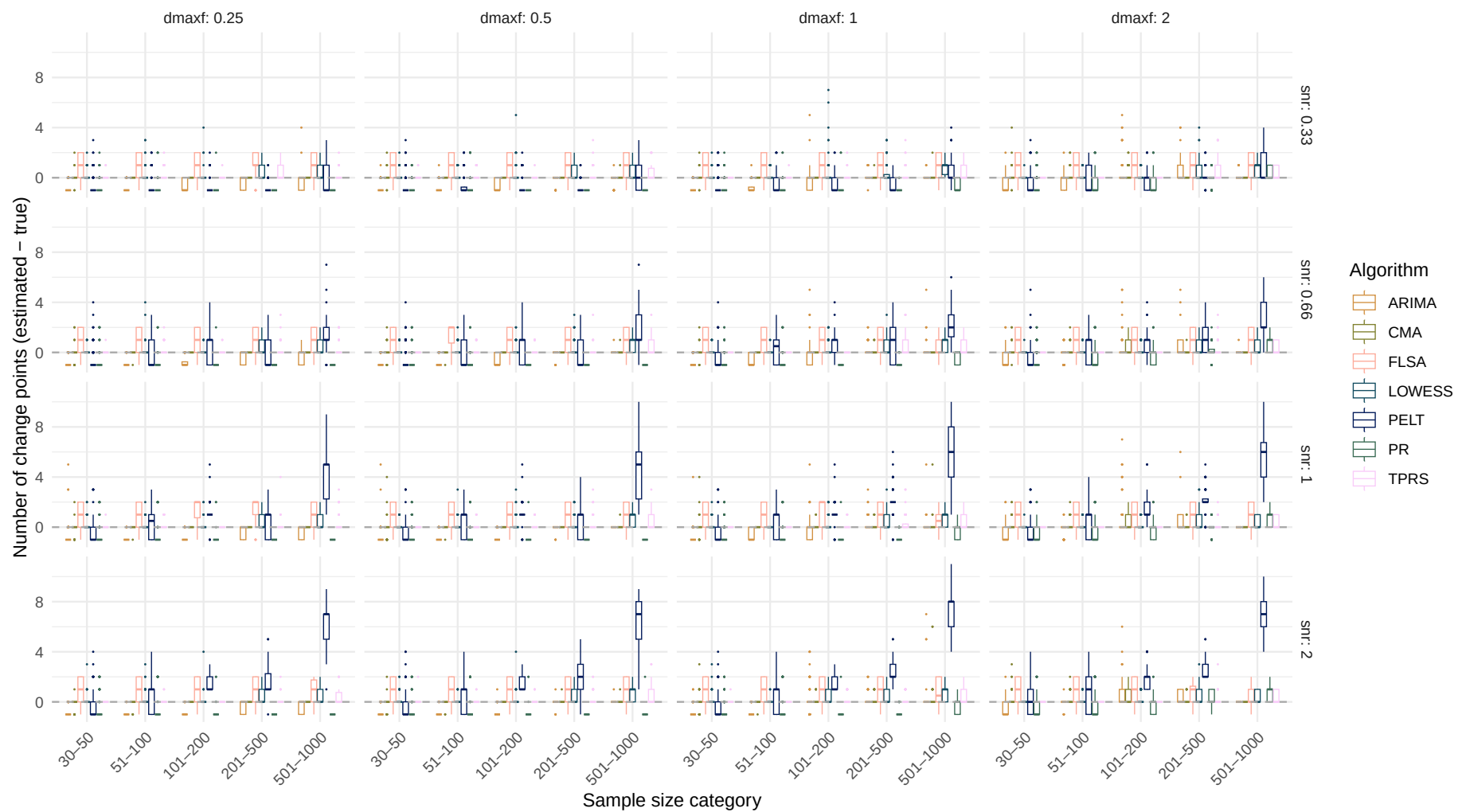


Systematic change pattern: Jump

Estimated number of change points over sample size
(Dashed line indicates the true number of change points)

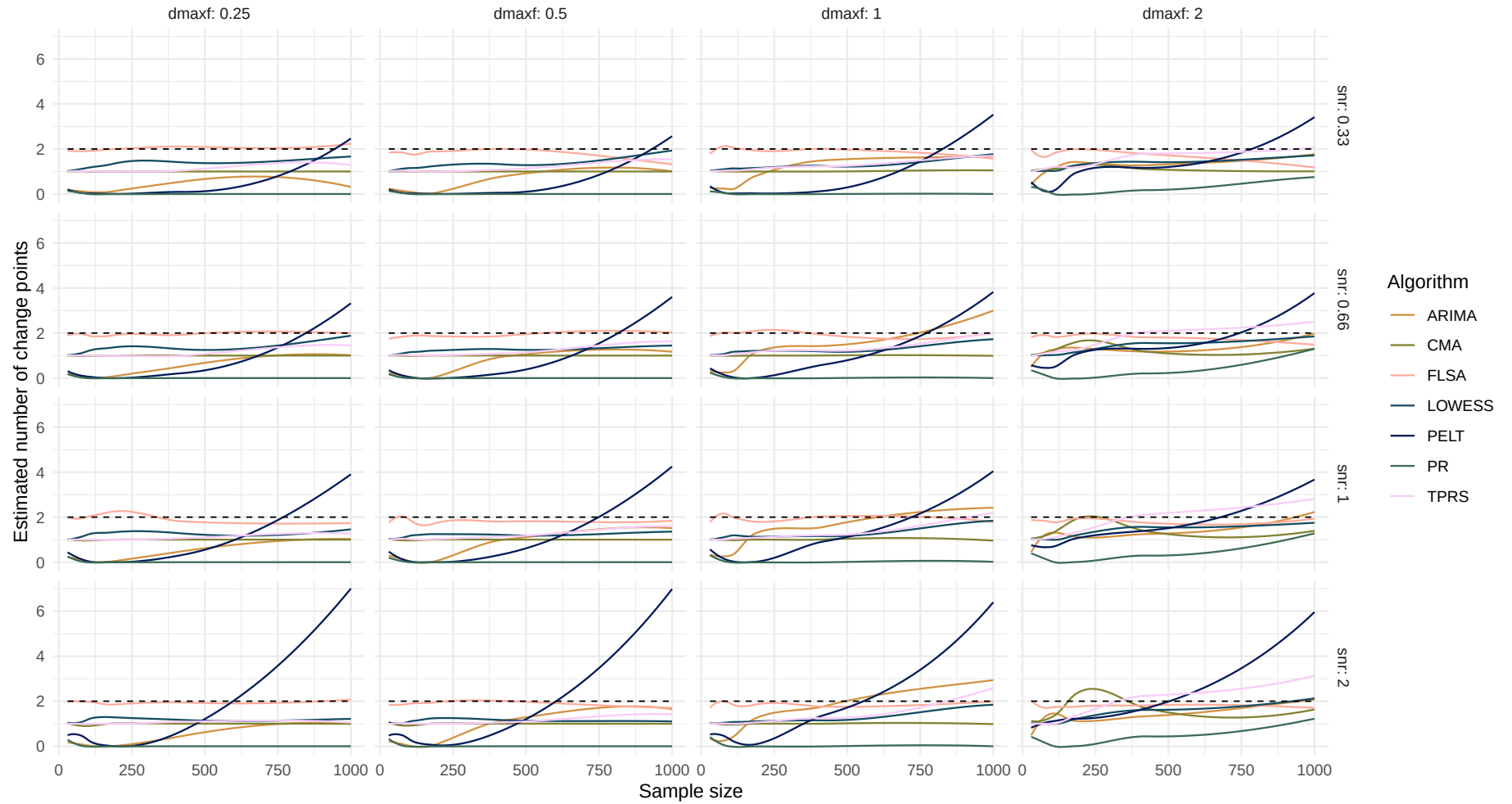


Boxplot of number of change points difference over sample size

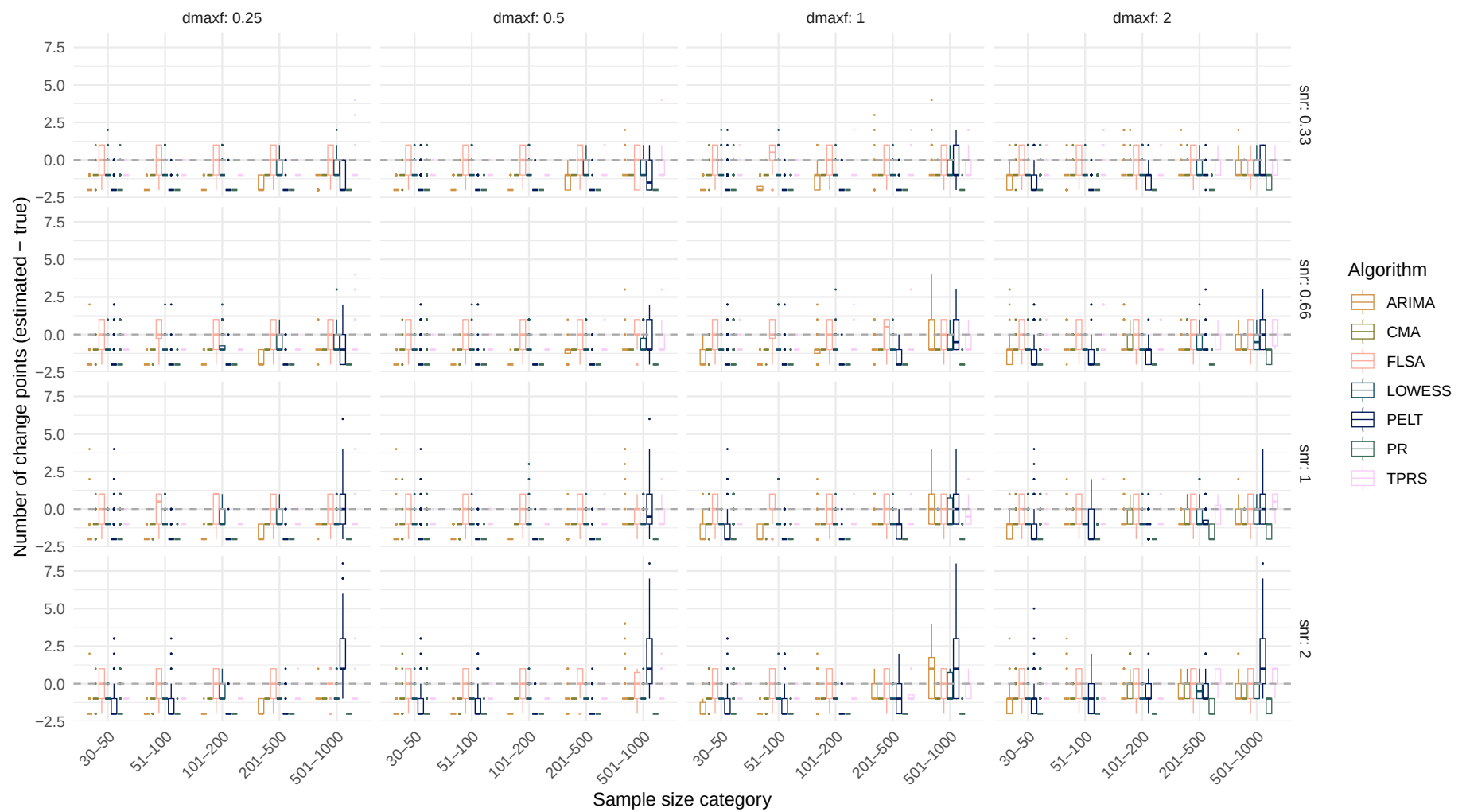


Systematic change pattern: Linear

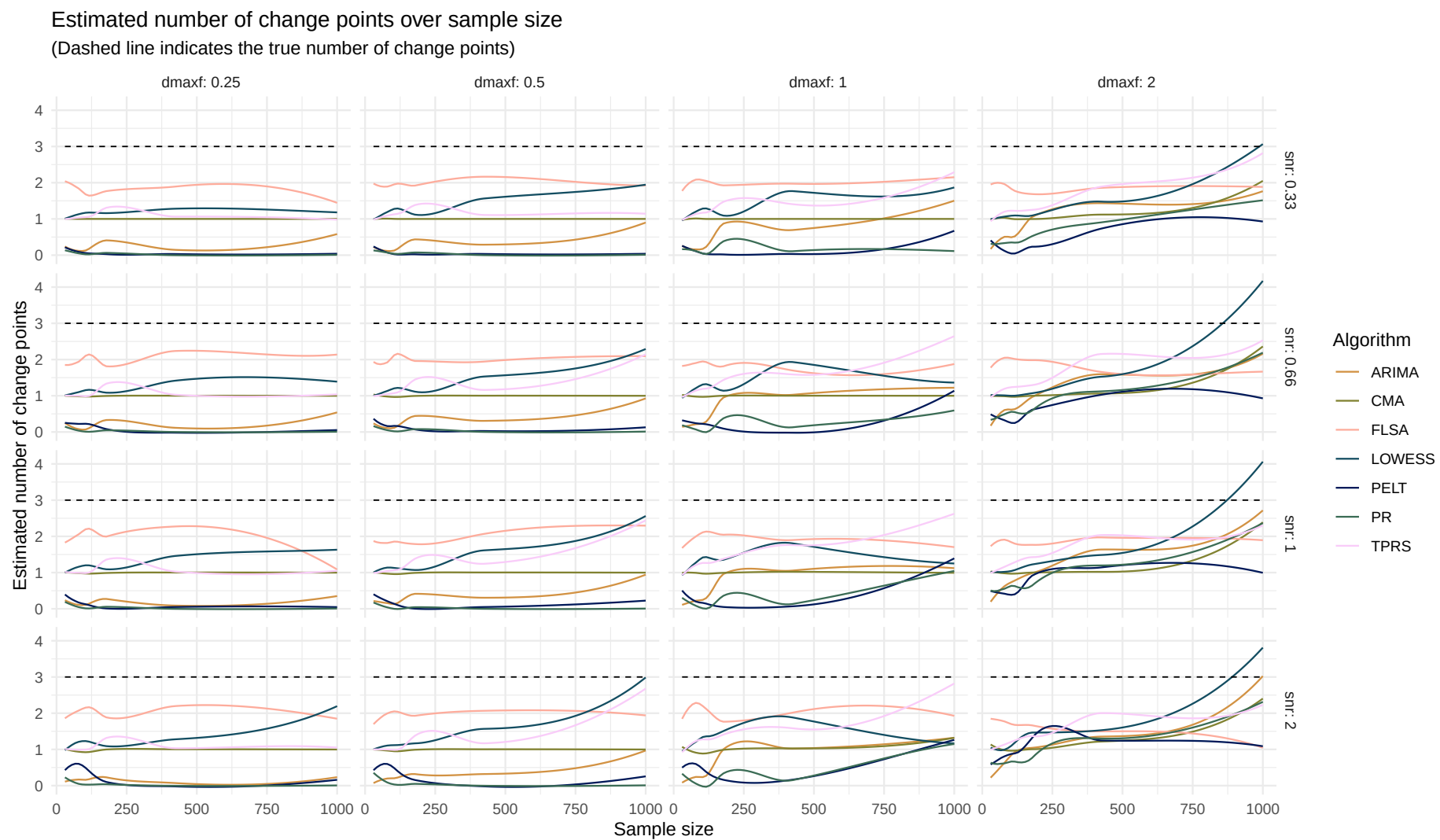
Estimated number of change points over sample size
(Dashed line indicates the true number of change points)



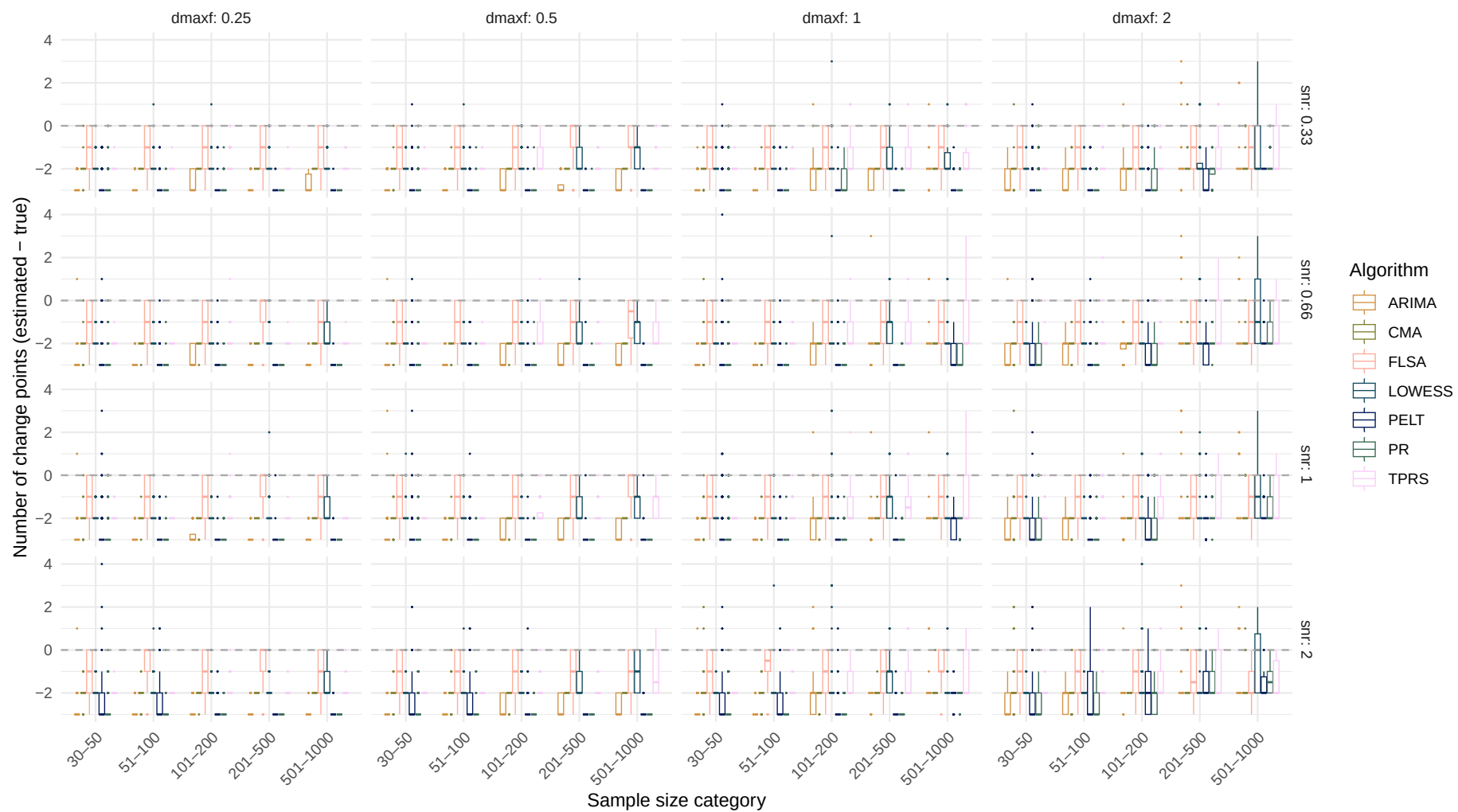
Boxplot of number of change points difference over sample size



Systematic change pattern: Quadratic

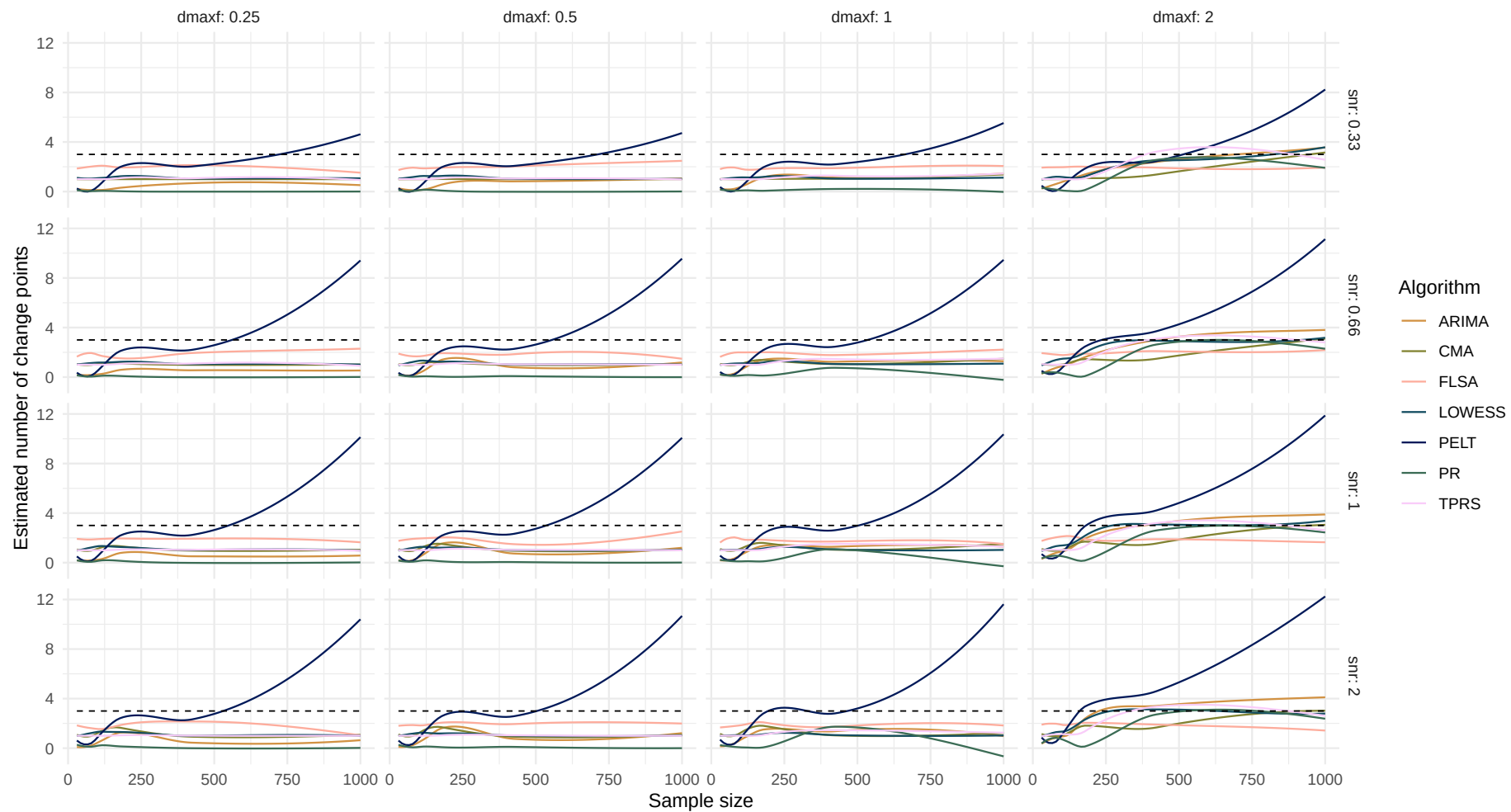


Boxplot of number of change points difference over sample size

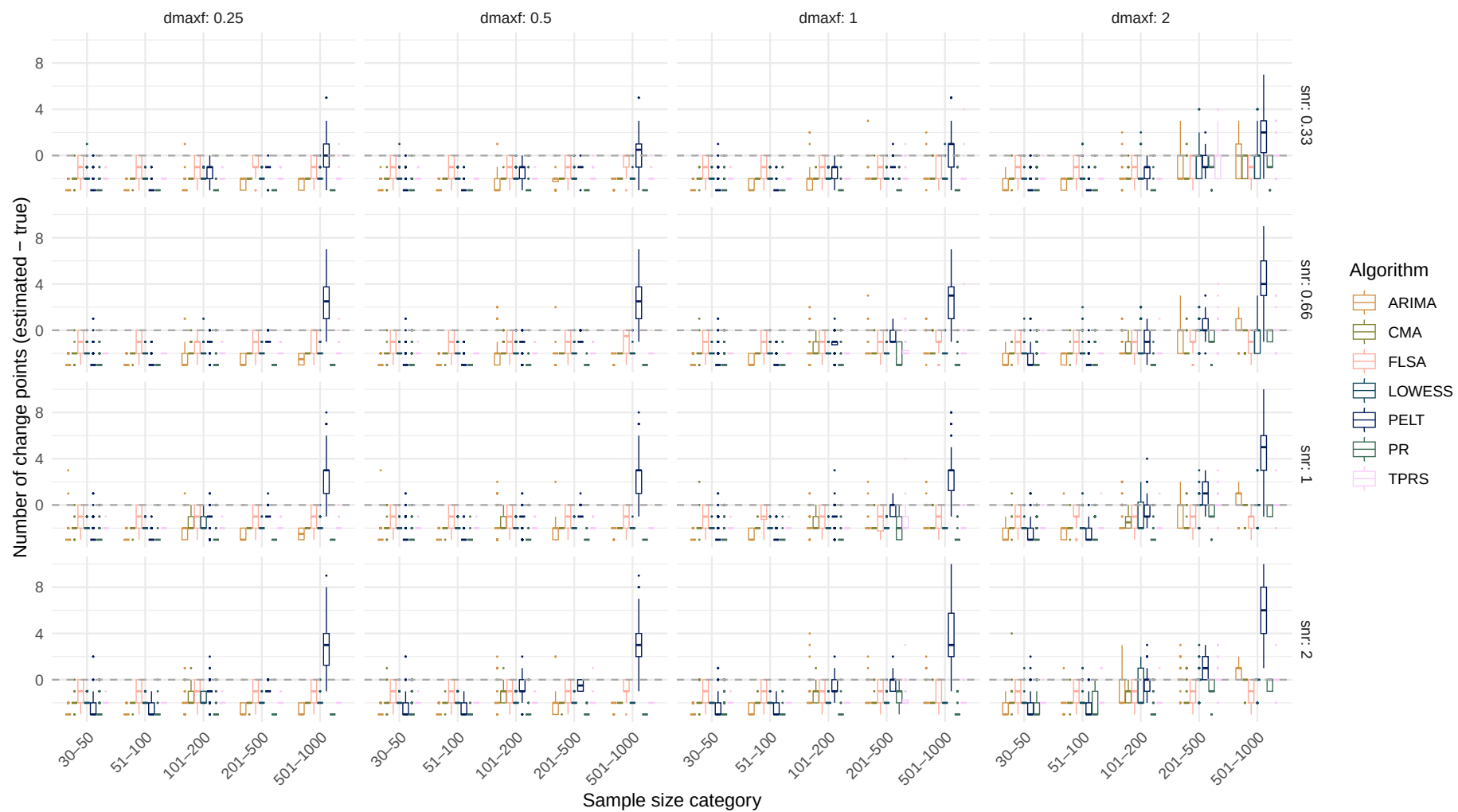


Systematic change pattern: Recurrent

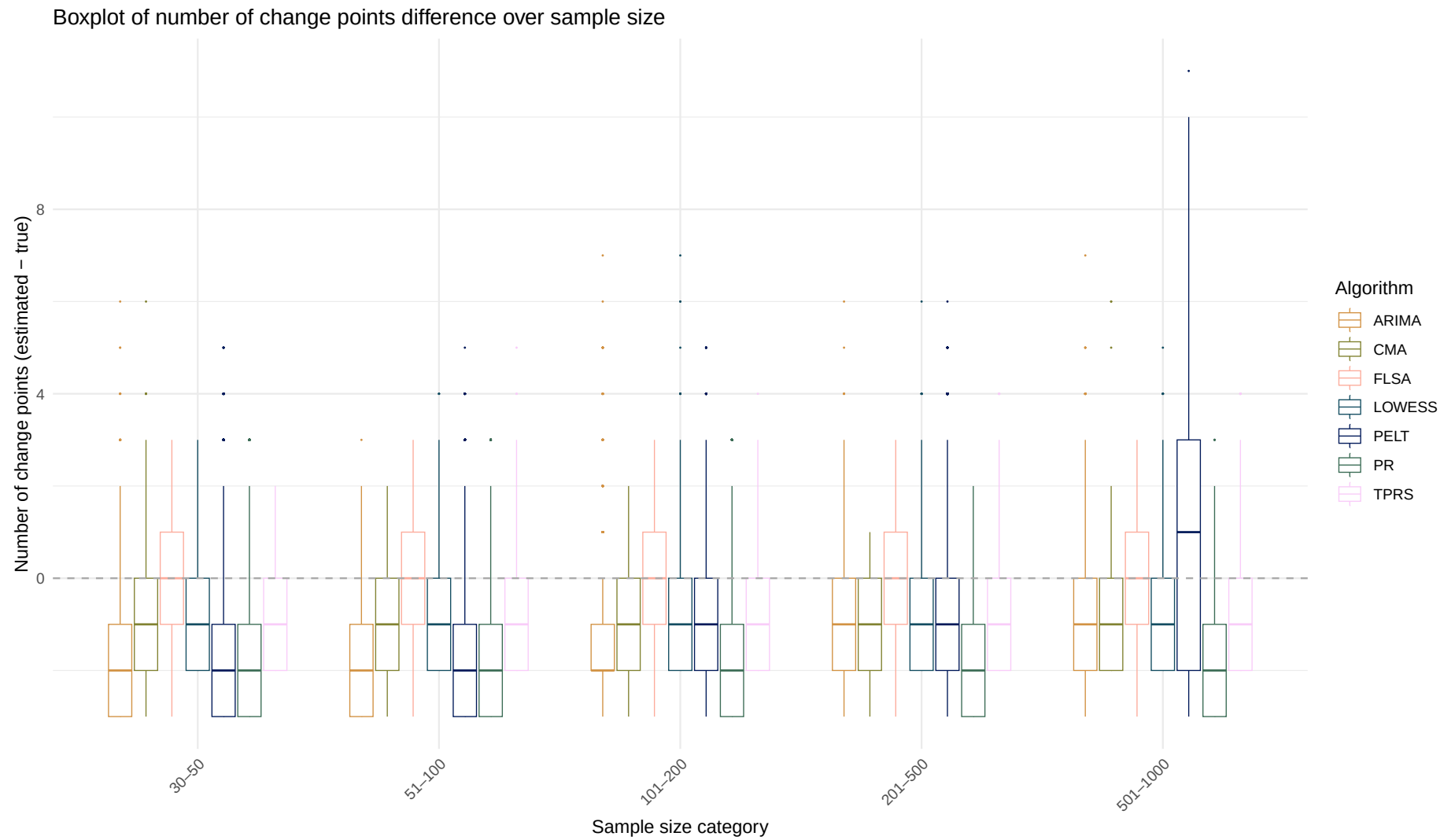
Estimated number of change points over sample size
(Dashed line indicates the true number of change points)



Boxplot of number of change points difference over sample size



Systematic change pattern: All patterns combined



Boxplot of number of change points difference over sample size

